

Placebo Discontinuity Design

Rahul Singh*

Harvard Society of Fellows and Department of Economics

Moses Stewart

Harvard Department of Economics

Original draft: July 2025. This draft: July 2026.

Abstract

Standard regression discontinuity design (RDD) models rely on the continuity of expected potential outcomes at the cutoff. The standard continuity assumption can be violated by strategic manipulation of the running variable, which is realistic when the cutoff is widely known and when the treatment of interest is a social program or government benefit. In this work, we identify the treatment effect despite such a violation, by leveraging a placebo treatment and a placebo outcome. We introduce a local instrumental variable estimator. Our estimator decomposes into two terms: the standard RDD estimator of the target outcome's discontinuity, and a new adjustment term based on the placebo outcome's discontinuity. We show that our estimator is consistent, and we justify a robust bias-corrected inference procedure. Our method expands the applicability of RDD to settings with strategic behavior around the cutoff, which commonly arise in social science.

Keywords: regression discontinuity design; placebo outcome; negative control; causal inference; bias-corrected inference

*The authors thank Joshua Angrist for helpful discussions.

1 Introduction

Regression discontinuity design (RDD) strategies are widely used to estimate the effect of a treatment on an outcome of interest, when the probability of treatment assignment exhibits a discontinuous jump at a known threshold (Thistlethwaite and Campbell, 1960). Applications of RDD are broad and include studies in electoral politics (Hall, 2015; Spenkuch and Toniatti, 2018), education (Cook and Kang, 2016; Goodman et al., 2019), housing markets (Kumar, 2018), and market design (Abdulkadiroğlu et al., 2017; Kong, 2021), among others.

The main identifying assumption in standard RDD is the continuity of expected potential outcomes at the cutoff (Hahn et al., 2001). This assumption ensures that, in the absence of treatment, units just above and below the cutoff are comparable. As discussed in Remark 1, this condition is closely related to the requirement that the conditional distribution of unobserved confounding given the running variable is continuous at the cutoff (Lee and Lemieux, 2010). To empirically assess the plausibility of this assumption, researchers often conduct falsification tests using placebo variables.¹ For example, Imbens and Lemieux (2008) and Cattaneo et al. (2024) recommend checking for discontinuities in placebo outcomes at the cutoff as a diagnostic for invalid RDD. However, when these falsification tests indicate a failure of the RDD continuity assumption, current methods offer no path forward for recovering the treatment effect using these falsification test results.

In this paper, we propose a new identification strategy that leverages a placebo treatment and a placebo outcome to recover the treatment effect even when the continuity of expected potential outcomes may fail. We introduce a local linear instrumental variable estimator that calculates the treatment effect as the difference between the right-limit of the observed outcome and an adjusted left-limit that accounts for unobserved confounding

¹The literature also considers predetermined covariates, of which the placebo treatment may be viewed as a special case.

via placebo variables. The estimator decomposes into a standard RDD estimate for the target outcome, $\hat{\tau}_{\text{rdd}}^y$, minus an adjustment term $(\hat{\tau}_{\text{rdd}}^w)^\top \hat{\gamma}_-$, where $\hat{\tau}_{\text{rdd}}^w$ is the discontinuity in the placebo outcome and $\hat{\gamma}_-$ is an appropriate weight.

Conceptually, we build on the insights and techniques of the placebo variable literature (also called the negative control literature) (Miao et al., 2018; Deaner, 2018; Tchetgen Tchetgen et al., 2024), to answer a salient question about RDD, which is an extremely popular method in social science. Whereas that literature uses placebo variables to adjust for unobserved confounders of random treatments, we use placebo variables to adjust for sorting and strategic manipulation with possibly nonrandom treatment rules. To the best of our knowledge, our identification result appears to be novel and motivates a new estimation problem. For this new estimation problem, we propose and analyze what appears to be a novel variation of the widely used local linear RDD estimator (Fan and Gijbels, 1992), and we derive bias-corrected inference (Calonico et al., 2014, 2019). Future work may develop bias-aware inference (Armstrong and Kolesár, 2018; Imbens and Wager, 2019; Noack and Rothe, 2024).

We contribute to a literature that studies estimation of treatment effects at a cutoff when the standard RDD continuity assumption fails. For a French regulation that binds on firms exceeding 50 employees, Garicano et al. (2016) recover the effect of crossing the threshold by imposing a parametric model of firm production; we instead recover the RDD treatment effect nonparametrically. Eckles et al. (2020) provide identification and inference results for the RDD treatment effect without the standard continuity assumption, in settings where the researcher *knows* the conditional distribution of the running variable given the unobserved variables, $f(d \mid u, \eta_y)$. We instead study settings in which this distribution is unknown, where using standard RDD strategies would require justifying that there is no unobserved confounding around the cutoff. Most closely related, Gerard et al. (2020) bound the treatment effect when the running variable is manipulated around the cutoff.

We complement their partial-identification approach by providing conditions under which the effect is point identified.

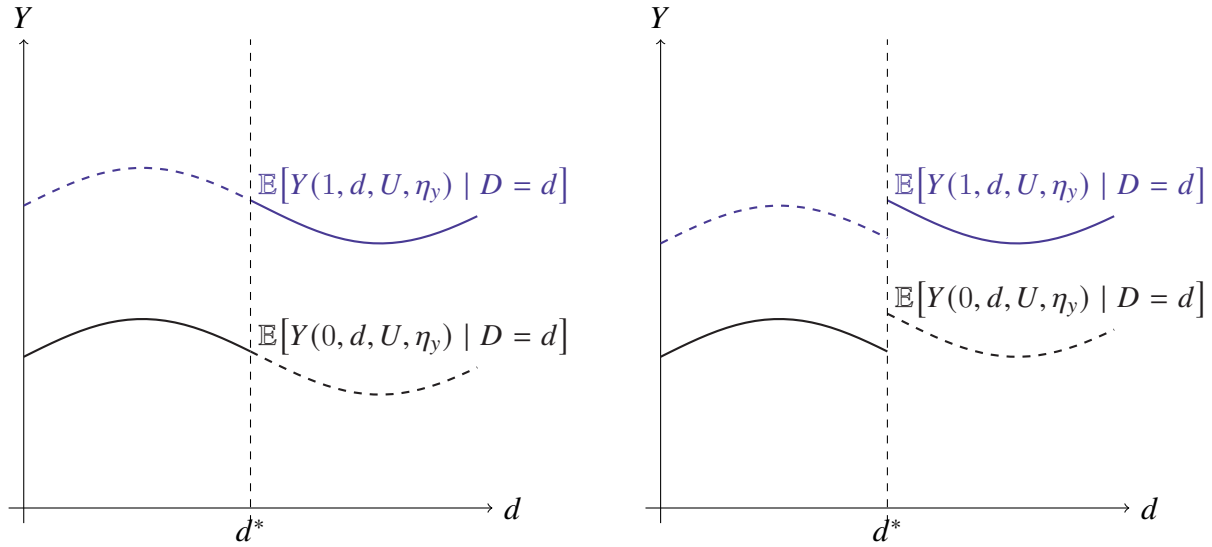
A related literature augments standard RDD inference using auxiliary plots and variables while maintaining the continuity of potential outcomes, and so rules out the unobserved confounding we study. A “difference-in-discontinuities” strategy combines RDD estimates across populations to isolate a treatment effect when several factors besides the treatment change at the threshold (Grembi et al., 2016; Galindo-Silva et al., 2018; Butts, 2023; Picchetti et al., 2024; Leventer and Nevo, 2024); this decoupling of co-occurring changes is sometimes called confounding, but it differs from the unobserved confounding around the cutoff that we address. When distinct cutoffs apply to different groups, Bertanha (2020) and Cattaneo et al. (2021) extend RDD identification to trace a dose-response curve in a neighborhood of the cutoffs. A third strand combines additional variables with auxiliary assumptions to extrapolate the RDD effect away from the cutoff (Angrist and Rokkanen, 2015; Dong and Lewbel, 2015; Deaner and Kwon, 2025). Like these papers, we use auxiliary structure (here placebo variables) together with additional assumptions to augment RDD inference; unlike them, we attain point identification and nonparametric inference precisely when the continuity assumption fails at the cutoff.

The remainder of the paper is organized as follows. Section 2 illustrates the main idea through a real-world application. Section 3 presents our identification assumptions and main result. Section 4 derives our estimator. Section 5 establishes consistency and asymptotic normality with bias-corrected inference. Section 6 concludes. Formal proofs are deferred to the appendix.

2 Example: Maimonides rule

To motivate our approach, consider an example adapted from Angrist et al. (2019). Suppose we are interested in estimating the effect of transitioning from a large class (39-40 pupils) to a small class (20-21 pupils), denoted by the binary treatment A . The outcome of interest is student achievement Y . A Maimonides-style rule determines class size: schools must split classes when total enrollment in a grade level D exceeds a statutory threshold $d^* = 40$. For instance, a school with $D = 42$ students in a grade level receives funding for two classes of 21 students ($A = 1$), while a school with exactly $D = 40$ students in a grade level keeps one class of 40 students ($A = 0$).

Figure 1: Valid and invalid regression discontinuity design



Notes: Figure 1 illustrates the standard continuity assumption on the expected potential outcomes for a valid RDD (on the left) and an invalid RDD (on the right). The Y -axis gives the expected test score $\mathbb{E}[Y(a, d, U, \eta_y) | D = d]$ for a student in a class of 20-21 students ($A = 1$) or 39-40 students ($A = 0$) given $D = d$ enrolled students in the school. The x -axis gives the number of students enrolled in the school. The dotted regions of the line indicate the unobserved, counterfactual expected outcomes.

To apply a conventional RDD estimator in this setting, one must assume that the expected potential outcomes $\mathbb{E}[Y(1, d, U, \eta_y) | D = d]$ and $\mathbb{E}[Y(0, d, U, \eta_y) | D = d]$ are continuous at the cutoff d^* . Here, $Y(A, D, U, \eta_y)$ denotes the test score of a student in a class

size of 20-21 students ($A = 1$) or 39-40 students ($A = 0$), given the total enrollment in the grade level (D). We model unobserved heterogeneity through two components—school administrator sophistication U , which may influence total enrollment near the cutoff, and student-level heterogeneity η_y , which is random. The left panel of Figure 1 depicts a valid RDD scenario under this assumption.

Remark 1 (Continuity and unobserved confounding). As noted by Lee and Lemieux (2010), the standard RDD assumption, i.e. continuity of the expected potential outcomes at the cutoff, is implied by the following two conditions: for all (u, η_y) , (i) the conditional distributions $d \mapsto \mathbb{P}(d \mid u, \eta_y)$ and $d \mapsto \mathbb{P}(d)$ are continuous at d^* ; and (ii) the potential outcome functions $d \mapsto Y(1, d, u, \eta_y)$ and $d \mapsto Y(0, d, u, \eta_y)$ are bounded and continuous at d^* .

To see this, note that continuity of $\mathbb{P}(d \mid u, \eta_y)$ and $\mathbb{P}(d)$ implies continuity of $\mathbb{P}(u, \eta_y \mid d)$ by Bayes' rule.² Then, applying the dominated convergence theorem, we can exchange the limit and expectation, so that

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \mathbb{E}[Y(1, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon] &= \lim_{\epsilon \downarrow 0} \int Y(1, d^* + \epsilon, u, \eta_y) \, d\mathbb{P}(u, \eta_y \mid d^* + \epsilon) \\ &= \int Y(1, d^*, U, \eta_y) \, d\mathbb{P}(u, \eta_y \mid d^*) = \mathbb{E}[Y(1, d^*, U, \eta_y) \mid D = d^*]. \end{aligned}$$

The same argument applies from the left. ◀

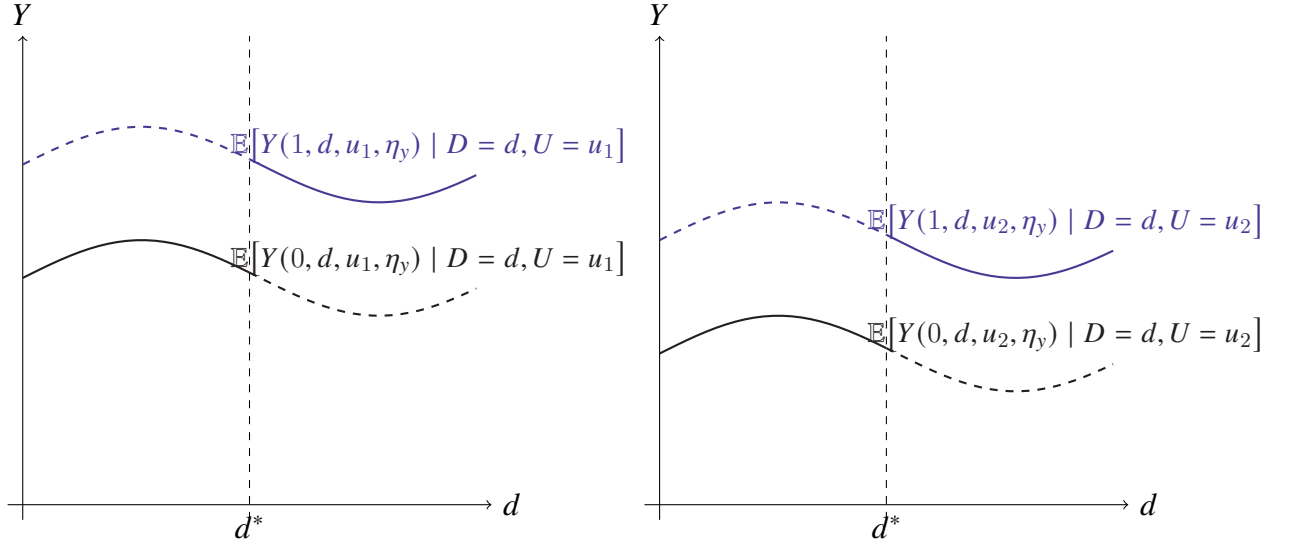
However, this assumption may fail if school administrators behave strategically. For instance, more sophisticated school leaders may recognize the budgetary and pedagogical advantages of exceeding the class size threshold, perhaps by enrolling just above the cutoff. These administrators may also tend to run more effective schools with higher-achieving students, thereby inducing a spurious upward jump in test scores around d^* that is not due to class size per se (Angrist et al., 2019). In such a case, the unobserved aptitude of school leaders (U) confounds the relationship between enrollment (D) and student achievement (Y) near the cutoff. As a result, schools just above d^* may perform better because they

²This holds only at points where $\mathbb{P}(d) > 0$.

are led by more capable administrators, invalidating the continuity assumption that would allow researchers to evaluate the effect of class size. This confounded scenario is depicted in the right panel of Figure 1.

Our identification strategy recovers the treatment effect in such confounded settings by positing that continuity holds after further conditioning on the unobserved school-level aptitude U . As shown in Figure 2, we require that, conditional on both $D = d$ and $U = u$, the expected potential outcomes become continuous at the cutoff. We then exploit the statistical relationship between test scores in the prior year W , which serve as placebo outcomes, and the availability of transfer-eligible students based on birth-dates Z , which act as placebo treatments, to appropriately adjust for the unobserved confounder U .

Figure 2: Valid placebo discontinuity design



Notes: Figure 1 illustrates the relaxed assumption we place on the expected potential outcomes. The Y -axis gives the test score $\mathbb{E}[Y(a, d, u, \eta_y) \mid D = d, U = u]$ for a student in a class of 20-21 students ($A = 1$) or 39-40 students ($A = 0$) enrolled in a school with $D = d$ pupils and administrator aptitude level $U = u$. The x -axis gives the number of students enrolled in the school. The dotted regions of the line indicate the unobserved, counterfactual outcome.

Intuitively, test scores in a previous year are unaffected by enrollment (D) in the current year, making it a valid placebo outcome (W). In this sense, previous test scores (W) serve as a proxy for the school administrator aptitude (U).

Meanwhile, the availability of students able to transfer across grades should only affect student test scores (Y) through its influence on total enrollment in the grade (D), making it a valid placebo treatment (Z). The availability of transfer students is like a local instrument, or a special type of pre-determined covariate, that can help to distill the variation due to school administrator aptitude.

3 Identification: Placebo discontinuity

In this section, we formalize the model, interpret the key assumptions, and present our main identification result: how to recover the treatment effect in an RDD setting with strategic behavior around the cutoff. Standard practice uses placebo variables to detect this issue; as our first contribution, we use placebo variables to correct this issue.

3.1 Notation

Consider the nonseparable model

$$Y = Y(A, D, U, \eta_y),$$

where A is the treatment, D is the running variable, U is unobserved confounding, and η_y is unobserved heterogeneity. Notice that we decompose the unobservable into the component U for which we will have a model, and the component η_y which will be exogenous conditional on U .

We will study the usual causal parameter in RDD,

$$\tau_0 = \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*],$$

which reflects the average causal effect of moving from untreated to treated at the cutoff d^* , conditional on the running variable.

In contrast with standard approaches, we allow for discontinuities in the conditional

distribution of unobservables $f(u, \eta_y \mid d)$ at d^* , acknowledging that strategic behavior or sorting can lead to non-smooth densities near the cutoff. However, we require the density to remain strictly positive in a neighborhood of d^* to maintain identifiability.

We employ placebo variables (Z, W) to handle unobserved confounding. The full structural system is written as,

$$Y = Y(A, D, Z, U, \eta_y), \quad W = W(A, D, Z, U, \eta_w), \quad A = A(D, Z, U, \eta_a).$$

We summarize the invariances of the structural system in the next section. A graphical summary of the causal structure is previewed in Figure 3.

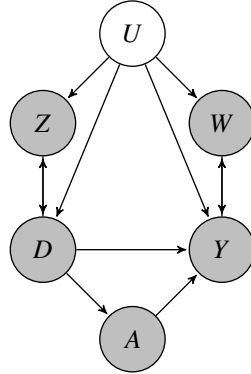


Figure 3: Placebo discontinuity design DAG

3.2 Assumptions

The following assumptions identify τ_0 even when the expected potential outcomes $d \mapsto \mathbb{E}[Y(0, D, U, \eta_y) \mid D = d]$ and $d \mapsto \mathbb{E}[Y(1, D, U, \eta_y) \mid D = d]$ are discontinuous at the cutoff $d = d^*$.

Throughout, we place assumptions in some neighborhood of the threshold d^* defined as,

$$\mathcal{D}_-(\epsilon) = (d^* - \epsilon, d^*), \quad \mathcal{D}_+(\epsilon) = (d^*, d^* + \epsilon), \quad \mathcal{D}(\epsilon) = \mathcal{D}_-(\epsilon) \cup \mathcal{D}_+(\epsilon).$$

Assumption 1 (RDD). *Assume that the following limits exist and are unequal:*

$$\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* - \epsilon]], \quad \lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon]].$$

The existence of this limit, for the expectation of the treatment assignment A given the running variable D , is standard in the RDD identification literature ([Hahn et al., 2001](#); [Lee and Lemieux, 2010](#)).

Definition 1 (Sharp design). Under a “sharp” design, we assume that there is no heterogeneity in treatment assignment given the running variable (η_a does not exist) and for all $\tilde{\epsilon} \in \mathcal{D}_+(\epsilon) - d^*$,

$$\mathbb{E}[A \mid D = d^* - \tilde{\epsilon}] = 0, \quad \mathbb{E}[A \mid D = d^* + \tilde{\epsilon}] = 1.$$

A “fuzzy” design is one that is not sharp, i.e. one that does not satisfy the equalities above.

In a sharp design, A is fully determined by D , implying no heterogeneity in treatment assignment (i.e., no η_a). In this case, Assumption 1 is trivially satisfied.

We define $\mathcal{Z}(\epsilon)$ such that $\mathbb{P}(Z \in \mathcal{Z}(\epsilon) \mid D \in \mathcal{D}(\epsilon)) = 1$.

Assumption 2 (Placebo variables). *Assume there exists some $\epsilon > 0$ such that for all $d \in \mathcal{D}(\epsilon)$ and for all $z \in \mathcal{Z}(\epsilon)$,*

a (Causal consistency): if $A = a$, $D = d$, and $Z = z$ then $Y = Y(a, d, z, U, \eta_y)$, and $W = W(a, d, z, U, \eta_w)$ almost surely. If $D = d$ and $Z = z$ then $A = A(d, z, U, \eta_a)$ almost surely.

b (Placebo selection on unobservables): $Z \perp\!\!\!\perp \eta_y, \eta_a \mid U, D$ and $\eta_w \perp\!\!\!\perp D, Z \mid U$.

c (Overlap): if $f(u) > 0$ then $f(a, d, z \mid u) > 0$, where we assume the densities exist.

d (Exclusion): $Y(a, d, z, U, \eta_y) = Y(a, d, U, \eta_y)$, $W(a, d, z, U, \eta_w) = W(U, \eta_w)$, and $A(d, z, U, \eta_a) = A(d, \eta_a)$ almost surely.

In summary, we have the following simplification local to the cutoff:

$$Y = Y(A, D, U, \eta_y), \quad W = W(U, \eta_w), \quad A = A(D, \eta_a).$$

Causal consistency rules out network effects near the cutoff.

The main condition within placebo selection on observables is the implication that $Z \perp\!\!\!\perp \eta_y, \eta_w | D, U$: conditional upon the running variable and unobserved confounder, the placebo treatment is as good as random for the target outcome and placebo outcome.³ In the sharp design, $Z \perp\!\!\!\perp \eta_a | D, U$ automatically holds; in the fuzzy design, it imposes that the placebo treatment is conditionally independent of the idiosyncratic aspect of treatment assignment. As shown in Appendix B, $Z \perp\!\!\!\perp \eta_y, \eta_a, \eta_w | D, U$ together with an additional continuity condition suffice for our argument to work. In the main text, to simplify the exposition, we strengthen $\eta_w \perp\!\!\!\perp Z | D, U$ to $\eta_w \perp\!\!\!\perp D, Z | U$ and thereby avoid the additional continuity.

Overlap ensures there is no stratum of unobserved confounding for which the treatment, running variable, or placebo treatment are deterministic. In the language of the RDD literature, we rule out complete control but allow imprecise control (Lee and Lemieux, 2010).

Exclusion encodes the idea that the placebo treatment does not directly cause the outcome of interest; instead, it affects the outcome via the running variable. Exclusion also encodes the idea that the placebo outcome is not caused by the treatment, running variable, or placebo treatment. Finally, exclusion imposes that the treatment is pinned down by only the running variable (and possibly idiosyncratic noise in the fuzzy setting).

Assumption 3 (Conditional independence). *There exists some $\epsilon > 0$ such that for all $d \in \mathcal{D}(\epsilon)$, $\eta_a \perp\!\!\!\perp \eta_y, U | D = d$.*

Assumption 3 is vacuously satisfied by the sharp design. In the fuzzy design, it is

³By weak union, $\eta_w \perp\!\!\!\perp D, Z | U$ implies $\eta_w \perp\!\!\!\perp Z | D, U$.

analogous to [Hahn et al. \(2001, Condition C3\)](#): it implies that conditional on D , the randomness in the treatment assignment mechanism is unrelated to the idiosyncratic aspect of potential outcomes and the unobserved confounding. This assumption in the fuzzy design may be relaxed by imposing homogeneity of effects; see [Appendix C](#).

Assumptions [2](#) and [3](#) can be summarized by the results below:

Lemma 1 (Necessary independences). *Assumption [2](#) implies that for all $d \in \mathcal{D}(\epsilon)$ and $z \in \mathcal{Z}(\epsilon)$,*

$$a \ Y \perp\!\!\!\perp Z \mid D, U$$

$$b \ W \perp\!\!\!\perp D, Z \mid U^4$$

Additionally under Assumption [3](#) it follows that,

$$c \ A \perp\!\!\!\perp U \mid D$$

Example 1 (Maimonides rule). Recall the Maimonides rule example from [Section 2](#), where class size A is determined by grade-level enrollment D , and we study its effect on student achievement Y in grade t . Administrator ability U is unobserved, and strategic manipulation around the cutoff induces a jump in $f(d \mid u, \eta_y)$ (and hence $f(u, \eta_y \mid d)$): sophisticated administrators who are aware of the threshold d^* may inflate enrollment to open an additional class. Let Z denote the availability of students to transfer across grade levels, and W denote achievement in grade $t - 1$.

$Y \perp\!\!\!\perp Z \mid D, U$ requires that student transfer availability is independent of test scores conditional on enrollment in grade t and administrator ability. This is plausible because transfer availability is determined by demographic factors which are unrelated to how

⁴The conditional independence restriction on our placebo outcome W resembles the conditional independence assumption imposed on the outcome Y in [Angrist and Rokkanen \(2015\)](#). Here, the unobserved confounder U in [Lemma 1b](#) plays the role of the covariates X , and allows us to attribute changes in the conditional distribution $f(w \mid d)$ at the cutoff to the unobserved confounder U .

students perform on exams under conditioning. Angrist et al. (2019) use Z as an instrument in their IV specification under the stronger condition $Y \perp\!\!\!\perp Z \mid D$.

$W \perp\!\!\!\perp D, Z \mid U$ requires that prior-year test scores in grade $t - 1$ are independent of current enrollment and transfer availability in grade t , conditional on administrator ability. A straightforward timing argument supports this: scores from earlier grades carry no information about current enrollment or transfer availability.

$A \perp\!\!\!\perp U \mid D$ requires that administrator heterogeneity can only affect class size through enrollment. This holds because the Maimonides rule is a deterministic function of D : once enrollment is fixed, the number of classes is mechanically determined by the rule, leaving no room for administrator discretion.

Finally, the overlap condition in Assumption 2c requires that school officials cannot perfectly control enrollment D . This is credible because enrollment depends on many demographic factors outside the administrator’s control, even if some manipulation occurs at the margin. This accommodates the jump in $d \mapsto f(d \mid u, \eta_y)$ that motivates our approach. ◀

Example 2 (Birth-weight classification). Consider the empirical setting of Almond et al. (2005), who study the effect of being classified as “critically underweight” on infant mortality, where the classification is triggered when an infant’s reported birth-weight falls below the 1500 gram threshold. The running variable D is the infant’s reported birth-weight; the treatment A is the indicator for being classified as critically underweight; the outcome Y is infant mortality; and the unobserved confounder U is healthcare quality. The concern motivating the design is that U confounds the running variable: better doctors may deliberately report birth-weights just below 1500 grams so that their patients receive the additional care reserved for critically underweight infants. This strategic manipulation around the cutoff induces a jump in $f(d \mid u, \eta_y)$, invalidating the standard continuity argument. To recover the treatment effect, let Z be an indicator for whether the infant was born at night (when

hospitals are understaffed), and W be the mother’s education level.

$Y \perp\!\!\!\perp Z \mid D, U$, requires that the time of day at which the infant is born is independent of mortality once we condition on healthcare quality and reported birth-weight. This is plausible: holding healthcare quality fixed, the time of birth should have no remaining effect on whether the infant survives.

$W \perp\!\!\!\perp D, Z \mid U$, requires that the mother’s education level is independent of both the reported birth-weight and the time of birth given healthcare quality. Holding the quality of care fixed, there is no plausible channel through which the mother’s education can influence the reported birth-weight or the timing of delivery, so this restriction likely holds.

$A \perp\!\!\!\perp U \mid D$, requires that healthcare quality affects the weight classification only through the reported birth-weight. This holds by construction: the classification A is a deterministic function of D , so once the reported birth-weight is fixed, whether the infant is labeled critically underweight is fully determined by the threshold rule, leaving no residual dependence on U .

Finally, the overlap condition in Assumption 2c requires that doctors cannot perfectly control the reported birth-weight D . This is reasonable: an infant weighing well above 1500 grams cannot credibly be reported as weighing below the threshold. $\blacktriangleleft$

Assumption 4 (Continuity). *For all $d \in D(\epsilon)$, assume that $d \mapsto \mathbb{E}[Y(1, D, U, \eta_y) \mid D = d, U = u]$ and $d \mapsto \mathbb{E}[Y(0, D, U, \eta_y) \mid D = d, U = u]$ are continuous for all u .*

Assumption 4 weakens the standard continuity assumption of the RDD literature: continuity of potential outcomes conditional upon the running variable $D = d$ near the cutoff d^* . Instead, it requires continuity of potential outcomes conditional upon both $D = d$ and $U = u$.

While the density of student enrollment may jump at d^* , we anticipate that potential student achievement is continuous when fixing the administrator’s aptitude $U = u$.

Lemma 2 (Limit). *Under a fuzzy design and Assumptions 1, 2, 3, and 4,*

$$\tau_0 = \frac{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^*]}{\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon] - \mathbb{E}[A \mid D = d^* - \epsilon]]}.$$

In a sharp design, and under Assumptions 1, 2, and 4,

$$\tau_0 = \mathbb{E} \left[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^* \right].$$

Lemma 2 solves for the treatment effect τ_0 in an interpretable way: for each value of the unobserved confounded $U = u$, conduct RDD conditional upon $U = u$, then average over U . In our running example, this is equivalent to averaging the treatment effect of a class size (A) on test scores in year- t (Y) for schools with enrollment levels at d^* over different administrator-ability levels U .

Clearly, this calculation is infeasible, since U is unobserved. We need additional assumptions to identify τ_0 in terms of observed variables.

3.3 Main identification result

Section 3.2 established that the causal estimand τ_0 can be written in terms of conditional expectations over unobserved confounding (U). The key challenge now is to express τ_0 in terms of observable variables. To do this, we adapt the “confounding bridge” approach (Tchetgen Tchetgen et al., 2024), which uses auxiliary variables to solve an integral equation that recovers the influence of U on Y .

Assumption 5 (Confounding bridge). *For all $d \in \mathcal{D}_+(\epsilon)$, there exists a bounded solution to the following operator equation almost surely,*

$$\mathbb{E}[Y \mid D = d, U] = \int h_+(d - d^*, w) \, d\mathbb{P}(w \mid D = d, U).$$

Similarly, for all $d \in \mathcal{D}_-(\epsilon)$ we have,

$$\mathbb{E}[Y \mid D = d, U] = \int h_-(d - d^*, w) \, d\mathbb{P}(w \mid D = d, U).$$

The confounding bridge assumption states that, for fixed values of the running variable (D) and unobserved confounder (U), the conditional expectation of the outcome (Y) can be expressed as a function of a proxy variable (W) that is associated with (U). The functions h_+ and h_- act as weighting kernels that summarize how W captures the influence of U on Y on either side of the threshold.

Example 1 (Maimonides rule). In the Maimonides rule example, for schools with the same number of enrolled students ($D = d$), there exists a function $h_+(d - d^*, \cdot)$ such that we can recover expected student achievement using a reweighting of the distribution of test scores in year $t - 1$ (W). The function h_+ does not depend on U directly, but its integral with respect to the distribution of W conditional on U reconstructs the confounded conditional expectation of Y . This formulation allows us to bypass the unobserved U by using an observed variable W that is a proxy for U . ◀

Example 2 (Birth-weight classification). In the birth-weight example, Assumption 5 posits the existence of a function h_+ that links the observed placebo outcome W (the mother's education) to the unobserved confounder U (healthcare quality). Concretely, among infants with the same reported birth-weight, we require that the confounded conditional expectation of mortality, $\mathbb{E}[Y \mid D = d, U]$, can be expressed as an average of $h_+(d - d^*, W)$ over the distribution of the mother's education given U . Crucially, h_+ does not depend on healthcare quality U directly; it is a function only of the observed quantities $d - d^*$ and W . Yet integrating it against $\mathbb{P}(w \mid D = d, U)$ reproduces the U -confounded conditional expectation. This allows us to recover the quantities of interest through the observed proxy W . ◀

First we show that h_0 is a solution to an integral equation in terms of observed variables, though it may not be unique.

Lemma 3 (Factuals I). *Lemma 1 and Assumption 5 imply that any h_+ that exists also solves, for all $d \in \mathcal{D}_+(\epsilon)$ and all $z \in \mathcal{Z}(\epsilon)$,*

$$\mathbb{E}[Y \mid D = d, Z = z] = \int h_+(d - d^*, w) \, d\mathbb{P}(w \mid D = d, Z = z).$$

Analogously, for all $d \in \mathcal{D}_-(\epsilon)$,

$$\mathbb{E}[Y \mid D = d, Z = z] = \int h_-(d - d^*, w) \, d\mathbb{P}(w \mid D = d, Z = z).$$

Lemma 3 translates the bridge equation from a conditional expectation over U to one over observed variables Z and W . It shows that the same h_0 function also solves a second integral equation where both sides of the equation are estimable from the data.

Assumption 6 (Completeness). *For all $d \in \mathcal{D}(\epsilon)$ and all $z \in \mathcal{Z}(\epsilon)$, $\mathbb{E}\{f(U) \mid D = d, Z = z\} = 0$ if and only if $f(U) = 0$ almost surely.*

Assumption 6 is necessary to guarantee a unique solution to the equation in Lemma 3 (factuals). Without this assumption, there could be multiple functions h that satisfy the same moment condition, making point identification of τ_0 more challenging. Future work may build on [Bennett et al. \(2022\)](#) to give weaker conditions under which τ_0 can still be identified without assuming completeness.

Example 3 (Partially linear DGP). Assumptions 5 and 6 are structural requirements on the informativeness of the placebo variables, and they are generally untestable. To aid interpretation, we introduce a partially linear DGP in which both conditions take on a transparent form. Consider

$$\begin{aligned} Y &= f(D - d^*) + \beta_u U + \eta_y, & \mathbb{E}[\eta_y \mid D, Z, U] &= 0 \\ W &= \alpha_0 + \alpha_u U + \eta_w, & \mathbb{E}[\eta_w \mid D, Z, U] &= 0, \end{aligned}$$

where $f: \mathcal{D}(\epsilon) \mapsto \mathbb{R}$ is continuous at all $d \in \mathcal{D}(\epsilon)$ such that $d \neq d^*$, $A = \mathbb{1}(D \geq d^*)$, and we assume that W is bounded for all $d \in \mathcal{D}(\epsilon)$.

Under this DGP, computing $\mathbb{E}[Y \mid D, U]$ and $\mathbb{E}[W \mid D, U]$ and eliminating U shows that Assumption 5 (confounding bridge) reduces to the condition $\alpha_u \neq 0$, i.e., the placebo outcome W must be genuinely informative about the confounder U .⁵ Although this condition is untestable since U is unobserved, any researcher who uses a placebo outcome to diagnose continuity violations must implicitly believe that W is related to U in this way; our framework makes that belief precise.

For Assumption 6 (completeness), suppose additionally that Z and U are both discrete. Newey and Powell (2003) show that a simple sufficient condition is that Z takes on at least as many distinct values as U , together with nondegeneracy of the relevant conditional distributions. More generally, completeness is a rank condition on the conditional distribution of Z given U , and a range of sufficient conditions and supporting examples are available in the econometrics literature (Andrews, 2017). ◀

Lemma 4 (Factuals II). *Lemma 1 and Assumption 6 imply that if there exists a function h_+ that, for all $d \in \mathcal{D}_+(\epsilon)$ and all $z \in \mathcal{Z}(\epsilon)$ satisfies,*

$$\mathbb{E}[Y \mid D = d, Z = z] = \int h_+(d - d^*, w) \, d\mathbb{P}(w \mid D = d, Z = z),$$

then Assumption 5 holds. An analogous result holds for a function h_- and all $d \in \mathcal{D}_-(\epsilon)$.

Lemma 4 completes the identification argument by showing that the existence of a function h_0 that solves the observed-data version of the bridge equation implies the existence of a confounding bridge in the structural model. This allows us to write τ_0 in terms of functions of observed variables.

Theorem 1 (Placebo identification). *Suppose Assumptions 1, 2, 3, 4, and 5 hold. Then*

$$\tau_0 = \frac{\lim_{\epsilon \downarrow 0} \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^*) - \lim_{\epsilon \downarrow 0} \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^*)}{\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon] - \mathbb{E}[A \mid D = d^* - \epsilon]]}$$

⁵See Lemma G.1 for a formal justification.

If in addition Assumption 6 holds then the integrals of h_0 in the numerator are identified. Therefore τ_0 is identified.

In sharp design, under Assumptions 1, 2, 4, and 5, then this simplifies to:

$$\tau_0 = \lim_{\epsilon \downarrow 0} \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^*) - \lim_{\epsilon \downarrow 0} \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^*)$$

Theorem 1 provides a complete identification strategy—it shows how the discontinuity in potential outcomes conditional on unobserved U can be corrected using an observable proxy (W) and an auxiliary variable (Z). The functions h_+ and h_- effectively “invert” the confounding effect of U using the relationship between U and W . This inversion is unique when completeness holds and yields a point-identified causal estimand.

Example 3 (Partially linear DGP). Recall the structural equations from Example 3, now written with treatment-specific regression functions on either side of the cutoff,

$$\begin{aligned} Y &= Af_+(D - d^*) + (1 - A)f_-(D - d^*) + \beta_u U + \eta_y, & \mathbb{E}[\eta_y \mid D, Z, U] &= 0 \\ W &= \alpha_0 + \alpha_u U + \eta_w, & \mathbb{E}[\eta_w \mid D, Z, U] &= 0, \end{aligned}$$

where $f_+, f_- : \mathcal{D}(\epsilon) \mapsto \mathbb{R}$ are continuous on either side of the cutoff.

Lemma G.1 verifies that the confounding bridges

$$\begin{aligned} h_+(D - d^*, W) &= A \left\{ f_+(D - d^*) - \frac{\beta_u}{\alpha_u} \alpha_0 + \frac{\beta_u}{\alpha_u} W \right\} \\ h_-(D - d^*, W) &= (1 - A) \left\{ f_-(D - d^*) - \frac{\beta_u}{\alpha_u} \alpha_0 + \frac{\beta_u}{\alpha_u} W \right\} \end{aligned}$$

satisfy the moment condition in Lemma 3 (factuals). This allows the moment condition in Lemma 2 (limit) to be expressed entirely in terms of observable variables, permitting estimation.

Applying Theorem 1 (identification), the RDD treatment effect is then identified as

$$\tau_0 = f_+(0) - f_-(0) = \mathbb{E} \left[\lim_{\epsilon \downarrow 0} [h_+(\epsilon, W) - h_-(-\epsilon, W)] \mid D = d^* \right].$$

That is, τ_0 is recovered by taking the limiting difference of the two bridge functions at the

cutoff, which removes the confounding contribution of U and leaves only the causal effect of the treatment. ◀

4 Estimation: Local instrumental variable regression

Theorem 1 identifies the RDD treatment effect by reweighting two confounding bridges $h_-(d, w)$ and $h_+(d, w)$. Recent literature to approximate these integral equations rely on kernel methods to non-parametrically estimate these confounding bridges. However, Fan and Gijbels (1992) show that the order of bias of these kernel estimators on boundary points (d^*) is high, and advocate the use of local linear regressions to estimate points on the boundary with less bias.

As our second contribution, we propose a local linear instrumental variable estimator, extending Fan and Gijbels (1992) to accommodate instruments. For tractability, we introduce a partially linear approximation to the confounding bridge in Assumption 5. The final estimator decomposes into two interpretable terms; it adjusts the discontinuity in the target outcome by the discontinuity in the placebo outcome.

For clarity, we focus on the sharp design, i.e. the numerator in Theorem 1. The fuzzy design is a straightforward extension, since the denominator in Theorem 1 is a standard RDD estimand.

For simplicity, we also focus on the exactly identified case where $\dim(z) = \dim(w)$. Our results naturally extend to the overidentified case $\dim(z) \geq \dim(w)$ by standard techniques.

4.1 Locally approximating the confounding bridge

In line with the identification strategy presented in Section 3, we estimate the treatment effect τ_0 using a two-step procedure. First, we solve for the confounding bridges $h_+(\cdot, \cdot)$ and $h_-(\cdot, \cdot)$ at the cutoff d^* using the moment condition from Lemma 3. Then, invoking Theo-

rem 1, we compute the difference in their limits, $\tau_0 = \lim_{\epsilon \downarrow 0} \left[\int h_+(\epsilon, w) d\mathbb{P}(w \mid D = d^*) - \int h_-(-\epsilon, w) d\mathbb{P}(w \mid D = d^*) \right]$.

To motivate our estimator, consider the class of partially linear functions,

$$h_+(d - d^*, w) = g_+(d - d^*) + w^\top \gamma_+, \quad h_-(d - d^*, w) = g_-(d - d^*) + w^\top \gamma_-, \quad (1)$$

where $g_+ : \mathcal{D}_+(\epsilon) \rightarrow \mathbb{R}$ and $g_- : \mathcal{D}_-(\epsilon) \rightarrow \mathbb{R}$ are any continuously differentiable functions, and γ_+ plays the role of the best linear projection of $h_+(\cdot, \cdot)$ onto W at $D = d^*$.

Appendix G shows that partially linear potential outcomes imply partially linear confounding bridges. More generally, a partially linear confounding bridge may be viewed as a best-in-class approximation to a nonlinear confounding bridge; see Proposition 2 below for a detailed characterization.

Within this class, the moment condition in Lemma 3 becomes

$$\mathbb{E}[Y - g_+(D - d^*) - W^\top \gamma_+ \mid D = d, Z = z] = 0. \quad (2)$$

This expression closely resembles a partially linear instrumental variable moment condition, where W plays the role of the endogenous regressor and Z serves as the instrument. Therefore, we propose a local instrumental variable regression approach to estimate the best approximation to $h_+(\cdot, \cdot)$ near d^* .

4.2 Algorithm

To define our estimator, we introduce some notation. Let $\alpha_+ = (\alpha_{+,0}, h_n \alpha_{+,1})^\top = (g_+(0), h_n g'_+(0))^\top$ concatenate the nonlinear component of h_+ and its derivative at the cutoff, where h_n is the kernel bandwidth. We concatenate all of the parameters to be estimated as $\nu_+^\top = (\alpha_{+,0}, \alpha_{+,1}, \gamma_+^\top)$.

We propose what appears to be a new local linear objective function, which we call local instrumental variable regression. Its first order condition, which yields the estimator

$\hat{\nu}_+$, is

$$\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \left(Y_i - R_{i,1}^\top \hat{\alpha}_+ - W_i^\top \hat{\gamma}_+ \right) = \mathbf{0},$$

where $\omega_{i,+} = \frac{1}{h_n} \mathbb{1}(D_i \geq d^*) K\left(\frac{|D_i - d^*|}{h_n}\right)$ are local kernel weights, $R_{i,1} = \left(1, \frac{D_i - d^*}{h_n}\right)^\top$ is a transformed vector of the running variable, and $Z_i \in \mathbb{R}^{\dim(w)}$ is a vector of placebo treatments. This empirical moment is clearly a sample analogue of the population moment implied by equation (2). The left limit objects are analogous.

We then use the coefficients estimated by local instrumental variable regression to construct our estimator of the treatment effect. The appropriate formula is immediate from Theorem 1 and equation (1). Formally, the approximate treatment effect and its estimator are

$$\begin{aligned} \tau_{\text{pdd}} &= g_+(0) + \lim_{\epsilon \downarrow 0} \mathbb{E}[W^\top \mid D = d^* + \epsilon] \gamma_+ - \left(g_-(0) + \lim_{\epsilon \downarrow 0} \mathbb{E}[W^\top \mid D = d^* + \epsilon] \gamma_- \right) \\ &= \alpha_{+,0} + (\beta_{+,0}^w)^\top \gamma_+ - \{ \alpha_{-,0} + (\beta_{+,0}^w)^\top \gamma_- \} \\ \hat{\tau}_{\text{pdd}} &= \hat{\alpha}_{+,0} + (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_+ - \{ \hat{\alpha}_{-,0} + (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_- \}, \end{aligned} \tag{3}$$

where $\hat{\beta}_{+,0}^w$ estimates the right-limit $\beta_{+,0}^w = \lim_{\epsilon \downarrow 0} \mathbb{E}[W_i \mid D = d^* + \epsilon]$ using the local linear estimator of Hahn et al. (2001):

$$H_1 \hat{\beta}_+^{wj} = \arg \min_{\beta} \sum_{i=1}^n \omega_{i,+} (W_{i,j} - R_{i,1}^\top \beta)^2.$$

Then, we let $\hat{\beta}_{+,0}^w = \left(\hat{\beta}_{+,0}^{w_1}, \dots, \hat{\beta}_{+,0}^{w_{\dim(w)}} \right)^\top \in \mathbb{R}^{\dim(w)}$ concatenate the first components of these estimated coefficients. The left limit objects are analogous.

Section 5 below shows that this estimator is consistent for τ_{pdd} . After bias correction, which we defer to Section 5, it is also asymptotically normal at the familiar rate of $n^{-2/5}$.

4.3 Equivalence: RDD with a new adjustment term

For any finite sample size, our proposed estimator is numerically equivalent to a highly interpretable procedure: take the standard RDD estimator based on discontinuity in the outcome (Y), and add an adjustment term based on the discontinuity in the placebo outcome (W).

Proposition 1 (Estimator equivalence). *Our proposed estimator in equation (3) is numerically equivalent to*

$$\hat{\tau}_{pdd} = \hat{\tau}_{rdd}^y - (\hat{\tau}_{rdd}^w)^\top \hat{\gamma}_-,$$

where $\hat{\tau}_{rdd}^y = \hat{\beta}_{+,0}^y - \hat{\beta}_{-,0}^y \in \mathbb{R}$ and $\hat{\tau}_{rdd}^w = (\hat{\beta}_{+,0}^w - \hat{\beta}_{-,0}^w) \in \mathbb{R}^{\dim(w)}$ are constructed from the familiar RDD local linear objective functions, namely

$$\hat{\beta}_+^y = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n \omega_{i,+} (Y_i - R_{i,1}^\top \beta)^2, \quad \hat{\beta}_+^w = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n \omega_{i,+} (W_{i,j} - R_{i,1}^\top \beta)^2,$$

and analogous objectives below the cutoff. Moreover, $\hat{\gamma}_-$ is the local instrumental variable regression of the residualized $Y^\perp$ on the residualized $W^\perp$, instrumenting with Z , below cutoff:

$$\hat{\gamma}_- = \left[\frac{1}{n} \sum_{i=1}^n \omega_{i,-} \{Z_i (W_i^\perp)^\top\} \right]^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n \omega_{i,-} (Z_i Y_i^\perp) \right\}.$$

Here, $(Y_i^\perp)$ are the residuals from a local linear regression of (Y_i) onto (D_i) , and $(W_i^\perp)$ are the residuals from a local linear regression of (W_i) onto (D_i) . In other words, we residualize the target outcome and the placebo outcome using the running variable. $\blacktriangleleft$

The RDD literature, summarized recently by e.g. Cattaneo et al. (2024), advocates for testing whether $\hat{\tau}_{rdd}^w$ is close to zero as a diagnostic for whether $f(d \mid u, \eta_y)$ (or, relatedly, $f(u, \eta_y \mid d)$) changes discontinuously at d^* . Our contribution is to show that this discontinuity is an essential ingredient for an adjustment term. If $\hat{\tau}_{rdd}^w$ is indeed negligible, our adjustment term is zero and our estimator $\hat{\tau}_{pdd}$ simplifies to the standard RDD estimate

$$\hat{\tau}_{rdd}^y.$$

Within our adjustment term, the placebo outcome discontinuity ($\hat{\tau}_{\text{rdd}}^w$) is multiplied by weights ($\hat{\gamma}_-$).

These weights are increasing in the correlation between the residualized outcome ($Y^\perp$) and the placebo treatment (Z). Intuitively, if this correlation is stronger, then there is more unobserved confounding on the target outcome, and hence we increase the weights.

The weights are decreasing in the correlation between the residualized placebo outcome ($W^\perp$) and the placebo treatment (Z). Intuitively, if this correlation is weaker, then the placebo outcome is a weaker proxy for the unobserved confounding, and hence we compensate by increasing the weights on its estimated discontinuity.

Proposition 1 clarifies the danger of a very weak proxy, which would lead to numerical instability in this final aspect of the weights. Future work may study the behavior of our estimator when using a weak proxy.

Remark 2 (Only left-adjustment). The decomposition in Proposition 1 only estimates γ_- , and not γ_+ , because our treatment effect of interest is the effect at the cutoff. As such, it is the effect on the treated, so only the untreated potential outcome needs adjustment.

In more detail, our estimator is anchored on the distribution of U just above the cutoff. It preserves the right-limit $\mathbb{E}[Y \mid D = d^* + \epsilon]$. However, it adjusts the left-limit using γ_- . The adjusted term allows us to estimate the counterfactual below the cutoff $\mathbb{E}[\mathbb{E}[Y \mid D = d^* - \epsilon, U] \mid D = d^*]$, in the thought experiment where U matches the distribution it would have followed above the cutoff.

An alternative causal parameter is $\tilde{\tau}_0 = \mathbb{E}[Y(1, d^*, U, \eta_y) - Y(0, d^*, U, \eta_y)]$, for which our estimator would have two adjustment terms: $\tilde{\tau}_{\text{pdd}} = \hat{\tau}_{\text{rdd}}^y + \left(\overline{W} - \hat{\beta}_{+,0}^w\right)^\top \hat{\gamma}_+ - \left(\overline{W} - \hat{\beta}_{-,0}^w\right)^\top \hat{\gamma}_-$.

◀

5 Bias corrected inference

As our third contribution, we study the large sample properties of the estimator in Proposition 1. We prove it is consistent. Then, we prove that it is asymptotically normal at the familiar rate $n^{-2/5}$ after bias correction.

5.1 Consistency

While prior work used placebo outcomes and predetermined covariates to detect assumption violations, we propose a method to directly correct for them and thereby recover τ_0 . Below we state sufficient conditions for consistency.

Let $\mu_{+,y}(d-d^*) = \mathbb{E}[Y \mid D = d]$ for $d \geq d^*$ and $\mu_{-,w_j}(d-d^*) = \mathbb{E}[W_{i,j} \mid D = d]$ for $d < d^*$. Similarly, define $\mu_{-,y}(d-d^*)$, $\mu_{+,w_j}(d-d^*)$, $\mu_{-,w_j}(d-d^*)$, $\mu_{+,z_j}(d-d^*)$, and $\mu_{-,z_j}(d-d^*)$. Let $\mu_{+,z_j w_k}(d-d^*) = \mathbb{E}[W_{i,k} Z_{i,j} \mid D = d]$. Let $\sigma_{+,y}^2(d-d^*) = \mathbb{E}\left[(Y_i - g_+(d-d^*) - W_i^\top \gamma_+)^2 \mid D_i = d\right]$ for $d \geq d^*$ and $\sigma_{-,w_j}^2(d-d^*) = \mathbb{E}\left[(W_{i,j} - \mu_{-,w_j}(d-d^*))^2 \mid D_i = d\right]$ for $d < d^*$. Similarly, define $\sigma_{-,y}^2(d-d^*)$, $\sigma_{+,w_j}^2(d-d^*)$, $\sigma_{+,z_j}^2(d-d^*)$, and $\sigma_{-,y}^2(d-d^*)$.

Assumption 7 (Regularity conditions). *Let the following conditions hold.*

a (Continuous mean) For all $d \in \mathcal{D}_+(\epsilon)$ and $j \in \{1, \dots, \dim(w)\}$, $g_+(\cdot)$ and $\mu_{+,w_j}(\cdot)$ are 3-times continuously differentiable and bounded. Additionally, $\mu_{+,z_j}(\cdot)$ and $\mu_{+,z_j w_k}(\cdot)$ are continuous and bounded.

For all $d \in \mathcal{D}_-(\epsilon)$ and $j \in \{1, \dots, \dim(w)\}$, $\mu_{-,y}(\cdot)$ and $\mu_{-,w_j}(\cdot)$ are 3-times continuously differentiable and bounded. Additionally, $\mu_{-,z_j}(\cdot)$ and $\mu_{-,z_j w_k}(\cdot)$ are continuous and bounded.

b (Continuous variance) For all $d \in \mathcal{D}_+(\epsilon)$ and $j, k \in \{1, \dots, \dim(w)\}$, $\sigma_{+,y}^2(\cdot)$, $\sigma_{+,w_j}^2(\cdot)$, and $\sigma_{+,z_j}^2(\cdot)$ are continuous and bounded away from 0. For all $d \in \mathcal{D}_-(\epsilon)$, $\sigma_{-,y}^2(\cdot)$,

$\sigma_{-,w_j}^2(\cdot)$, and $\sigma_{-,z_j}^2(\cdot)$ are continuous and bounded away from 0.

For all $d \in \mathcal{D}_+(\epsilon)$, $\rho_{+,y^2z}(d - d^*)$, $\rho_{+,y^2zz^\top}(d - d^*)$, $\rho_{+,yw_jz}(d - d^*)$, $\rho_{+,yw_j}(d - d^*)$, $\rho_{+,z_kw_j}(d - d^*)$, and $\rho_{+,w_jw_k}(d - d^*)$ are continuous. For all $d \in \mathcal{D}_-(\epsilon)$, $\rho_{-,y^2z}(d - d^*)$, $\rho_{-,y^2zz^\top}(d - d^*)$, $\rho_{-,yw_jz}(d - d^*)$, $\rho_{-,yw_j}(d - d^*)$, $\rho_{-,z_kw_j}(d - d^*)$, and $\rho_{-,w_jw_k}(d - d^*)$ are continuous.⁶

c (Continuous distribution) For all $d \in \mathcal{D}(\epsilon) \setminus \{d^*\}$, $f_d(d)$ is bounded, continuous, and nonzero, where $f_d(d)$ is the distribution of D .

d (Bounded higher moments) For all $d \in \mathcal{D}(\epsilon)$ and $j \in \{1, \dots, \dim(w)\}$, there exists some $\zeta > 0$ and $M \in \mathbb{R}$ such that we have $\mathbb{E}\left[|Y_i - g_+(d - d^*) - W_i^\top \gamma_+|^{2+\zeta} \mid D_i = d\right]$, $\mathbb{E}\left[|Y_i - g_-(d - d^*) - W_i^\top \gamma_-|^{2+\zeta} \mid D_i = d\right]$, $\mathbb{E}\left[|W_{i,j} - \mu_{+,w_j}(d - d^*)|^{2+\zeta} \mid D_i = d\right]$, $\mathbb{E}\left[|W_{i,j} - \mu_{-,w_j}(d - d^*)|^{2+\zeta} \mid D_i = d\right]$, $\mathbb{E}\left[|Z_{i,j}(Y_i - g_+(d - d^*) - W_i^\top \gamma_+)|^{2+\zeta} \mid D_i = d\right] < M$.

The regularity conditions in Assumption 7 are variations of standard assumptions in the RDD literature (Porter, 2003; Calonico et al., 2019).

Assumption 8 (Kernel conditions). Assume that the kernel function $K: \mathbb{R} \mapsto \mathbb{R}^+$ is symmetric, bounded, and continuous at 0 with bounded support.

Assumption 8 is satisfied by the window kernel $K(u) = \mathbb{1}(|u| \leq 1)$ and the triangle kernel $K(u) = (1 - |u|)\mathbb{1}(|u| \leq 1)$, which are the most frequently used kernels in the RDD literature. It also allows for the Epanechnikov kernel $K(u) = \frac{3}{4}(1 - u^2)\mathbb{1}(|u| \leq 1)$. Under these conditions we can characterize the probability limit of our local instrumental variable estimator for any vanishing bandwidth $1 \gg h_n \gg n^{-1}$.

Proposition 2 (Probability limit). Let Assumptions 7a-c and 8 hold, and assume that

⁶The $\rho_{\pm,x}(\cdot)$ are covariance terms, with formulas given in Appendix F.

$nh_n \rightarrow \infty$ and $h_n \rightarrow 0$. Then, it follows that,

$$\hat{\tau}_{pdd} \xrightarrow{P} \tau_{rdd}^y - (\tau_{rdd}^w)^\top \gamma_-,$$

where $\gamma_- = \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* - \epsilon)^{-1} \text{Cov}(Z_i, Y_i \mid D_i = d^* - \epsilon)$ is a best local linear approximation, i.e. it solves the local projection problem

$$Y = c + W^\top \gamma_- + \eta, \quad \lim_{\epsilon \downarrow 0} \mathbb{E}[\eta Z_i \mid D_i = d^* - \epsilon] = 0, \quad \lim_{\epsilon \downarrow 0} \mathbb{E}[\eta \mid D_i = d^* - \epsilon] = 0.$$

◀

Building on Proposition 1 (estimator equivalence), which characterizes the finite sample estimate, Proposition 2 characterizes the asymptotic limit of the local estimator. Now, at the population level, we show that if the placebo outcome does not jump at the cutoff ($\tau_{rdd}^w = 0$), then the local instrumental variable estimand reduces to the traditional RDD estimand.

The population limit of the weights admits a similar interpretation as before. It also sheds light on the nature of our approximation. In particular, Proposition 2 shows that γ_- can be interpreted as a linear projection of Y onto the distribution of W , instrumenting with Z , when D approaches the cutoff from the left. In other words, γ_- is the parameter which best approximates the confounding bridge in Lemma 3 (factuals) as a partially linear function of the placebo outcome W .

If the confounding bridge $h_-(0, w)$ is very nonlinear in W then this approximation can be poor. However, we advocate for this approach because full nonparametric estimation over (D_i, W_i) would require us to uniformly estimate $h_-(0, w)$ as a function of w to characterize the asymptotic bias of the local instrumental variable estimator. There would be $\dim(w)$ additional kernels and bandwidths in the estimator. As discussed by Calonico et al. (2019), localizing weights for many variables would suffer from a curse of dimensionality, and hence empirical applications would become challenging. Our estimator does not suffer from this

curse of dimensionality, and is therefore practical for empirical researchers. Future work may attempt to overcome these challenges, perhaps by placing additional structure on the effective dimension or sparsity of W (Noack et al., 2021).

Corollary 1 (Consistency). *Let Assumptions 1, 2, 3, 4, 5, 7a, and 8 hold. Additionally, suppose that equation (1) holds almost surely. Then, if $nh_n \rightarrow \infty$ and $h_n \rightarrow 0$, it follows that $\hat{\tau}_{pdd} \xrightarrow{P} \tau_0$.* $\blacktriangleleft$

Corollary 1 shows that when the confounding bridge lies in the class of partially linear functions in equation (1), then the local instrumental variable estimator is consistent for the treatment effect τ_0 . As shown in Appendix G, partially linear potential outcomes imply partially linear confounding bridges. More generally, partial linearity of the confounding bridge may be a reasonable approximation.

5.2 Bias correction and asymptotic normality

As shown in Proposition 1, our estimator can be decomposed into a sum of two components: (i) the RDD estimate of the outcome discontinuity $\hat{\tau}_{\text{rdd}}^y$, and (ii) the RDD estimate of the placebo outcome discontinuity, appropriately weighted as $(\hat{\tau}_{\text{rdd}}^w)^\top \hat{\gamma}_-$. A key insight from Calonico et al. (2014) is that, when the bandwidth h_n is selected to optimize mean squared error (MSE), these discontinuity estimators suffer from non-negligible bias in their sampling distributions. This bias persists in large samples and leads to invalid inference, particularly when constructing confidence intervals.

Proposition 3 shows the asymptotic bias of the naive PDD estimator at the MSE-optimal bandwidth rate of $h_n \asymp n^{-1/5}$.

Proposition 3 (PDD bias). *Let Assumptions 1, 2, 3, 4, 5, 7, and 8 hold, and assume that the bandwidths satisfy $h_n = c_h n^{-1/5}$ and $b_n = c_b n^{-1/5}$, where b_n is the bandwidth of the bias correction. Additionally, suppose that equation (1) holds almost surely. Then, it follows*

that,

$$\mathbb{E}\left[\sqrt{nh_n}(\hat{\tau}_{pdd} - \tau_{pdd})\right] \rightarrow \mathbf{e}_{+,w}^\top \left(\Psi_{+,1}^{-1} \eta_{+,IV} - \Psi_{-,1}^{-1} \eta_{-,IV} \right) + (\gamma_+ - \gamma_-)^\top \begin{pmatrix} \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_1} \\ \vdots \\ \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_q} \end{pmatrix}.$$

Formulas for $\mathbf{e}_{+,w}$, $\Psi_{+,1}$, $\eta_{+,IV}$, $\Gamma_{+,1}$, and η_{+,w_j} are given in Appendix [F](#). $\blacktriangleleft$

To address this issue, we extend the robust bias correction technique developed by [Calonico et al. \(2014\)](#). In particular, we construct bias-corrected estimators that remain consistent and asymptotically normal, even when using MSE-optimal bandwidths. We then define the bias-corrected PDD estimator $\hat{\tau}_{pdd}^{\text{rbc}}$ as,

$$\hat{\tau}_{pdd}^{\text{rbc}} = \hat{\tau}_{pdd} - \left(\hat{\mathbf{e}}_{+,w}^\top \left(\hat{\Psi}_{+,1}^{-1} \hat{\eta}_{+,IV} - \hat{\Psi}_{-,1}^{-1} \hat{\eta}_{-,IV} \right) + (\hat{\gamma}_{+,1} - \hat{\gamma}_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\eta}_{+,w_1} + \cdots + (\hat{\gamma}_{+,q} - \hat{\gamma}_{-,q}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\eta}_{+,w_q} \right). \quad (4)$$

The theorem below establishes the asymptotic normality of our estimator.

Theorem 2 (Limit distribution). *Let Assumptions [1](#), [2](#), [3](#), [4](#), [5](#), [7](#), and [8](#) hold, and assume that the bandwidths satisfy $h_n = c_h n^{-1/5}$ and $b_n = c_b n^{-1/5}$, where b_n is the bandwidth of the bias correction. Additionally, suppose that equation [\(1\)](#) holds almost surely. Then, it follows that,*

$$\sqrt{nh_n}(\hat{\tau}_{pdd}^{\text{rbc}} - \tau_0) \xrightarrow{d} \mathcal{N}(0, V),$$

where V and a consistent estimator for V are given in Appendix [F](#). $\blacktriangleleft$

Theorem [2](#) establishes the asymptotic normality of the bias-corrected local instrumental variable estimator under the commonly used MSE-optimal bandwidth rate of $h_n \asymp n^{-1/5}$ ([Imbens and Lemieux, 2008](#)). The bias correction bandwidth may be of the same order, i.e. $b_n \asymp h_n$.

In practice, we implement a data-driven bandwidth choice $\hat{h}_n$ along the lines of e.g. [Imbens and Kalyanaraman \(2012\)](#). Future work may develop a data-driven bandwidth

algorithm to minimize the length of confidence intervals along the lines of [Calonico et al. \(2018, 2020\)](#).

Theorem 2 justifies the use of standard Wald-type inference for $\hat{\tau}_{\text{pdd}}^{\text{rbc}}$, e.g. the construction of confidence intervals and hypothesis tests.

6 Discussion

This paper develops a new identification and estimation framework for regression discontinuity designs when the standard continuity of potential outcomes assumption fails due to unobserved confounding. We show that by leveraging a placebo treatment and a placebo outcome, it is possible to recover the causal parameter even when the running variable’s distribution exhibits discontinuities at the threshold. Our identification argument relies on conditional continuity given an unobserved confounder, and it employs the concept of a confounding bridge to encode the relationship between the unobserved confounder and its observed proxy. Under a completeness condition, we show that the integrated bridge function is identified from observed data, and the treatment effect at the cutoff can be recovered using integrals of observable variables.

To operationalize this strategy, we propose a local instrumental variable estimator that approximates the confounding bridge using a partially linear specification. We demonstrate that the estimator decomposes into a standard RDD term and an adjustment term driven by the discontinuity in the placebo outcome. This decomposition yields an interpretable adjustment that allows researchers to correct for violations of the standard RDD assumptions, rather than discarding such designs altogether. We establish conditions for consistency and bias-corrected inference, allowing optimal bandwidths. Our results extend the applicability of RDD methods to settings with strategic behavior that is detected via placebo discontinuities, providing researchers with a tractable and theoretically grounded

approach for handling unobserved confounding near the cutoff.

References

- Abdulkadiroğlu, A., Angrist, J. D., Narita, Y., Pathak, P. A., and Zarate, R. A. (2017). Regression Discontinuity in Serial Dictatorship: Achievement Effects at Chicago’s Exam Schools. *American Economic Review*, 107(5):240–245.
- Almond, D., Chay, K. Y., and Lee, D. S. (2005). The costs of low birth weight. *The Quarterly Journal of Economics*, 120(3):1031–1083.
- Andrews, D. W. (2017). Examples of l2-complete and boundedly-complete distributions. *Journal of econometrics*, 199(2):213–220.
- Angrist, J. D., Lavy, V., Leder-Luis, J., and Shany, A. (2019). Maimonides’ Rule Redux. *American Economic Review: Insights*, 1(3):309–324.
- Angrist, J. D. and Rokkanen, M. (2015). Wanna get away? regression discontinuity estimation of exam school effects away from the cutoff. *Journal of the American Statistical Association*, 110(512):1331–1344.
- Armstrong, T. B. and Kolesár, M. (2018). Optimal inference in a class of regression models. *Econometrica*, 86(2):655–683.
- Bennett, A., Kallus, N., Mao, X., Newey, W., Syrgkanis, V., and Uehara, M. (2022). Inference on strongly identified functionals of weakly identified functions. *arXiv:2208.08291*.
- Bertanha, M. (2020). Regression discontinuity design with many thresholds. *Journal of Econometrics*, 218(1):216–241.
- Butts, K. (2023). Geographic difference-in-discontinuities. *Applied Economics Letters*, 30(5):615–619.
- Calonico, S., Cattaneo, M. D., and Farrell, M. H. (2018). On the effect of bias estimation on coverage accuracy in nonparametric inference. *Journal of the American Statistical Association*, 113(522):767–779.
- Calonico, S., Cattaneo, M. D., and Farrell, M. H. (2020). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal*, 23(2):192–210.
- Calonico, S., Cattaneo, M. D., Farrell, M. H., and Titiunik, R. (2019). Regression Discontinuity Designs Using Covariates. *The Review of Economics and Statistics*, 101(3):442–451.
- Calonico, S., Cattaneo, M. D., and Titiunik, R. (2014). Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs. *Econometrica*, 82(6):2295–2326.
- Cattaneo, M. D., Idrobo, N., and Titiunik, R. (2024). *A Practical Introduction to Regression Discontinuity Designs*. Cambridge Elements: Quantitative and Computational Methods for Social Science. Cambridge University Press., Cambridge University Press.

- Cattaneo, M. D., Keele, L., Titiunik, R., and Vazquez-Bare, G. (2021). Extrapolating treatment effects in multi-cutoff regression discontinuity designs. *Journal of the American Statistical Association*, 116(536):1941–1952.
- Cook, P. J. and Kang, S. (2016). Birthdays, Schooling, and Crime: Regression-Discontinuity Analysis of School Performance, Delinquency, Dropout, and Crime Initiation. *American Economic Journal: Applied Economics*, 8(1):33–57.
- Deaner, B. (2018). Proxy controls and panel data. *arXiv:1810.00283*.
- Deaner, B. and Kwon, S. (2025). Extrapolation in regression discontinuity design using comonotonicity. *arXiv preprint arXiv:2507.00289*.
- Dong, Y. and Lewbel, A. (2015). Identifying the effect of changing the policy threshold in regression discontinuity models. *Review of Economics and Statistics*, 97(5):1081–1092.
- Eckles, D., Ignatiadis, N., Wager, S., and Wu, H. (2020). Noise-induced randomization in regression discontinuity designs. *arXiv preprint arXiv:2004.09458*.
- Fan, J. and Gijbels, I. (1992). Variable Bandwidth and Local Linear Regression Smoothers. *The Annals of Statistics*, 20(4).
- Galindo-Silva, H., Some, N. H., and Tchente, G. (2018). Fuzzy difference-in-discontinuities: Identification theory and application to the affordable care act. *arXiv preprint arXiv:1812.06537*.
- Garicano, L., Lelarge, C., and Van Reenen, J. (2016). Firm size distortions and the productivity distribution: Evidence from france. *American Economic Review*, 106(11):3439–79.
- Gerard, F., Rokkanen, M., and Rothe, C. (2020). Bounds on treatment effects in regression discontinuity designs with a manipulated running variable. *Quantitative Economics*, 11(3):839–870.
- Goodman, J., Melkers, J., and Pallais, A. (2019). Can Online Delivery Increase Access to Education? *Journal of Labor Economics*, 37(1):1–34.
- Grembi, V., Nannicini, T., and Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*, 8(3):1–30.
- Hahn, J., Todd, P., and Van der Klaauw, W. (2001). Identification and estimation of treatment effects with a regression-discontinuity design. *Econometrica*, 69(1):201–209.
- Hahn, J., Todd, P. E., and van der Klaauw, W. H. (1999). Evaluating the effect of an antidiscrimination law using a regression-discontinuity design.
- Hall, A. B. (2015). What Happens When Extremists Win Primaries? *American Political Science Review*, 109(1):18–42.
- Imbens, G. and Kalyanaraman, K. (2012). Optimal bandwidth choice for the regression discontinuity estimator. *The Review of Economic Studies*, 79(3):933–959.

- Imbens, G. and Wager, S. (2019). Optimized regression discontinuity designs. *Review of Economics and Statistics*, 101(2):264–278.
- Imbens, G. W. and Lemieux, T. (2008). Regression discontinuity designs: A guide to practice. *Journal of Econometrics*, 142(2):615–635.
- Kong, Y. (2021). Sequential Auctions with Synergy and Affiliation across Auctions. *Journal of Political Economy*, 129(1):148–181.
- Kumar, A. (2018). Do Restrictions on Home Equity Extraction Contribute to Lower Mortgage Defaults? Evidence from a Policy Discontinuity at the Texas Border. *American Economic Journal: Economic Policy*, 10(1):268–297.
- Lee, D. S. and Lemieux, T. (2010). Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2):281–355.
- Leventer, D. and Nevo, D. (2024). Correcting invalid regression discontinuity designs with multiple time period data. *arXiv preprint arXiv:2408.05847*.
- Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.
- Newey, W. K. and Powell, J. L. (2003). Instrumental variable estimation of nonparametric models. *Econometrica*, 71(5):1565–1578.
- Noack, C., Olma, T., and Rothe, C. (2021). Flexible covariate adjustments in regression discontinuity designs. *arXiv:2107.07942*.
- Noack, C. and Rothe, C. (2024). Bias-aware inference in fuzzy regression discontinuity designs. *Econometrica*, 92(3):687–711.
- Picchetti, P., Pinto, C. C., and Shinoki, S. T. (2024). Difference-in-discontinuities: estimation, inference and validity tests. *arXiv preprint arXiv:2405.18531*.
- Porter, J. (2003). Estimation in the regression discontinuity model. *Unpublished Manuscript, Department of Economics, University of Wisconsin at Madison*, 2003:5–19.
- Spenkuch, J. L. and Toniatti, D. (2018). Political Advertising and Election Results. *The Quarterly Journal of Economics*, 133(4):1981–2036.
- Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. (2024). An introduction to proximal causal inference. *Statistical Science*, 39(3):375–390.
- Thistlethwaite, D. L. and Campbell, D. T. (1960). Regression-discontinuity analysis: An alternative to the ex post facto experiment. *Journal of Educational Psychology*, 51(6):309.

A Identification proof

A.1 Necessary independences

Proof of Lemma 1 (necessary independences). We proceed in steps.

1. First, we show that $A \perp\!\!\!\perp Z \mid D, U$. We write,

$$\begin{aligned}\mathbb{P}[A = a \mid D = d, Z = z, U] &= \mathbb{P}[A(d, \eta_a) = a \mid D = d, Z = z, U] \\ &= \mathbb{P}[A(d, \eta_a) = a \mid D = d, U],\end{aligned}$$

where the first line follows from Assumption 2d (exclusion) and the second line follows from Assumption 2b (placebo selection on unobservables).

To show that $Y \perp\!\!\!\perp Z \mid D, U$, first note that,

$$\begin{aligned}\mathbb{P}[Y = y \mid D = d, Z = z, U] &= \mathbb{P}[Y(A, d, U, \eta_y) = y \mid D = d, Z = z, U] \\ &= \mathbb{P}[Y(A, d, U, \eta_y) = y \mid D = d, U],\end{aligned}$$

where the first line follows from Assumption 2d (exclusion) and the second line follows from Assumption 2b (placebo selection on unobservables) and the result we have already shown: $(A \perp\!\!\!\perp Z \mid D, U)$.

2. To show the second claim $(W \perp\!\!\!\perp D, Z \mid U)$, we have that,

$$\begin{aligned}\mathbb{P}[W = w \mid D = d, Z = z, U] &= \mathbb{P}[W(U, \eta_w) = w \mid D = d, Z = z, U] \\ &= \mathbb{P}[W(U, \eta_w) = w \mid U],\end{aligned}$$

where the first line follows from Assumption 2d (exclusion) and the second line follows from Assumption 2b (placebo selection on unobservables).

3. Similarly, we prove the last claim $(A \perp\!\!\!\perp U \mid D)$:

$$\begin{aligned}\mathbb{P}[A \mid D = d, U = u] &= \mathbb{P}[A(d, \eta_a) \mid D = d, U = u] \\ &= \mathbb{P}[A(d, \eta_a) \mid D = d].\end{aligned}$$

where the first line follows from Assumption 2d (exclusion) and the second line follows from Assumption 3b ($\eta_a \perp\!\!\!\perp U \mid D$).

□

A.2 Limit

Proof of Lemma 2 (limit). Assume a fuzzy design. We proceed in steps.

1. Notice that under Assumption 2 (placebo variables), we are able to write,

$$Y = Y(0, D, U, \eta_y) + A(D, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)).$$

2. Then, we have that,

$$\begin{aligned} & \mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U] \\ &= \mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U] \\ & \quad + \mathbb{E}[A(D, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)) \mid D = d^* + \epsilon, U] \\ & \quad - \mathbb{E}[A(D, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)) \mid D = d^* - \epsilon, U] \quad (\text{I}) \\ &= \mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U] \\ & \quad + \mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon, U] \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* + \epsilon, U] \\ & \quad - \mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon, U] \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* - \epsilon, U] \quad (\text{II}) \\ &= \mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U] \\ & \quad + \mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon] \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* + \epsilon, U] \\ & \quad - \mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon] \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* - \epsilon, U] \quad (\text{III}), \end{aligned}$$

where (I) follows from the expansion in step 1. The factoring of $A(D, \eta_a)$ in (II) follows from Assumption 3 ($\eta_a \perp\!\!\!\perp \eta_y, U \mid D = d$) and weak union, whereby $\eta_a \perp\!\!\!\perp \eta_y \mid D = d, U$. Dropping the conditioning on U in (III) follows from Lemma 1c ($A \perp\!\!\!\perp U \mid D$).

3. By Assumptions 1 (RDD) and 4 (continuity), we can take the limit of the equality above to show,

$$\begin{aligned}
& \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \\
&= \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U]] \\
&\quad + \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon]] \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* + \epsilon, U]] \\
&\quad - \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon]] \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^* - \epsilon, U]] \\
&= \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon] - \mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon]] \\
&\quad \times \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*, U].
\end{aligned}$$

Specifically, to go from the second line to the third line, the initial terms cancel, and the later terms factorize.

Rearranging the above, we then find that,

$$\begin{aligned}
& \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*, U] \\
&= \frac{\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]]}{\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon] - \mathbb{E}[A \mid D = d^* - \epsilon]]}.
\end{aligned}$$

This implies

$$\begin{aligned}
& \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*] \\
&= \mathbb{E}[\mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*, U] \mid D = d^*] \\
&= \frac{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^*]}{\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon] - \mathbb{E}[A \mid D = d^* - \epsilon]]}
\end{aligned}$$

where first equality follows from the law of iterated expectations and the second equality substitutes in our result above.

The argument for the sharp design is the same, recognizing that Assumption 3 vacuously holds and simplifying the denominator. $\square$

A.3 Factuals

Lemma A.1 (Towards factuals). *Lemma 1 implies for all $d \in \mathcal{D}(\epsilon)$ and all $z \in \mathcal{Z}(\epsilon)$,*

$$\begin{aligned}\mathbb{P}(y \mid d, z) &= \int \mathbb{P}(y \mid d, u) \, d\mathbb{P}(u \mid d, z) \\ \mathbb{P}(w \mid d, z) &= \int \mathbb{P}(w \mid d, u) \, d\mathbb{P}(u \mid d, z) = \int \mathbb{P}(w \mid u) \, d\mathbb{P}(u \mid d, z)\end{aligned}$$

Proof of Lemma A.1 (towards factuals). To lighten notation, we abbreviate $(D = d, Z = z) = (d, z)$ and similarly for other variables. Then, we have that,

$$\begin{aligned}\mathbb{P}(y \mid d, z) &= \int \mathbb{P}(y, u \mid d, z) \, du = \int \mathbb{P}(y \mid d, z, u) \, d\mathbb{P}(u \mid d, z) = \int \mathbb{P}(y \mid d, u) \, d\mathbb{P}(u \mid d, z) \\ \mathbb{P}(w \mid d, z) &= \int \mathbb{P}(w, u \mid d, z) \, du = \int \mathbb{P}(w \mid d, z, u) \, d\mathbb{P}(u \mid d, z) = \int \mathbb{P}(w \mid d, u) \, d\mathbb{P}(u \mid d, z) \\ &= \int \mathbb{P}(w \mid u) \, d\mathbb{P}(u \mid d, z),\end{aligned}$$

where we use that $Y \perp\!\!\!\perp Z \mid U, D$ and $W \perp\!\!\!\perp D, Z \mid U$ from Lemma 1. $\square$

Proof of Lemma 3 (factuals I). We have that,

$$\begin{aligned}\mathbb{E}[Y \mid d, z] &= \int y \, d\mathbb{P}(y \mid d, z) \\ &= \int y \, d\mathbb{P}(y \mid d, u) \, d\mathbb{P}(u \mid d, z) \quad (\text{I}) \\ &= \int \mathbb{E}[Y \mid d, u] \, d\mathbb{P}(u \mid d, z) \\ &= \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, u) \, d\mathbb{P}(u \mid d, z) \quad (\text{II}) \\ &= \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, z) \quad (\text{III}),\end{aligned}$$

where (I) and (III) follow from Lemma A.1 (towards factuals), and (II) follows from Assumption 5 (confounding bridge). $\square$

Proof of Lemma 4 (factuals II). Suppose a solution to $\mathbb{E}[Y \mid d, z] = \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, z)$ exists. Then by Lemma A.1 (towards factuals),

$$\begin{aligned}\mathbb{E}[Y \mid d, z] &= \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, z) \\ &= \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, u) \, d\mathbb{P}(u \mid d, z).\end{aligned}$$

Furthermore, using Lemma A.1 (towards factuality), we can also write that,

$$\begin{aligned}\mathbb{E}[Y \mid d, z] &= \int y \, d\mathbb{P}(y \mid d, z) \\ &= \int y \, d\mathbb{P}(y \mid d, u) \, d\mathbb{P}(u \mid d, z) \\ &= \int \mathbb{E}(Y \mid d, u) \, d\mathbb{P}(u \mid d, z).\end{aligned}$$

Therefore by Assumption 6 (completeness), we can equate the objects within the integrals. $\square$

A.4 Main result

Proof of Theorem 1 (placebo identification). We proceed in steps.

1. First, note that

$$\begin{aligned}\mathbb{E}\left[\int h_0(d - d^*, w) \, d\mathbb{P}(w \mid D = d, U) \mid D = d'\right] &= \mathbb{E}\left[\int h_0(d - d^*, w) \, d\mathbb{P}(w \mid U) \mid D = d'\right] \\ &= \mathbb{E}\left[\int h_0(d - d^*, w) \, d\mathbb{P}(w \mid D = d', U) \mid D = d'\right] \\ &= \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid D = d'),\end{aligned}$$

where the first and second line follow from Lemma 1b ($W \perp\!\!\!\perp D \mid U$) and the last line uses the law of iterated expectations.

2. Also recall from the proof of Lemma 2 (limit) we showed that,

$$\begin{aligned}&\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \\ &= \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon] - \mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon]] \\ &\quad \times \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*, U].\end{aligned}$$

The limits on the right hand side exist due to Assumption 1 (RDD). Therefore the limit on the left hand side also exists.

3. Therefore, we can write the numerator of the expression from Lemma 2 (limit) as,

$$\begin{aligned}
& \mathbb{E} \left[\lim_{\epsilon \downarrow 0} \{ \mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U] \mid D = d^* \} \right] \\
&= \mathbb{E} \left[\lim_{\epsilon \downarrow 0} \left\{ \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^* + \epsilon, U) - \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^* - \epsilon, U) \right\} \mid D = d^* \right] \\
&= \lim_{\epsilon \downarrow 0} \mathbb{E} \left[\int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^* + \epsilon, U) - \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^* - \epsilon, U) \mid D = d^* \right] \\
&= \lim_{\epsilon \downarrow 0} \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^*) - \lim_{\epsilon \downarrow 0} \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^*)
\end{aligned}$$

where the first equality follows from Assumption 5 (confounding bridge), the second equality uses the dominated convergence theorem and the boundedness of Assumption 5 (confounding bridge), and the last equality follows from step 1.

4. Finally, we argue that the overall expression is unique. To begin, note that $\mathbb{E}[Y \mid d, U]$ is unique. Furthermore, in the proof of Lemma 4 (factuals II), we showed that $\mathbb{E}[Y \mid d, U] = \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, U)$ almost surely by appealing to Assumption 6, so the latter integral is unique.

□

B Relaxing Assumption 2b

Assumption 2b implies the necessary condition $Z \perp\!\!\!\perp \eta_y, \eta_a, \eta_w \mid D, U$.

In this appendix, we show that this necessary condition, together with an additional continuity condition, lead to the same main result. Intuitively, we trade off an assumption on placebo exogeneity for an assumption on placebo continuity, i.e. independence for functional form.

Our goal is to define an alternative set of assumptions under which the conclusion of Theorem 1 continues to hold, replacing Assumption 2b with weaker conditions. We summarize the invariances of the full structural system in the next section. A graphical

summary of the causal structure is provided in Figure 4.

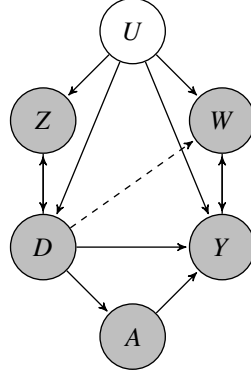


Figure 4: Additional continuity DAG

B.1 Extension

Assumption B.1 (Placebo variables). *Assume there exists some $\epsilon > 0$ such that for all $d \in \mathcal{D}(\epsilon)$ and for all $z \in \mathcal{Z}(\epsilon)$,*

a (Causal consistency): if $A = a$, $D = d$, and $Z = z$ then $Y = Y(a, d, z, U, \eta_y)$, and $W = W(a, d, z, U, \eta_w)$ almost surely. If $D = d$ and $Z = z$ then $A = A(d, z, U, \eta_a)$ almost surely.

b (Placebo selection on unobservables): $Z \perp\!\!\!\perp \eta_y, \eta_a, \eta_w \mid U, D$.

c (Overlap): if $f(u) > 0$ then $f(a, d, z \mid u) > 0$, where we assume the densities exist.

d (Exclusion): $Y(a, d, z, U, \eta_y) = Y(a, d, U, \eta_y)$, $W(a, d, z, U, \eta_w) = W(U, \eta_w)$, and $A(d, z, U, \eta_a) = A(d, \eta_a)$ almost surely.

Assumptions B.1a,c,d are the same as those in Assumption 2. However, Assumption B.1b is weaker than Assumption 2b by weak union. This relaxation makes placebo selection on observables a condition that is exclusively about the placebo treatment.

Lemma B.1 (Necessary independences). *Assumption B.1 implies that for all $d \in \mathcal{D}(\epsilon)$ and $z \in \mathcal{Z}(\epsilon)$,*

$$a \ Y \perp\!\!\!\perp Z \mid D, U$$

$$b \ W \perp\!\!\!\perp Z \mid D, U$$

Additionally under Assumption 3 it follows that,

$$c \ A \perp\!\!\!\perp U \mid D$$

Lemma B.1b is weaker than Lemma 1b by weak union. The rest of the lemma remains the same.

Assumption B.2 (Confounding bridge continuity). *Assume that the mapping $d \mapsto \mathbb{E}[h_+(d - d^*, W) \mid D = d, U = u]$ is continuous for all $d \in \mathcal{D}_+(\epsilon)$ and all u . Similarly, assume that $d \mapsto \mathbb{E}[h_-(d - d^*, W) \mid D = d, U = u]$ is continuous for all $d \in \mathcal{D}_-(\epsilon)$ and all u .*

This additional continuity condition generalizes the standard RDD continuity condition (Assumption 4). In particular, it replaces the potential outcome function with the confounding bridge function. Intuitively, if confounding is continuous, then we can allow more complex forms of it.

Theorem B.1 (Placebo identification). *Suppose Assumptions 1, 3, 4, 5, B.1, and B.2 hold. Then*

$$\tau_0 = \frac{\int h_+(0, w) \, d\mathbb{P}(w \mid D = d^*) - \int h_-(0, w) \, d\mathbb{P}(w \mid D = d^*)}{\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon] - \mathbb{E}[A \mid D = d^* - \epsilon]]}$$

If in addition Assumption 6 holds then the integrals of h_0 in the numerator are identified.

Therefore τ_0 is identified.

In sharp design, under Assumptions 1, 4, 5, B.1, and B.2, then this simplifies to:

$$\tau_0 = \lim_{\epsilon \downarrow 0} \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^*) - \lim_{\epsilon \downarrow 0} \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^*)$$

Theorem B.1 recovers our main identification result (Theorem 1) under an alternative set of assumptions, as desired.

B.2 Proof of extension

Proof of Lemma B.1. Compared to the proof of Lemma 1, the only difference is in the second claim ($W \perp\!\!\!\perp Z \mid D, U$). We have that,

$$\begin{aligned} \mathbb{P}[W = w \mid D = d, Z = z, U] &= \mathbb{P}[W(U, \eta_w) = w \mid D = d, Z = z, U] \\ &= \mathbb{P}[W(U, \eta_w) = w \mid D, U], \end{aligned}$$

where the first line follows from Assumption B.1d (exclusion) and the second line follows from Assumption B.1b (placebo selection on unobservables). $\square$

Lemma B.2 (Towards factuais). *Lemma B.1 implies for all $d \in \mathcal{D}(\epsilon)$ and all $z \in \mathcal{Z}(\epsilon)$,*

$$\begin{aligned} \mathbb{P}(y \mid d, z) &= \int \mathbb{P}(y \mid d, u) \, d\mathbb{P}(u \mid d, z) \\ \mathbb{P}(w \mid d, z) &= \int \mathbb{P}(w \mid d, u) \, d\mathbb{P}(u \mid d, z). \end{aligned}$$

Proof. The argument is identical to Lemma A.1, excluding the final step. $\square$

Proof of Theorem B.1. Notice that Lemma 2 remains unchanged under the conditions of this appendix, since it only appeals to Lemma 1c, which remains unchanged. Therefore, we proceed in steps similar to the proof of Theorem 1.

1. From the proof of Lemma 2 (limit) we showed that,

$$\begin{aligned} &\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \\ &= \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, \eta_a) \mid D = d^* + \epsilon] - \mathbb{E}[A(D, \eta_a) \mid D = d^* - \epsilon]] \\ &\quad \times \mathbb{E}[Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y) \mid D = d^*, U]. \end{aligned}$$

The limits on the right hand side exist due to Assumption 1 (RDD). Therefore the limit on the left hand side also exists.

2. We can write the numerator of the expression from Lemma 2 (limit) as,

$$\begin{aligned}
& \mathbb{E} \left[\lim_{\epsilon \downarrow 0} \{ \mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U] \} \mid D = d^* \right] \\
&= \mathbb{E} \left[\lim_{\epsilon \downarrow 0} \left\{ \int h_+(\epsilon, w) \, d\mathbb{P}(w \mid D = d^* + \epsilon, U) - \int h_-(-\epsilon, w) \, d\mathbb{P}(w \mid D = d^* - \epsilon, U) \right\} \mid D = d^* \right] \\
&= \mathbb{E} \left[\int h_+(0, w) \, d\mathbb{P}(w \mid D = d^*, U) \mid D = d^* \right] - \mathbb{E} \left[\int h_-(0, w) \, d\mathbb{P}(w \mid D = d^*, U) \mid D = d^* \right] \\
&= \int h_+(0, w) \, d\mathbb{P}(w \mid D = d^*) - \int h_-(0, w) \, d\mathbb{P}(w \mid D = d^*)
\end{aligned}$$

where the first equality follows from Assumption 5 (confounding bridge), the second equality uses Assumption B.2, and the third uses the law of iterated expectations.

3. Finally, we argue that the overall expression is unique. To begin, note that $\mathbb{E}[Y \mid d, U]$ is unique. Furthermore, in the proof of Lemma 4 (factuals II), we showed that $\mathbb{E}[Y \mid d, U] = \int h_0(d - d^*, w) \, d\mathbb{P}(w \mid d, U)$ almost surely by appealing to Assumption 6, so the latter integral is unique.

□

C Relaxing Assumption 3

Assumption 3 holds trivially in the sharp design. In the fuzzy design, however, it imposes a nontrivial restriction on the treatment assignment mechanism.

In this appendix, we show that Assumption 3 can be relaxed in the fuzzy design if treatment effects are homogeneous. Intuitively, we trade off a restriction on the treatment mechanism for an assumption on the outcome model, i.e. independence for functional form.

Our goal is to define an alternative set of assumptions under which the conclusion of Lemma 2 continues to hold, replacing Assumption 3 with weaker conditions. We summarize

the invariances of the full structural system in the next section. A graphical summary of the causal structure is provided in Figure 5.

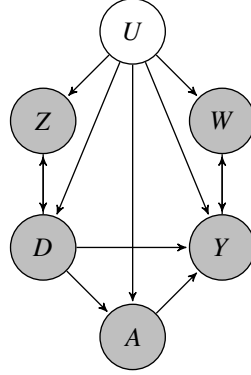


Figure 5: Homogeneous effect DAG

C.1 Extension

Assumption C.1 (RDD). *Assume that the following limits exist,*

$$\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* - \epsilon, U]], \quad \lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon, U]].$$

This condition modifies the standard RDD treatment assignment assumption (Assumption 1). It requires that a discontinuity in expected treatment probability at the cutoff after conditioning on U .

Assumption C.2 (Placebo variables). *Assume there exists some $\epsilon > 0$ such that for all $d \in \mathcal{D}(\epsilon)$ and for all $z \in \mathcal{Z}(\epsilon)$,*

a (Causal consistency): if $A = a$, $D = d$, and $Z = z$ then $Y = Y(a, d, z, U, \eta_y)$, $W = W(a, d, z, U, \eta_w)$. If $D = d$ and $Z = z$ then $A = A(d, z, U, \eta_A)$ almost surely.

b (Selection on unobservables): $Z \perp\!\!\!\perp \eta_y, \eta_a \mid U, D$ and $\eta_w \perp\!\!\!\perp Z, D \mid U$.

c (Overlap): if $f(u) > 0$ then $f(a, d, z \mid u) > 0$, where we assume the densities exist.

d (Exclusion): $Y(a, d, z, U, \eta_y) = Y(a, d, U, \eta_y)$, $W(a, d, z, U, \eta_w) = W(U, \eta_w)$, and $A(d, z, U, \eta_a) = A(d, U, \eta_a)$ almost surely.

Assumptions C.2a–c are equivalent to those in Assumption 2. However, Assumption C.2d relaxes Assumption 2d, allowing for unobserved confounding in the treatment mechanism. Under Assumption C.2, locally, our notation simplifies to

$$Y = Y(A, D, U, \eta_y), \quad W = W(U, \eta_w), \quad A = A(D, U, \eta_a),$$

and can be summarized by the result below.

Lemma C.1 (Necessary independences). *Assumption C.2 implies that for all $d \in \mathcal{D}(\epsilon)$ and $z \in \mathcal{Z}(\epsilon)$,*

$$a \quad A, Y \perp\!\!\!\perp Z \mid D, U$$

$$b \quad W \perp\!\!\!\perp D, Z \mid U$$

Assumption C.3 (Homogeneity). *Under homogenous treatment effects, we assume that for $d \in \mathcal{D}(\epsilon)$ the treatment effect is constant, i.e.*

$$\tau_0 = Y(1, d, U, \eta_y) - Y(0, d, U, \eta_y)$$

almost surely.

We use Assumption C.3 in place of Assumption 3 in Section 3. We impose that treatment effects are homogeneous across individuals near the cutoff, while allowing treatment assignment to depend on unobserved confounders that affect the potential outcomes.

Together with Assumptions C.1, C.2, and 4, this yields the following identification result:

Lemma C.2 (Limit). *Under Assumptions C.1, C.2, C.3, and 4,*

$$\tau_0 = \frac{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^*]}{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon, U] - \mathbb{E}[A \mid D = d^* - \epsilon, U]] \mid D = d^*]},$$

when $\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon, U] - \mathbb{E}[A \mid D = d^* - \epsilon, U]] \mid D = d^*] \neq 0$.

Lemma C.2 expresses the treatment effect τ_0 as the ratio of two discontinuities, each computed after conditioning on U and then averaging over U . This formulation permits identification even when the treatment (A) is confounded by unobservables.

For estimation, we can apply the same confounding bridge approach from Section 4 to recover the counterfactual term $\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* - \epsilon, U]] \mid D = d^*]$ by using the observed proxies for U . The final estimator will look like a ratio of estimators in Section 4, using Y in the numerator and A in the denominator.

C.2 Proof of extension

Proof of Lemma C.1. To show that $A \perp\!\!\!\perp Z \mid D, U$, we write

$$\begin{aligned} \mathbb{P}[A = a \mid D = d, Z = z, U] &= \mathbb{P}[A(d, U, \eta_a) = a \mid D = d, Z = z, U] \\ &= \mathbb{P}[A(d, U, \eta_a) = a \mid D = d, U], \end{aligned}$$

where the second line follows from Assumption C.2b (placebo selection on unobservables).

Otherwise, the result is the same as Lemma 1. $\square$

Proof of Lemma C.2. Assume a fuzzy design. We proceed in steps.

1. Notice that under Assumption C.2 (proxy controls), we are able to write,

$$Y = Y(0, D, U, \eta_y) + A(D, U, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)).$$

2. Using the decomposition above, we can write

$$\begin{aligned}
& \mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U] \\
&= \mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U] \\
&\quad + \mathbb{E}[A(d^* + \epsilon, U, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)) \mid D = d^* + \epsilon, U] \\
&\quad - \mathbb{E}[A(d^* - \epsilon, U, \eta_a)(Y(1, D, U, \eta_y) - Y(0, D, U, \eta_y)) \mid D = d^* - \epsilon, U] \\
&= \mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U] \\
&\quad + \tau_0 \mathbb{E}[A(d^* + \epsilon, U, \eta_a) \mid D = d^* + \epsilon, U] - \tau_0 \mathbb{E}[A(d^* - \epsilon, U, \eta_a) \mid D = d^* - \epsilon, U],
\end{aligned}$$

where we use Assumption C.3 to derive the second equality.

3. By Assumptions C.1 (RDD) and 4 (continuity), we can take the limit of the equality above to show,

$$\begin{aligned}
& \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \\
&= \lim_{\epsilon \downarrow 0} [\mathbb{E}[Y(0, d^* + \epsilon, U, \eta_y) \mid D = d^* + \epsilon, U] - \mathbb{E}[Y(0, d^* - \epsilon, U, \eta_y) \mid D = d^* - \epsilon, U]] \\
&\quad + \tau_0 \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, U, \eta_a) \mid D = d^* + \epsilon, U] - \mathbb{E}[A(D, U, \eta_a) \mid D = d^* - \epsilon, U]] \\
&= \tau_0 \lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, U, \eta_a) \mid D = d^* + \epsilon, U] - \mathbb{E}[A(D, U, \eta_a) \mid D = d^* - \epsilon, U]].
\end{aligned}$$

Specifically, to go from the second line to the third line, the initial terms cancel, and the later terms factorize.

4. Taking conditional expectations of both sides, we have that

$$\begin{aligned}
& \mathbb{E} \left[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^* \right] \\
&= \tau_0 \mathbb{E} \left[\lim_{\epsilon \downarrow 0} [\mathbb{E}[A(D, U, \eta_a) \mid D = d^* + \epsilon, U] - \mathbb{E}[A(D, U, \eta_a) \mid D = d^* - \epsilon, U]] \mid D = d^* \right].
\end{aligned}$$

Rearranging and gives

$$\tau_0 = \frac{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[Y \mid D = d^* + \epsilon, U] - \mathbb{E}[Y \mid D = d^* - \epsilon, U]] \mid D = d^*]}{\mathbb{E}[\lim_{\epsilon \downarrow 0} [\mathbb{E}[A \mid D = d^* + \epsilon, U] - \mathbb{E}[A \mid D = d^* - \epsilon, U]] \mid D = d^*]}.$$

□

D Equivalence proof

D.1 Notation

Let $\mathbf{e}_0 = (1, 0)^\top$ and $\mathbf{e}_1 = (0, 1)^\top$. Furthermore, let $q = \dim(w)$ and define

$$\begin{aligned} \omega_{i,+} &= \frac{1}{h_n} \mathbb{1}(D_i \geq d^*) K\left(\frac{|D_i - d^*|}{h_n}\right) \in \mathbb{R} & \mathbf{K}_+ &= \text{Diag}(\omega_{1,+}, \dots, \omega_{n,+}) \in \mathbb{R}^{n \times n} \\ R_{i,1} &= \left(1, \left(\frac{D_i - d^*}{h_n}\right)\right)^\top \in \mathbb{R}^2 & \mathbf{R}_p &= (R_{1,p}, \dots, R_{n,p})^\top \in \mathbb{R}^{n \times (p+1)} \\ R_{i,2} &= \left(1, \left(\frac{D_i - d^*}{b_n}\right), \left(\frac{D_i - d^*}{b_n}\right)^2\right)^\top \in \mathbb{R}^3 & \mathbf{Z} &= (Z_1, \dots, Z_n)^\top \in \mathbb{R}^{n \times q} \\ \mathbf{H}_p &= \text{Diag}(1, \dots, h_n^p) \in \mathbb{R}^{(p+1) \times (p+1)} & \mathbf{W} &= (W_1, \dots, W_n)^\top \in \mathbb{R}^{n \times q}. \end{aligned}$$

D.2 Estimator recap

We estimate $\mathbf{v}_+^\top = (\alpha_{+,0}, \alpha_{+,1}, \gamma_+^\top)$ by solving a local linear IV objective function with first order conditions,

$$\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \left(Y_i - R_{i,1}^\top \mathbf{H}_1 \hat{\alpha}_+ - W_i^\top \hat{\gamma}_+ \right) = \mathbf{0}.$$

Then we have that,

$$\hat{\mathbf{v}}_+ = \left(\begin{bmatrix} \mathbf{R}_1^\top \\ \mathbf{Z}^\top \end{bmatrix} \mathbf{K}_+ \begin{bmatrix} \mathbf{R}_1 & \mathbf{W} \end{bmatrix} \right)^{-1} \left(\begin{bmatrix} \mathbf{R}_1^\top \\ \mathbf{Z}^\top \end{bmatrix} \mathbf{K}_+ \mathbf{Y} \right) = \begin{bmatrix} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1 & \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \\ \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 & \mathbf{Z}^\top \mathbf{K}_+ \mathbf{W} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} \\ \mathbf{Z}^\top \mathbf{K}_+ \mathbf{Y} \end{bmatrix}.$$

We estimate $\beta_+^{w_j}$ using the local linear estimator of [Hahn et al. \(2001\)](#),

$$\mathbf{H}_1 \hat{\beta}_+^{w_j} = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n \omega_{i,+} \left(W_{i,j} - R_{i,1}^\top \beta \right)^2 = (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W}_{\cdot,j}.$$

Then, we let $\beta_{+,0}^w = (\beta_{+,0}^{w_1}, \dots, \beta_{+,0}^{w_q})^\top \in \mathbb{R}^q$ and $\beta_+^w = (\beta_+^{w_1}, \dots, \beta_+^{w_q}) \in \mathbb{R}^{2 \times q}$.

D.3 Main result

Proof of Proposition 1 (estimator decomposition). We proceed in steps, using regression algebra.

1. Using the formula for the inverse of a block matrix,

$$\begin{aligned}\hat{\mathbf{v}}_+ &= \begin{bmatrix} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1 & \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \\ \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 & \mathbf{Z}^\top \mathbf{K}_+ \mathbf{W} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} \\ \mathbf{Z}^\top \mathbf{K}_+ \mathbf{Y} \end{bmatrix} \\ &= \begin{bmatrix} (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} + (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} & -(\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \mathbf{Q}_+^{-1} \\ -\mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} & \mathbf{Q}_+^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} \\ \mathbf{Z}^\top \mathbf{K}_+ \mathbf{Y} \end{bmatrix}\end{aligned}$$

where $\mathbf{Q}_+ = \mathbf{Z}^\top \mathbf{K}_+ \{\mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+\} \mathbf{W}$ is the Schur complement. Then, it follows that for $\hat{\alpha}_+ = (\hat{g}_+(0), h_n \hat{g}_+^{(1)}(0))^\top$, we have that,

$$\begin{aligned}\hat{\alpha}_+ &= (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} \\ &\quad + (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} \\ &\quad - (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{W} \mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{Y} \\ &= \hat{\beta}_+^y + \hat{\beta}_+^w \{ \mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \mathbf{Y} - \mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \mathbf{Y} \} \\ &= \hat{\beta}_+^y - \hat{\beta}_+^w [\mathbf{Q}_+^{-1} \mathbf{Z}^\top \mathbf{K}_+ \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_+ \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_+ \} \mathbf{Y}] \\ &= \hat{\beta}_+^y - \hat{\beta}_+^w \hat{\gamma}_+, \end{aligned}$$

where $\hat{\beta}_+^y$ and $\hat{\beta}_+^{wj}$ solve the local linear objective functions,

$$\hat{\beta}_+^y = \arg \min_{\beta} \sum_{i=1}^n \omega_{i,+} \left(Y_i - R_{i,1}^\top \beta \right)^2, \quad \hat{\beta}_+^{wj} = \arg \min_{\beta} \sum_{i=1}^n \omega_{i,+} \left(W_{i,j} - R_{i,1}^\top \beta \right)^2.$$

2. We can go through the same steps to show that $\hat{\alpha}_- = \hat{\beta}_-^y - \hat{\beta}_-^w \hat{\gamma}_-$.

3. Plugging these expressions into our estimator for $\hat{\tau}_{\text{pdd}}$ in Equation (3), we find that,

$$\begin{aligned}\hat{\tau}_{\text{pdd}} &= \hat{\alpha}_{+,0} + \left(\hat{\beta}_{+,0}^w \right)^\top \hat{\gamma}_+ - \left\{ \hat{\alpha}_{-,0} + \left(\hat{\beta}_{+,0}^w \right)^\top \hat{\gamma}_- \right\} \\ &= \hat{\beta}_{+,0}^y - \left(\hat{\beta}_{+,0}^w \right)^\top \hat{\gamma}_+ + \left(\hat{\beta}_{+,0}^w \right)^\top \hat{\gamma}_+ - \left(\hat{\beta}_{-,0}^y - \left(\hat{\beta}_{-,0}^w \right)^\top \hat{\gamma}_- + \left(\hat{\beta}_{+,0}^w \right)^\top \hat{\gamma}_- \right) \\ &= \hat{\beta}_{+,0}^y - \hat{\beta}_{-,0}^y - \left(\hat{\beta}_{+,0}^w - \hat{\beta}_{-,0}^w \right)^\top \hat{\gamma}_- \\ &= \hat{\tau}_{\text{rdd}}^y - \left(\hat{\tau}_{\text{rdd}}^w \right)^\top \hat{\gamma}_-.\end{aligned}$$

4. Finally, we show the claim that $\hat{\gamma}_- = \left[\frac{1}{n} \sum_{i=1}^n \omega_{i,-} \{ Z_i (W_i^\perp)^\top \} \right]^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n \omega_{i,-} (Z_i Y_i^\perp) \right\}$.

Expanding out our expression for $\hat{\gamma}_-$, we have that,

$$\begin{aligned}
\hat{\gamma}_- &= \mathbf{Q}_-^{-1} \mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{Y} \\
&= \left[\mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{W} \right]^{-1} \left[\mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{Y} \right] \\
&= \left[\sum_{i=1}^n \omega_{i,-} Z_i \{ W_i^\top - R_{i,1}^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \} \right]^{-1} \left[\sum_{i=1}^n \omega_{i,-} Z_i \{ Y_i - R_{i,1}^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{Y} \} \right] \\
&= \left[\frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left\{ W_i^\top - R_{i,1}^\top \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,-} R_{i,1} R_{i,1}^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n \omega_{i,-} R_{i,1} W_i^\top \right\} \right]^{-1} \\
&\quad \times \left[\frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left\{ Y_i - R_{i,1}^\top \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,-} R_{i,1} R_{i,1}^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n \omega_{i,-} R_{i,1} Y_i \right\} \right] \\
&= \left[\frac{1}{n} \sum_{i=1}^n \omega_{i,-} \{ Z_i (W_i^\perp)^\top \} \right]^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n \omega_{i,-} (Z_i Y_i^\perp) \right\}.
\end{aligned}$$

□

E Consistency proof

E.1 Main result

Proof of Proposition 2 (estimand). We proceed in steps.

1. By Proposition 1,

$$\hat{\tau}_{\text{pdd}} = \hat{\beta}_{+,0}^y - \hat{\beta}_{-,0}^y - \left(\hat{\beta}_{+,0}^w - \hat{\beta}_{-,0}^w \right)^\top \hat{\gamma}_-.$$

Standard results from e.g. [Hahn et al. \(2001, Section 4.1\)](#) show that under our conditions,

$$\begin{aligned}
\hat{\beta}_{+,0}^y - \hat{\beta}_{-,0}^y &\xrightarrow{P} \lim_{\epsilon \downarrow 0} \{ \mathbb{E}[Y \mid D = d^* + \epsilon] - \mathbb{E}[Y \mid D = d^* - \epsilon] \} \\
\hat{\beta}_{+,0}^w - \hat{\beta}_{-,0}^w &\xrightarrow{P} \lim_{\epsilon \downarrow 0} \{ \mathbb{E}[W \mid D = d^* + \epsilon] - \mathbb{E}[W \mid D = d^* - \epsilon] \}.
\end{aligned}$$

Therefore, appealing to the continuous mapping theorem, it suffices to show that

$$\hat{\gamma}_- \xrightarrow{P} \gamma_-.$$

2. Within the proof of Proposition 1, we have shown that for

$$\mathbf{Q}_- = \mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{W},$$

$$\hat{\gamma}_- = \mathbf{Q}_-^{-1} \mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{Y}.$$

Using Lemma E.1 ($\mathbf{Q}_-$ -limit) and the continuous mapping theorem, we can write that,

$$n \mathbf{Q}_-^{-1} \xrightarrow{P} \kappa^{-1} \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* - \epsilon)^{-1}, \quad \kappa = f_{d,+}(d^*) \int_0^\infty K(u) \, du.$$

By Lemma E.1 ($\mathbf{Q}_-$ -limit), replacing $\mathbf{W}$ with $\mathbf{Y}$, we can analogously show that,

$$\frac{1}{n} \mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{Y} \xrightarrow{P} \kappa \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, Y_i \mid D_i = d^* - \epsilon).$$

Therefore, applying the continuous mapping and dominated convergence theorems, it follows that,

$$\hat{\gamma}_- \xrightarrow{P} \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* - \epsilon)^{-1} \text{Cov}(Z_i, Y_i \mid D_i = d^* - \epsilon).$$

3. We verify the claim that γ_- is the best linear predictor for

$$Y = c + W^\top \gamma_- + \eta, \quad \lim_{\epsilon \downarrow 0} \mathbb{E}[\eta Z_i \mid D_i = d^* - \epsilon] = 0, \quad \lim_{\epsilon \downarrow 0} \mathbb{E}[\eta \mid D_i = d^* - \epsilon] = 0.$$

Substituting for Y in the probability limit gives

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D_i = d^* - \epsilon)^{-1} \text{Cov}(Z_i, Y_i \mid D_i = d^* - \epsilon) \\ &= \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D_i = d^* - \epsilon)^{-1} \{ \text{Cov}(Z_i, c + W_i^\top \gamma_- + \eta \mid D_i = d^* - \epsilon) \} \\ &= \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D_i = d^* - \epsilon)^{-1} \{ \text{Cov}(Z_i, W_i^\top \mid D_i = d^* - \epsilon) \gamma_- + \text{Cov}(Z_i, \eta \mid D_i = d^* - \epsilon) \} \\ &= \gamma_-. \end{aligned}$$

□

Proof of Corollary 1 (consistency). We proceed in steps. We focus only on showing convergence above the cutoff, and convergence below the cutoff follows from the same argument

applied to $\alpha_{-,0}$ and γ_- .⁷ We want to show that,

$$\hat{\alpha}_{+,0} + (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_+ \xrightarrow[\epsilon \downarrow 0]{P} \lim_{\epsilon \downarrow 0} \mathbb{E}[h_+(\epsilon, W) \mid D = d^*].$$

1. We express the limit as a sum.

Substituting in $h_+(d - d^*, w)$ from Equation (1),

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \mathbb{E}[h_+(\epsilon, W) \mid D = d^*] &= g_+(0) + \mathbb{E}[W^\top \mid D = d^*] \gamma_+ \\ &= \alpha_{+,0} + (\beta_{+,0}^w)^\top \gamma_+. \end{aligned}$$

2. It is helpful to characterize a partially linear representation for Y .

By Lemma 3 (factuals) and Equation (1), for all $d \in \mathcal{D}_+(\epsilon)$ recall that we can write,

$$\mathbb{E}[Y - h_+(D - d^*, W) \mid D = d, Z = z] = \mathbb{E}[Y - g_+(D - d^*) - W^\top \gamma_+ \mid D = d, Z = z] = 0.$$

Using this moment condition, it follows that for some $\delta \in [0, D - d^*]$ we can use a Taylor expansion to represent,

$$\begin{aligned} Y &= g_+(D - d^*) + W^\top \gamma_+ + \tilde{\eta} \\ &= g_+(0) + g_+^{(1)}(0)(D - d^*) + W^\top \gamma_+ + \tilde{\eta} + \frac{1}{2} g_+^{(2)}(\delta)(D - d^*)^2 \\ &= \alpha_{+,0} + \alpha_{+,1}(D - d^*) + W^\top \gamma_+ + \tilde{\eta} + \frac{1}{2} g_+^{(2)}(\delta)(D - d^*)^2, \end{aligned}$$

where $\mathbb{E}[\tilde{\eta} \mid D = d, Z = z] = 0$ for all $d \in \mathcal{D}_+(\epsilon)$ and $z \in \mathcal{Z}_+(\epsilon)$.

3. Using the representation in step 2, we now show that $\alpha_{+,0} = \beta_{+,0}^y - (\beta_{+,0}^w)^\top \gamma_+$.

Applying the conditional expectation $\mathbb{E}[\cdot \mid Z, D = d^* + \epsilon]$ on both sides,

$$\mathbb{E}[Y \mid Z, D = d^* + \epsilon] = \alpha_{+,0} + \alpha_{+,1}\epsilon + \mathbb{E}[W^\top \mid Z, D = d^* + \epsilon] \gamma_+ + \frac{1}{2} g_+^{(2)}(\delta) \epsilon^2.$$

Further taking the conditional expectation $\mathbb{E}[\cdot \mid D = d^* + \epsilon]$ on both sides and appealing to the law of iterated expectations,

$$\mathbb{E}[Y \mid D = d^* + \epsilon] = \alpha_{+,0} + \alpha_{+,1}\epsilon + \mathbb{E}[W^\top \mid D = d^* + \epsilon] \gamma_+ + \frac{1}{2} g_+^{(2)}(\delta) \epsilon^2.$$

⁷We will not use cancellation, so that the same argument gives convergence below the cutoff.

Finally, taking the limit on both sides,

$$\lim_{\epsilon \downarrow 0} \mathbb{E}[Y \mid D_i = d^* + \epsilon] = \alpha_{+,0} + \lim_{\epsilon \downarrow 0} \mathbb{E}[W_i^\top \mid D_i = d^* + \epsilon] \gamma_+.$$

Rearranging gives $\alpha_{+,0} = \beta_{+,0}^y - (\beta_{+,0}^w)^\top \gamma_+$.

4. Next, we show that $\hat{\gamma}_+ \xrightarrow{P} \gamma_+$.

As argued in the proof of Proposition 2 (estimand) we have that,

$$\begin{aligned} \hat{\gamma}_+ &\xrightarrow{P} \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* + \epsilon)^{-1} \text{Cov}(Z_i, Y_i \mid D_i = d^* + \epsilon) \\ &= \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* + \epsilon)^{-1} \text{Cov}\left(Z_i, \alpha_{+,0} + \alpha_{+,1}(D_i - d^*) + W_i^\top \gamma_+ + \tilde{\eta} + \frac{1}{2} g_+^{(2)}(\delta)(D_i - d^*)^2 \mid D_i = d^* + \epsilon\right) \\ &= \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* + \epsilon)^{-1} \text{Cov}(Z_i, W_i^\top \mid D_i = d^* + \epsilon) \gamma_+ \\ &\quad + \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* + \epsilon)^{-1} \text{Cov}(Z_i, \tilde{\eta} \mid D_i = d^* + \epsilon) \\ &= \gamma_+, \end{aligned}$$

where the first equality uses step 2 and the third equality uses that

$$\begin{aligned} \text{Cov}(Z_i, \tilde{\eta} \mid D_i = d^* + \epsilon) &= \mathbb{E}[Z_i \tilde{\eta} \mid D_i = d^* + \epsilon] - \mathbb{E}[Z_i \mid D_i = d^* + \epsilon] \mathbb{E}[\tilde{\eta} \mid D_i = d^* + \epsilon] \\ &= \mathbb{E}[Z_i \mathbb{E}[\tilde{\eta} \mid Z_i, D_i = d^* + \epsilon]] - \mathbb{E}[Z_i \mid D_i = d^* + \epsilon] \mathbb{E}[\mathbb{E}[\tilde{\eta} \mid Z_i, D_i = d^* + \epsilon]] \\ &= 0. \end{aligned}$$

5. We also show that $\hat{\alpha}_{+,0} \xrightarrow{P} \alpha_{+,0}$.

In the proof of Proposition 1 (estimator decomposition), we showed that $\hat{\alpha}_{+,0} =$

$$\hat{\beta}_{+,0}^y - (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_+.$$

In step 3, we showed that $\alpha_{+,0} = \beta_{+,0}^y - (\beta_{+,0}^w)^\top \gamma_+$.

Recall that $\hat{\beta}_{+,0}^y \xrightarrow{P} \beta_{+,0}^y$ and $\hat{\beta}_{+,0}^w \xrightarrow{P} \beta_{+,0}^w$ by standard arguments (Hahn et al., 2001,

Section 4.1). Using that $\hat{\gamma}_+ \xrightarrow{P} \gamma_+$ from step 4, the continuous mapping theorem

implies that $\hat{\beta}_{+,0}^y - (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_+ \xrightarrow{P} \beta_{+,0}^y - (\beta_{+,0}^w)^\top \gamma_+$. Therefore, $\hat{\alpha}_{+,0} \xrightarrow{P} \alpha_{+,0}$.

6. Collecting results and applying the continuous mapping theorem we have shown that,

$$\begin{aligned}\hat{\alpha}_{+,0} + (\hat{\beta}_{+,0}^w)^\top \hat{\gamma}_+ &\xrightarrow{p} \alpha_{+,0} + (\beta_{+,0}^w)^\top \gamma_+ \\ &= \lim_{\epsilon \downarrow 0} \mathbb{E}[h_+(\epsilon, W) \mid D = d^*]\end{aligned}$$

where the convergence uses steps 4 and 5, and the equality uses step 1.

□

E.2 Technical lemma

Lemma E.1 ($\mathbf{Q}_-$ -limit). *Let Assumptions 7a-c and 8 hold, and assume that $h \rightarrow 0$ and $nh \rightarrow \infty$. Then,*

$$\frac{1}{n} \mathbf{Q}_- \xrightarrow{p} \kappa \lim_{\epsilon \downarrow 0} \text{Cov}(Z_i, W_i^\top \mid D = d^* - \epsilon),$$

for $\kappa = f_{d,-}(d^*) \int_0^\infty K(u) \, du$.

◀

Proof of Lemma E.1. We proceed in steps.

1. As shown in the proof of Proposition 1,

$$\mathbf{Q}_- = \mathbf{Z}^\top \mathbf{K}_- \{ \mathbf{I} - \mathbf{R}_1 (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \} \mathbf{W}.$$

Therefore, we have the decomposition

$$\begin{aligned}\frac{1}{n} \mathbf{Q}_- &= \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \{ W_i^\top - R_{i,1}^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \} \\ &= \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left(W_i^\top - \mathbf{e}_0^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \right) \\ &\quad - \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left\{ \left(\frac{D_i - d^*}{h_n} \right) \mathbf{e}_1^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \right\}.\end{aligned}$$

We analyze each term below.

2. Under Assumptions 7a and 8, results from Hahn et al. (2001, Section 4.1) imply that

$$\mathbf{e}_0^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \xrightarrow{p} (\beta_{-,0}^w)^\top = \lim_{\epsilon \downarrow 0} \mathbb{E}[W_i^\top \mid D_i = d^* - \epsilon]$$

$$\frac{1}{h_n} \mathbf{e}_1^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \xrightarrow{p} (\beta_{-,1}^w)^\top.$$

3. Here, we show that,

$$\left(\frac{1}{nh_n} \sum_{i=1}^n \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (W_i - \beta_{-,0}^w)^\top \right) = \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (W_i - \beta_{-,0}^w)^\top \right] + o_p(1)$$

Set $\mathbf{X}_i = \frac{1}{h_n} \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (W_i - \beta_{-,0}^w)^\top \in \mathbb{R}^{q \times q}$. For each component $\ell \in \{(1,1), \dots, (q,q)\}$, we show $\frac{1}{n} \sum_i X_{i,\ell} = \mathbb{E}[X_{i,\ell}] + o_p(1)$ by bounding $\text{Var}(\frac{1}{n} \sum_i X_{i,\ell}) = \frac{1}{n} \text{Var}(X_{1,\ell}) \leq \frac{1}{n} \|X_{1,\ell}\|_{L_2}^2$ and applying Chebyshev's inequality. Computing the L_2 bound, by the law of iterated expectations we have,

$$\begin{aligned} h_n \mathbb{E}[X_{i,\ell}^2] &= \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_{i,k}^2 (W_{i,j} - \beta_{-,0}^{w_j})^2 \right] \\ &= \int_{-\infty}^0 K(u)^2 f_d(uh_n + d^*) \mathbb{E} \left[Z_{i,k}^2 (W_{i,j} - \beta_{-,0}^{w_j})^2 \mid D = uh_n + d^* \right] du \\ &\rightarrow \rho_{-,z_k w_j}(0) f_{d,-}(d^*) \int_{-\infty}^0 K(u)^2 du < \infty, \end{aligned}$$

where the squaring is element-wise and the limit/integral swap uses Assumptions 7 and 8. This shows that $\|X_{1,\ell}\|_{L_2}^2 = O(1/h_n)$ and thus $\frac{1}{n} \|X_{1,\ell}\|_{L_2}^2 = O\left(\frac{1}{nh_n}\right) \rightarrow 0$. Then from Chebyshev's inequality,

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n (X_{i,\ell} - \mathbb{E}[X_{i,\ell}]) \right| > \epsilon \right) \leq \frac{1}{\epsilon^2} \frac{1}{n} \|X_{1,\ell}\|_{L_2}^2 \rightarrow 0,$$

and the conclusion follows.

4. Consider the first term in the decomposition. We will show that

$$\frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left(W_i^\top - \mathbf{e}_0^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \right) \xrightarrow[p]{\kappa \lim_{\epsilon \downarrow 0}} \text{Cov}(Z_i, W_i^\top \mid D_i = d^* - \epsilon).$$

By steps 2 and 3,

$$\begin{aligned}
& \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left(W_i^\top - \mathbf{e}_0^\top (\mathbf{R}_1^\top \mathbf{K} - \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K} - \mathbf{W} \right) \\
&= \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i (W_i - \beta_{-,0}^w)^\top + o_p(1) \\
&= \left(\frac{1}{nh_n} \sum_{i=1}^n \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (W_i - \beta_{-,0}^w)^\top \right) + o_p(1) \\
&= \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (W_i - \beta_{-,0}^w)^\top \right] + o_p(1).
\end{aligned}$$

We use the law of iterated expectations and u -substitution with $u = \frac{d-d^*}{h_n}$ to express the former term as,

$$\begin{aligned}
& \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{E}[Z_i (W_i - \beta_{-,0}^w)^\top \mid D_i] \right] \\
&= \frac{1}{h_n} \int_{-\infty}^{d^*} f_d(d) \mathbb{1}(d < d^*) K\left(\frac{d - d^*}{h_n}\right) \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d] \, dd \\
&= \frac{1}{h_n} \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d^* + uh_n] h_n \, du \\
&= \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d^* + uh_n] \, du \\
&= \kappa \lim_{\epsilon \downarrow 0} \text{Cov}(Z, W^\top \mid D = d^* - \epsilon) + o_p(1).
\end{aligned}$$

The last line uses the dominated convergence theorem and the boundedness of $\mathbb{E}[Z_i W_i^\top \mid D_i = \cdot]$ and $\mu_{-,z}(\cdot)$ from Assumption 7, taking the limit as $h_n \rightarrow 0$. Specifically, we argue that

$$\begin{aligned}
& \lim_{h_n \downarrow 0} \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d^* + uh_n] \, du \\
&= \int_{-\infty}^0 f_d(d^*) K(-u) \lim_{\epsilon \downarrow 0} \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d^* - \epsilon] \, du \\
&= f_d(d^*) \lim_{\epsilon \downarrow 0} \mathbb{E}[Z(W - \beta_{-,0}^w)^\top \mid D = d^* - \epsilon] \int_{-\infty}^0 K(-u) \, du \\
&= \kappa \lim_{\epsilon \downarrow 0} \text{Cov}(Z, W^\top \mid D = d^* - \epsilon).
\end{aligned}$$

5. Consider the second term in the decomposition. We will show that

$$\frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left\{ \left(\frac{D_i - d^*}{h_n} \right) \mathbf{e}_1^\top (\mathbf{R}_1^\top \mathbf{K} - \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K} - \mathbf{W} \right\} \xrightarrow{p} 0.$$

By step 2 and an analogous argument as in step 3,

$$\begin{aligned}
& \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i \left\{ \left(\frac{D_i - d^*}{h_n} \right) \mathbf{e}_1^\top (\mathbf{R}_1^\top \mathbf{K}_- \mathbf{R}_1)^{-1} \mathbf{R}_1^\top \mathbf{K}_- \mathbf{W} \right\} \\
&= \frac{1}{n} \sum_{i=1}^n \omega_{i,-} Z_i (D_i - d^*) (\beta_{-,1}^w)^\top + o_p(1) \\
&= \frac{1}{nh_n} \sum_{i=1}^n \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (D_i - d^*) (\beta_{-,1}^w)^\top + o_p(1) \\
&= \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_i (D_i - d^*) (\beta_{-,1}^w)^\top \right] + o_p(1).
\end{aligned}$$

We use the law of iterated expectations and u -substitution with $u = \frac{d-d^*}{h_n}$ to express the former term as

$$\begin{aligned}
& \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{E}[Z_i | D_i] (D_i - d^*) \right] (\beta_{-,1}^w)^\top \\
&= \frac{1}{h_n} \int_{-\infty}^{d^*} f_d(d) \mathbb{1}(d < d^*) K\left(\frac{d - d^*}{h_n}\right) \mathbb{E}[Z | D = d] (d - d^*) \, dd (\beta_{-,1}^w)^\top \\
&= \frac{1}{h_n} \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z | D = d^* + uh_n] u h_n \, du (\beta_{-,1}^w)^\top \\
&= h_n \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z | D = d^* + uh_n] u \, du (\beta_{-,1}^w)^\top \\
&= o_p(1).
\end{aligned}$$

The last line uses the dominated convergence theorem and the boundedness of $\mu_{-,z}(\cdot)$ from Assumption 7, taking the limit as $h_n \rightarrow 0$. Specifically, we argue that

$$\begin{aligned}
\lim_{h_n \downarrow 0} \int_{-\infty}^0 f_d(d^* + uh_n) K(-u) \mathbb{E}[Z | D = d^* + uh_n] u \, du &= \int_{-\infty}^0 f_{d,+}(d^*) K(-u) \mu_{-,z}(0) u \, du \\
&= f_{d,-}(d^*) \mu_{-,z}(0) \int_{-\infty}^0 K(-u) u \, du
\end{aligned}$$

□

F Bias corrected inference

F.1 Notation

Here, we compile all the notation used in the appendix. When we introduce a term for the first time, we try to redefine it in the proofs and lemmas in this section.

We write the ratio of the two bandwidths as

$$\xi := \frac{h_n}{b_n}, \quad \text{equivalently} \quad b_n = \frac{1}{\xi} h_n,$$

so that $h_n^p/b_n^p = \xi^p$ for every p . Under the MSE-optimal rates $h_n = c_h n^{-1/5}$ and $b_n = c_b n^{-1/5}$, the ratio $\xi = c_h/c_b$ is a fixed constant.

For the kernel constants, define

$$\gamma_{k,j} = \int_0^\infty u^k K(u)^j du, \quad \bar{\gamma}_{k,2}(\xi) = \int_0^\infty u^k K(u) K(\xi u) du.$$

We note that $\bar{\gamma}_{k,2}(1) = \gamma_{k,2}$, and when ξ is clear from context we write $\bar{\gamma}_{k,2} = \bar{\gamma}_{k,2}(\xi)$.

Throughout, $g_+(\cdot)$ and $g_-(\cdot)$ denote the outcome regression functions just above and below the cutoff, and $\mu_{+,w_j}(\cdot)$, $\mu_{+,z_j}(\cdot)$ the corresponding conditional-mean functions of $W_{i,j}$ and $Z_{i,j}$; superscript (m) denotes the m^{th} derivative, evaluated at 0 for the one-sided limit at the cutoff. We also write $C := c_h^{5/2}$. We let $\mathbf{e}_k$ be a conformable matrix of zeros, with a 1 in the $(k+1)^{\text{th}}$ position. For example, written $\mathbf{e}_2^\top \Gamma_{+,2}$, we have $\mathbf{e}_2 = (0, 0, 1)^\top$, and witting $\mathbf{e}_2^\top \Psi_{+,2}$, we have $\mathbf{e}_2 = (0, 0, 1, 0, \dots, 0) \in \mathbb{R}^{3+q}$.

We collect the basic objects below.

$$\begin{aligned}
f_{d,+}(d^*) &:= \lim_{\epsilon \downarrow 0^+} f_d(d^* + \epsilon) \\
f_{d,-}(d^*) &:= \lim_{\epsilon \downarrow 0^+} f_d(d^* - \epsilon) \\
\mu_{+,w}(0) &:= (\mu_{+,w_1}(0), \dots, \mu_{+,w_q}(0))^\top \in \mathbb{R}^q \\
\mu_{+,z}(0) &:= (\mu_{+,z_1}(0), \dots, \mu_{+,z_q}(0))^\top \in \mathbb{R}^q \\
\mu_{+,zw^\top}(0) &= \begin{pmatrix} \mu_{+,z_1 w_1}(0) & \cdots & \mu_{+,z_1 w_q}(0) \\ \vdots & \ddots & \vdots \\ \mu_{+,z_q w_1}(0) & \cdots & \mu_{+,z_q w_q}(0) \end{pmatrix} \in \mathbb{R}^{q \times q} \\
\hat{\beta}_{+,0}^{w_j} &:= \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{+,i} R_{i,1} W_{i,j} \right) \in \mathbb{R} \\
\hat{\beta}_{+,0}^w &:= \left(\hat{\beta}_{+,0}^{w_1}, \dots, \hat{\beta}_{+,0}^{w_q} \right)^\top \in \mathbb{R}^q \\
\mathbf{e}_{+,w} &:= (1, 0, \lim_{\epsilon \rightarrow 0^+} \mathbb{E}[W_i^\top \mid D = d^* + \epsilon])^\top \in \mathbb{R}^{2+q} \\
&= (1, 0, \beta_{+,0}^{w_1}, \dots, \beta_{+,0}^{w_q}) \\
\mathbf{H}_1 &:= \text{Diag}(1, h_n) \in \mathbb{R}^2 \\
\mathbf{B}_2 &:= \text{Diag}(1, b_n, b_n^2) \in \mathbb{R}^3 \\
\omega_{+,i} &:= \frac{1}{h_n} \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \\
\delta_{+,i} &:= \frac{1}{b_n} \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right)
\end{aligned}$$

Here $R_{i,1} = \left(1, \frac{D_i - d^*}{h_n}\right)^\top$ is localized at the bandwidth h_n , whereas $R_{i,2} = \left(1, \frac{D_i - d^*}{b_n}, \left(\frac{D_i - d^*}{b_n}\right)^2\right)^\top$ and $\delta_{+,i}$ are localized at the bandwidth b_n . For a vector $\mathbf{v}$ we use $\mathbf{v}^2 = (v_1^2, \dots, v_n^2)^\top$, and $\mathbf{e}_p$ denotes a conformable vector of zeros with a 1 in the $(p+1)^{\text{th}}$ element.

The “ $-$ ” analogues of every object below are defined by replacing each “ $+$ ” subscript, and the associated one-sided limits and weights, with their “ $-$ ” counterparts. The structural coefficients $\gamma_+, \gamma_- \in \mathbb{R}^q$ on W_i (Equation (3)) are estimated by $\hat{\gamma}_+, \hat{\gamma}_-$, with components $\gamma_{\pm,j}$. The remaining regression coefficients and their estimators are

$$\begin{aligned}
\alpha_+ &:= \mathbf{H}_1(g_+(0), g_+^{(1)}(0))^\top = (\alpha_{+,0}, \alpha_{+,1})^\top \\
\nu_+ &:= (\alpha_{+,0}, \alpha_{+,1}, \gamma_+^\top)^\top, \quad \hat{\nu}_+ := \hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_i \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} Y_i \right) \\
\beta_{+,0}^{w_j} &:= \mu_{+,w_j}(0), \quad \beta_+^{w_j} := \mathbf{H}_1(\mu_{+,w_j}(0), \mu_{+,w_j}^{(1)}(0))^\top \\
\pi_+ &:= \mathbf{B}_2 \left(g_+(0), g_+^{(1)}(0), \frac{1}{2} g_+^{(2)}(0) \right)^\top \\
\kappa_+^{w_j} &:= \mathbf{B}_2 \left(\mu_{+,w_j}(0), \mu_{+,w_j}^{(1)}(0), \frac{1}{2} \mu_{+,w_j}^{(2)}(0) \right)^\top \\
\hat{V} &:= \hat{\mathbf{s}}_+^\top \left(\frac{1}{nh_n} \sum_i X_{i,+} X_{i,+}^\top \right) \hat{\mathbf{s}}_+ + \hat{\mathbf{s}}_-^\top \left(\frac{1}{nh_n} \sum_i \tilde{X}_{i,-} \tilde{X}_{i,-}^\top \right) \hat{\mathbf{s}}_- \\
\mathbf{e}_2^\top \hat{\pi}_+ &:= \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{n} \sum_i \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} Y_i \right) \\
\mathbf{e}_2^\top \hat{\kappa}_+^{w_j} &:= \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{n} \sum_i \delta_{+,i} R_{i,2} W_{i,j} \right) \\
\hat{g}_+^{(2)}(0) &:= \frac{2}{b_n^2} \mathbf{e}_2^\top \hat{\pi}_+ \\
\hat{\mu}_{+,w_j}^{(2)}(0) &:= \frac{2}{b_n^2} \mathbf{e}_2^\top \hat{\kappa}_+^{w_j} \\
\hat{\mathbf{e}}_{+,w} &:= \left(1, 0, \left(\hat{\beta}_{+,0}^w \right)^\top \right)^\top \in \mathbb{R}^{2+q} \\
V &:= \mathbf{s}_+^\top \Sigma_{x,+} \mathbf{s}_+ + \mathbf{s}_-^\top \Sigma_{x,-} \mathbf{s}_-
\end{aligned}$$

The bias-correction design matrices and the curvature weights, bias terms, and their estimators are

$$\begin{aligned}
\hat{\Gamma}_{+,1} &:= \frac{1}{n} \sum_{i=1}^n \omega_{i,+} R_{i,1} R_{i,1}^\top \in \mathbb{R}^{2 \times 2} \\
\hat{\Psi}_{+,1} &:= \frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} R_{i,1}^\top & R_{i,1} W_i^\top \\ Z_i R_{i,1}^\top & Z_i W_i^\top \end{bmatrix} \in \mathbb{R}^{(2+q) \times (2+q)} & \hat{\Omega}_+ &:= \frac{1}{n} \sum_i \omega_{+,i} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \left(\frac{D_i - d^*}{h_n} \right)^2 \in \mathbb{R}^{2+q} \\
\hat{\Gamma}_{+,2} &:= \frac{1}{n} \sum_i \delta_{+,i} R_{i,2} R_{i,2}^\top \in \mathbb{R}^{3 \times 3} & \hat{\Lambda}_+ &:= \frac{1}{n} \sum_i \omega_{+,i} R_{i,1} \left(\frac{D_i - d^*}{h_n} \right)^2 \in \mathbb{R}^2 \\
\hat{\Psi}_{+,2} &:= \frac{1}{n} \sum_i \delta_{+,i} \begin{bmatrix} R_{i,2} R_{i,2}^\top & R_{i,2} W_i^\top \\ Z_i R_{i,2}^\top & Z_i W_i^\top \end{bmatrix} \in \mathbb{R}^{(3+q) \times (3+q)} & \Omega_+ &:= f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1}, \gamma_{2,1} \mu_{+,z}(0)^\top)^\top \in \mathbb{R}^{2+q} \\
\Gamma_{+,1} &:= f_{d,+}(d^*) \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} \\ \gamma_{1,1} & \gamma_{2,1} \end{bmatrix} \in \mathbb{R}^{2 \times 2} & \Lambda_+ &:= f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1})^\top \in \mathbb{R}^2 \\
\Gamma_{+,2} &:= f_{d,+}(d^*) \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} & \gamma_{2,1} \\ \gamma_{1,1} & \gamma_{2,1} & \gamma_{3,1} \\ \gamma_{2,1} & \gamma_{3,1} & \gamma_{4,1} \end{bmatrix} \in \mathbb{R}^{3 \times 3} & \eta_{+,IV} &:= \frac{C}{2} g_+^{(2)}(0) \Omega_+ \\
& & \hat{\eta}_{+,IV} &:= \frac{h_n^2}{2} \hat{\Omega}_+ \hat{g}_+^{(2)}(0) \\
& & \eta_{+,w_j} &:= \frac{C}{2} \Lambda_+ \mu_{+,w_j}^{(2)}(0) \\
& & \hat{\eta}_{+,w_j} &:= \frac{h_n^2}{2} \hat{\Lambda}_+ \hat{\mu}_{+,w_j}^{(2)}(0)
\end{aligned}$$

$\Psi_{+,1}$ and $\Psi_{+,2}$ denote the probability limits of $\hat{\Psi}_{+,1}$ and $\hat{\Psi}_{+,2}$, with explicit forms in Lemmas F.1 and F.8; τ_{pdd} is the PDD estimand; $\hat{\tau}_{\text{pdd}}$ is its estimator in Equation (3); and $\hat{\tau}_{\text{pdd}}^{\text{rbc}}$ is the bias-corrected estimator in Equation (6).

The conditional second-moment functions of the residuals that enter Lemma F.11 are defined as follows. Writing $\varepsilon_{i,y} = Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+$ and $\varepsilon_{i,w_j} = W_{i,j} - \mu_{+,w_j}(D_i - d^*)$ for the above-cutoff residuals (with “ $-$ ” analogues replacing $g_+, \gamma_+, \mu_{+,w_j}$ by $g_-, \gamma_-, \mu_{-,w_j}$), and taking one-sided limits at the cutoff,

$$\begin{aligned}
\sigma_{+,y}^2(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,y}^2 \mid D_i = d^* + \epsilon] \in \mathbb{R} & \rho_{+,y^2z}(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,y}^2 Z_i \mid D_i = d^* + \epsilon] \in \mathbb{R}^q \\
\sigma_{+,w_j}^2(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,w_j}^2 \mid D_i = d^* + \epsilon] \in \mathbb{R} & \rho_{+,y^2zz^\top}(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,y}^2 Z_i Z_i^\top \mid D_i = d^* + \epsilon] \in \mathbb{R}^{q \times q} \\
\rho_{+,yw_j}(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,y} \varepsilon_{i,w_j} \mid D_i = d^* + \epsilon] \in \mathbb{R} & \rho_{+,yw_jz}(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[\varepsilon_{i,y} \varepsilon_{i,w_j} Z_i \mid D_i = d^* + \epsilon] \in \mathbb{R}^q \\
\rho_{+,zkw_j}(0) &:= \lim_{\epsilon \downarrow 0^+} \mathbb{E}[Z_{i,k} \varepsilon_{i,w_j} \mid D_i = d^* + \epsilon] \in \mathbb{R}
\end{aligned}$$

Since $\mathbb{E}[\varepsilon_{i,y} \mid D_i, Z_i] = 0$ and $\mathbb{E}[\varepsilon_{i,w_j} \mid D_i] = 0$, these coincide with the corresponding one-sided conditional variances and covariances at the cutoff.

F.2 Characterizing asymptotic bias

Let $\hat{\mathbf{e}}_{+,w} = \left(1, 0, \left(\hat{\beta}_{+,0}^w\right)^\top\right)^\top$ and $\nu_+ = (\alpha_{+,0}, \alpha_{+,1}, \gamma_+^\top)^\top \in \mathbb{R}^{2+q}$, where $\alpha_+ = \mathbf{H}_1(g_+(0), g_+^{(1)}(0))^\top$.

Using this notation, we can express our estimator from Equation (3) as,

$$\begin{aligned} \hat{\tau}_{\text{pdd}} - \tau_{\text{pdd}} &= \left\{ \hat{\alpha}_{+,0} + \left(\hat{\beta}_{+,0}^w\right)^\top \hat{\gamma}_+ - \alpha_{+,0} - \left(\beta_{+,0}^w\right)^\top \gamma_+ \right\} - \left\{ \hat{\alpha}_{-,0} + \left(\hat{\beta}_{+,0}^w\right)^\top \hat{\gamma}_- - \alpha_{-,0} - \left(\beta_{+,0}^w\right)^\top \gamma_- \right\} \\ &= \hat{\mathbf{e}}_{+,w}^\top \{\hat{\nu}_+ - \nu_+\} + \nu_+^\top (\hat{\mathbf{e}}_{+,w} - \mathbf{e}_{+,w}) - \hat{\mathbf{e}}_{+,w}^\top \{\hat{\nu}_- - \nu_-\} - \nu_-^\top (\hat{\mathbf{e}}_{+,w} - \mathbf{e}_{+,w}) \\ &= \hat{\mathbf{e}}_{+,w}^\top \{\hat{\nu}_+ - \nu_+\} + \gamma_+^\top (\hat{\beta}_{+,0}^w - \beta_{+,0}^w) - \hat{\mathbf{e}}_{+,w}^\top \{\hat{\nu}_- - \nu_-\} - \gamma_-^\top (\hat{\beta}_{+,0}^w - \beta_{+,0}^w) \\ \hat{\nu}_+ &:= \hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} Y_i \right). \end{aligned}$$

With these definitions, to characterize the asymptotic bias we decompose the differences $(\hat{\nu}_+ - \nu_+)$ and $(\hat{\beta}_{+,0}^w - \beta_{+,0}^w)^\top$ as follows,

$$\begin{aligned} (\hat{\nu}_+ - \nu_+) &= \hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - R_{i,1}^\top \alpha_+ - W_i^\top \gamma_+) \right) \\ &= \hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \\ &\quad + \hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (g_+(D_i - d^*) - R_{i,1}^\top \alpha_+) \right) \\ (\hat{\beta}_{+,0}^w - \beta_{+,0}^w)^\top &= \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_i^\top - \mu_{+,w}(D_i - d^*)^\top\} \right) \\ &\quad + \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{\mu_{+,w}(D_i - d^*)^\top - R_{i,1}^\top \beta_+^w\} \right) \end{aligned} \tag{5}$$

The two lemmas below characterize the limit of the leading terms above at the MSE-optimal rate $h_n \asymp c_h n^{-1/5}$.

Lemma F.1 ($\hat{\Psi}_{+,1}$ convergence). *Under the conditions of Theorem 2, $\hat{\Psi}_{+,1} \xrightarrow{P} \Psi_{+,1}$, where*

$$\Psi_{+,1} := f_{d,+}(d^*) \begin{pmatrix} \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} \\ \gamma_{1,1} & \gamma_{2,1} \end{bmatrix} & \begin{bmatrix} \gamma_{0,1} \\ \gamma_{1,1} \end{bmatrix} \mu_{+,w}(0)^\top \\ \mu_{+,z}(0) \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} \end{bmatrix} & \gamma_{0,1} \mu_{+,zw}(0) \end{pmatrix}.$$

◀

Lemma F.2 ($\hat{\Gamma}_{+,1}$ convergence). *Under the conditions of Theorem 2, $\hat{\Gamma}_{+,1} \xrightarrow{P} \Gamma_{+,1} := f_{d,+}(d^*) \begin{pmatrix} \gamma_{0,1} & \gamma_{1,1} \\ \gamma_{1,1} & \gamma_{2,1} \end{pmatrix}$.*

◀

Lemma F.3 (IV bias convergence). *Under the conditions of Theorem 2, for $C = c_h^{5/2}$, we have that,*

$$\begin{aligned} \sqrt{nh_n}(\hat{\nu}_+ - \nu_+) &= \hat{\Psi}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) + \Psi_{+,1}^{-1} \eta_{+,IV} + o_p(1) \\ \eta_{+,IV} &:= \frac{C}{2} g_+^{(2)}(0) \Omega_+, \quad \Omega_+ := f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1}, \gamma_{2,1} \mu_{+,z}(0)^\top)^\top. \end{aligned}$$

◀

Lemma F.4 (Local linear bias convergence). *Under the conditions of Theorem 2, for $C = c_h^{5/2}$, we have that,*

$$\begin{aligned} \sqrt{nh_n}(\hat{\beta}_{+,0}^{w_j} - \beta_{+,0}^{w_j}) &= \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_{i,j} - \mu_{+,w_j}(D_i - d^*)\} \right) + \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j} + o_p(1) \\ \eta_{+,w_j} &:= \frac{C}{2} \Lambda_+ \mu_{+,w_j}^{(2)}(0), \quad \Lambda_+ := f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1})^\top. \end{aligned}$$

◀

F.3 Bias correction

To correct the asymptotic bias above, we estimate the bias terms $\eta_{+, \cdot}$ from Proposition 3 (PDD bias) and subtract them from $\hat{\tau}_{\text{pdd}}$.

Recall $\eta_{+,IV} = \frac{C}{2} g_+^{(2)}(0) \Omega_+$. To estimate $\frac{1}{2} g_+^{(2)}(0)$, define $\pi_+ = \mathbf{B}_2 \left(g_+(0), g_+^{(1)}(0), \frac{1}{2} g_+^{(2)}(0) \right)^\top$

and estimate $\hat{\pi}_+$ by solving,

$$0 = \frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \left(Y_i - R_{i,2}^\top \hat{\pi}_+ - W_i^\top \tilde{\gamma}_+ \right) \Rightarrow \mathbf{e}_2^\top \hat{\pi}_+ = \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} Y_i \right),$$

where $\hat{\Psi}_{+,2} := \frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{pmatrix} R_{i,2} R_{i,2}^\top & R_{i,2} W_i^\top \\ Z_i R_{i,2}^\top & Z_i W_i^\top \end{pmatrix}$. Letting $\mathbf{e}_2 = (0, 0, 1)^\top \in \mathbb{R}^3$, we define,

$$\hat{\eta}_{+,IV} := \frac{h_n^2}{2} \hat{\Omega}_+ \hat{g}_+^{(2)}(0), \quad \hat{g}_+^{(2)}(0) := \frac{2}{b_n^2} \mathbf{e}_2^\top \hat{\pi}_+, \quad \hat{\Omega}_+ := \frac{1}{n} \sum_{i=1}^n \omega_{+,i} \begin{pmatrix} R_{i,1} \\ Z_i \end{pmatrix} \left(\frac{D_i - d^*}{h_n} \right)^2.$$

For $\eta_{+,w_j} = \frac{C}{2} \Lambda_+ \mu_{+,w_j}^{(2)}(0)$, we apply the same correction as [Calonico et al. \(2014\)](#). For $\kappa_+^{w_j} = \mathbf{B}_2 \left(\mu_{+,w_j}(0), \mu_{+,w_j}^{(1)}(0), \frac{1}{2} \mu_{+,w_j}^{(2)}(0) \right)^\top$, estimate $\hat{\kappa}_+^{w_j}$ by solving,

$$0 = \frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(W_{i,j} - R_{i,2}^\top \hat{\kappa}_+^{w_j} \right) \Rightarrow \mathbf{e}_2^\top \hat{\kappa}_+^{w_j} = \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} W_{i,j} \right),$$

where $\hat{\Gamma}_{+,2} := \frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} R_{i,2}^\top$. We then define,

$$\hat{\eta}_{+,w_j} := \frac{h_n^2}{2} \hat{\Lambda}_+ \hat{\mu}_{+,w_j}^{(2)}(0), \quad \hat{\mu}_{+,w_j}^{(2)}(0) := \frac{2}{b_n^2} \mathbf{e}_2^\top \hat{\kappa}_+^{w_j}, \quad \hat{\Lambda}_+ := \frac{1}{n} \sum_{i=1}^n \omega_{+,i} R_{i,1} \left(\frac{D_i - d^*}{h_n} \right)^2.$$

The lemmas below characterize the limits of the estimators above.

Lemma F.5 (Λ -convergence). *Under the conditions of Theorem 2, we have that $\hat{\Lambda}_+ \xrightarrow{p} \Lambda_+$, where,*

$$\hat{\Lambda}_+ := \frac{1}{n} \sum_{i=1}^n \omega_{+,i} R_{i,1} \left(\frac{D_i - d^*}{h_n} \right)^2, \quad \Lambda_+ := f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1})^\top.$$

◀

Lemma F.6 (Ω -convergence). *Under the conditions of Theorem 2, we have that $\hat{\Omega}_+ \xrightarrow{p} \Omega_+$, where,*

$$\hat{\Omega}_+ := \frac{1}{n} \sum_{i=1}^n \omega_{+,i} \begin{pmatrix} R_{i,1} \\ Z_i \end{pmatrix} \left(\frac{D_i - d^*}{h_n} \right)^2, \quad \Omega_+ := f_{d,+}(d^*) (\gamma_{2,1}, \gamma_{3,1}, \gamma_{2,1} \mu_{+,z}(0)^\top)^\top.$$

◀

Lemma F.7 ($\Gamma_{+,2}$ -convergence). *Under the conditions of Theorem 2, we have that $\hat{\Gamma}_{+,2} \xrightarrow{p} \Gamma_{+,2}$, where,*

$$\hat{\Gamma}_{+,2} := \frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} R_{i,2}^\top, \quad \Gamma_{+,2} := f_{d,+}(d^*) \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} & \gamma_{2,1} \\ \gamma_{1,1} & \gamma_{2,1} & \gamma_{3,1} \\ \gamma_{2,1} & \gamma_{3,1} & \gamma_{4,1} \end{bmatrix}.$$

◀

Lemma F.8 ($\Psi_{+,2}$ -convergence). *Under the conditions of Theorem 2, we have that $\hat{\Psi}_{+,2} \xrightarrow{p} \Psi_{+,2}$, where*

$$\hat{\Psi}_{+,2} := \frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{pmatrix} R_{i,2} R_{i,2}^\top & R_{i,2} W_i^\top \\ Z_i R_{i,2}^\top & Z_i W_i^\top \end{pmatrix}, \quad \Psi_{+,2} := f_{d,+}(d^*) \begin{pmatrix} \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} & \gamma_{2,1} \\ \gamma_{1,1} & \gamma_{2,1} & \gamma_{3,1} \\ \gamma_{2,1} & \gamma_{3,1} & \gamma_{4,1} \end{bmatrix} & \begin{bmatrix} \gamma_{0,1} \\ \gamma_{1,1} \\ \gamma_{2,1} \end{bmatrix} \mu_{+,w}(0)^\top \\ \mu_{+,z}(0) \begin{bmatrix} \gamma_{0,1} & \gamma_{1,1} & \gamma_{2,1} \end{bmatrix} & \gamma_{0,1} \mu_{+,zw}^\top(0) \end{pmatrix}.$$

◀

We define the bias-corrected PDD estimator $\hat{\tau}_{\text{pdd}}^{\text{rbc}}$ as,

$$\hat{\tau}_{\text{pdd}}^{\text{rbc}} = \hat{\tau}_{\text{pdd}} - \left(\hat{\mathbf{e}}_{+,w}^\top \left(\hat{\Psi}_{+,1}^{-1} \hat{\eta}_{+,IV} - \hat{\Psi}_{-,1}^{-1} \hat{\eta}_{-,IV} \right) + \sum_{j=1}^q (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\eta}_{+,w_j} \right). \quad (6)$$

F.4 Central limit theorem

In this section, we state the final results necessary to reach the conclusion of Theorem 2.

Lemma F.9 ($\hat{\eta}_{IV}$ -decomposition). *Under the conditions of Theorem 2,*

$$\begin{aligned} \hat{\Psi}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,IV} &= \sqrt{nh_n} h_n^2 \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \\ &\quad + \Psi_{+,1}^{-1} \eta_{+,IV} + o_p(1). \end{aligned}$$

◀

Lemma F.10 ($\hat{\eta}_w$ -decomposition). *Under the conditions of Theorem 2,*

$$\begin{aligned} \hat{\Gamma}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,w_j} &= \sqrt{nh_n} h_n^2 \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(W_{i,j} - \mu_{+,w_j}(D_i - d^*) \right) \right) \\ &\quad + \Gamma_{+,1}^{-1} \eta_{+,w_j} + o_p(1). \end{aligned}$$

◀

We collect the scores of the terms above into the vectors below. The blocks paired with $\hat{\Psi}_{+,2}$ and $\hat{\Gamma}_{+,2}$ use $R_{i,2}$ and the kernel $K((D_i - d^*)/b_n)$; the remaining blocks use $R_{i,1}$ and $K((D_i - d^*)/h_n)$.

$$\begin{aligned} \hat{\mathbf{s}}_+ &:= \begin{pmatrix} \left\{ \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \right\}^\top \\ - \left\{ \xi^3 \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \right\}^\top \\ \left\{ (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \right\}^\top \\ \vdots \\ \left\{ (\gamma_{+,q} - \gamma_{-,q}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \right\}^\top \\ - \left\{ \xi^3 (\hat{\gamma}_{+,1} - \hat{\gamma}_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \right\}^\top \\ \vdots \\ - \left\{ \xi^3 (\hat{\gamma}_{+,q} - \hat{\gamma}_{-,q}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \right\}^\top \end{pmatrix} \\ \hat{\mathbf{s}}_- &:= \begin{pmatrix} - \left\{ \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{-,1}^{-1} \right\}^\top \\ \left\{ \xi^3 \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{-,1}^{-1} \hat{\Omega}_- \mathbf{e}_2^\top \hat{\Psi}_{-,2}^{-1} \right\}^\top \end{pmatrix} \\ \hat{\mathbf{s}} &= (\hat{\mathbf{s}}_+^\top, \hat{\mathbf{s}}_-^\top)^\top \\ X_i &:= (X_{i,+}^\top, X_{i,-}^\top)^\top \\ \mathbf{X}_n &:= \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n X_i \end{aligned} \quad \begin{aligned} X_{i,+} &:= \begin{pmatrix} \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \\ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \\ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) R_{i,1} \{W_{i,1} - \mu_{+,w_1}(D_i - d^*)\} \\ \vdots \\ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) R_{i,1} \{W_{i,q} - \mu_{+,w_q}(D_i - d^*)\} \\ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \{W_{i,1} - \mu_{+,w_1}(D_i - d^*)\} \\ \vdots \\ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \{W_{i,q} - \mu_{+,w_q}(D_i - d^*)\} \end{pmatrix} \\ X_{i,-} &:= \begin{pmatrix} \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_-(D_i - d^*) - W_i^\top \gamma_-) \\ \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_-(D_i - d^*) - W_i^\top \gamma_-) \end{pmatrix} \end{aligned}$$

Corollary 2 ($\hat{\tau}_{\text{pdd}}^{\text{rbc}}$ vector characterization). *Under the conditions of Theorem 2, $\sqrt{nh_n}(\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau) = \hat{\mathbf{s}}^\top \mathbf{X}_n + o_p(1)$.*

◀

Using the results above, we are ready to prove Theorem 2.

Proof of Theorem 2. Without loss of generality, assume $\dim(W) = q = 1$; for $q > 1$ the argument is identical with additional cross terms. By Corollary 2, $\sqrt{nh_n}(\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau) = \hat{\mathbf{s}}^\top \mathbf{X}_n + o_p(1)$, where (suppressing the “ $-$ ” terms) $\hat{\mathbf{s}}_+ \in \mathbb{R}^{12}$ and $X_{i,+} \in \mathbb{R}^{12}$ are as below,

$$\begin{aligned} \hat{\mathbf{s}}_+ &:= \begin{pmatrix} \left\{ \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \right\}^\top \\ -\left\{ (\sqrt{nh_n} h_n^2 \xi) \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \right\}^\top \\ \left\{ (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \right\}^\top \\ -\left\{ (\sqrt{nh_n} h_n^2 \xi) (\hat{\gamma}_{+,1} - \hat{\gamma}_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \right\}^\top \end{pmatrix} \\ X_{i,+} &:= \begin{pmatrix} K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{1}(D_i \geq d^*) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \\ K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{1}(D_i \geq d^*) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \\ K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{1}(D_i \geq d^*) R_{i,1} \{W_{i,1} - \mu_{+,w_1}(D_i - d^*)\} \\ K\left(\frac{D_i - d^*}{h_n}\right) \mathbb{1}(D_i \geq d^*) R_{i,2} \{W_{i,1} - \mu_{+,w_1}(D_i - d^*)\} \end{pmatrix} \\ \mathbf{X}_{n,+} &:= \sum_{i=1}^n \frac{1}{\sqrt{nh_n}} X_{i,+} \end{aligned}$$

We proceed in three steps.

- We show $\mathbf{X}_{n,+} \xrightarrow{d} \mathcal{N}(0, \Sigma_{x,+})$ via the Cramér–Wold device, verifying the Lyapunov condition.
- We show $\hat{\mathbf{s}}_+ \xrightarrow{p} \mathbf{s}_+$, so Slutsky gives $\hat{\mathbf{s}}_+^\top \mathbf{X}_{n,+} \xrightarrow{d} \mathcal{N}(0, \mathbf{s}_+^\top \Sigma_{x,+} \mathbf{s}_+)$.
- By symmetry $\hat{\mathbf{s}}_-^\top \mathbf{X}_{n,-} \xrightarrow{d} \mathcal{N}(0, \mathbf{s}_-^\top \Sigma_{x,-} \mathbf{s}_-)$, and by independence across the cutoff we conclude.

i Let $\mathbf{t} = (t_1, \dots, t_{12})^\top$ be a nonzero finite vector and $\zeta > 0$ the constant in Assumption 7d. Group $\mathbf{t}^\top X_{i,+}$ into its four blocks, the second and fourth of which are localized at b_n :

$$\mathbf{t}^\top X_{i,+} = \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) A_i^{(1)} + \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) A_i^{(2)}$$

$$+ \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) B_i^{(1)} + \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) B_i^{(2)},$$

where, writing $u = \frac{D_i - d^*}{h_n}$ and $v = \frac{D_i - d^*}{b_n} = \xi u$,

$$\begin{aligned} A_i^{(1)} &= (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y}, & A_i^{(2)} &= (t_4 + t_5 v + t_6 v^2 + t_7 Z_i) \varepsilon_{i,y}, \\ B_i^{(1)} &= (t_8 + t_9 u) \varepsilon_{i,w_1}, & B_i^{(2)} &= (t_{10} + t_{11} v + t_{12} v^2) \varepsilon_{i,w_1}, \end{aligned}$$

with $\varepsilon_{i,y} = Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+$ and $\varepsilon_{i,w_1} = W_{i,1} - \mu_{+,w_1}(D_i - d^*)$. By the c_r -inequality, which states that $|a + b + c + d|^r \leq 4^{r-1}(|a|^r + |b|^r + |c|^r + |d|^r)$ for $r \geq 1$,

$$|\mathbf{t}^\top X_{i,+}|^{2+\zeta} \leq 4^{1+\zeta} \left(\left| K(u) A_i^{(1)} \right|^{2+\zeta} + \left| K(v) A_i^{(2)} \right|^{2+\zeta} + \left| K(u) B_i^{(1)} \right|^{2+\zeta} + \left| K(v) B_i^{(2)} \right|^{2+\zeta} \right) \mathbb{1}(D_i \geq d^*).$$

We bound the $\frac{1}{h_n} \mathbb{E}[\cdot]$ of each term, starting with $A_i^{(1)} = (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y}$. By the law of iterated expectations, conditioning on D_i , and the substitution $u = \frac{D_i - d^*}{h_n}$,

$$\begin{aligned} & \frac{1}{h_n} \mathbb{E} \left[K(u)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \left| A_i^{(1)} \right|^{2+\zeta} \right] \\ &= \frac{1}{h_n} \mathbb{E} \left[K\left(\frac{D_i - d^*}{h_n}\right)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \mathbb{E} \left[\left| (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y} \right|^{2+\zeta} \mid D_i \right] \right] \\ &= \int_0^\infty K(u)^{2+\zeta} \mathbb{E} \left[\left| (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y} \right|^{2+\zeta} \mid D_i = u h_n + d^* \right] f_d(u h_n + d^*) \, du. \end{aligned}$$

By the c_r -inequality, $\left| (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y} \right|^{2+\zeta} \leq 3^{1+\zeta} (|t_1|^{2+\zeta} |\varepsilon_{i,y}|^{2+\zeta} + |t_2|^{2+\zeta} |u|^{2+\zeta} |\varepsilon_{i,y}|^{2+\zeta} + |t_3|^{2+\zeta} |Z_i \varepsilon_{i,y}|^{2+\zeta})$, so, using Assumption 7d to bound the conditional $(2 + \zeta)$ -moments of Z_i and $\varepsilon_{i,y}$ by M , the conditional expectation is bounded by $M'(1 + |u|^{2+\zeta})$ for a constant M' depending only on $\mathbf{t}$ and M . Since the kernel constants $\gamma_{k,2+\zeta} = \int_0^\infty u^k K(u)^{2+\zeta} \, du$ are finite under Assumption 8, the integrand is dominated by the integrable function $M' f_d(u h_n + d^*) K(u)^{2+\zeta} (1 + |u|^{2+\zeta})$. By continuity of the conditional moments and of f_d (Assumption 7), dominated convergence then gives $\frac{1}{h_n} \mathbb{E} \left[K(u)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \left| A_i^{(1)} \right|^{2+\zeta} \right] \rightarrow q_A^h(\mathbf{t})$, where

$$q_A^h(\mathbf{t}) := f_{d,+}(d^*) \int_0^\infty K(u)^{2+\zeta} \mathbb{E} \left[\left| (t_1 + t_2 u + t_3 Z_i) \varepsilon_{i,y} \right|^{2+\zeta} \mid D_i = d^* \right] \, du < \infty.$$

The term in $B_i^{(1)} = (t_8 + t_9 u) \varepsilon_{i,w_1}$ is handled identically, giving $\frac{1}{h_n} \mathbb{E} \left[K(u)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \left| B_i^{(1)} \right|^{2+\zeta} \right] \rightarrow q_B^h(\mathbf{t}) < \infty$. Both limits are finite linear combinations of the kernel constants $\gamma_{k,2+\zeta}$ and the conditional moments $\mathbb{E}[|Z_i|^{2+\zeta} \mid D_i = d^*]$, $\mathbb{E}[|\varepsilon_{i,y}|^{2+\zeta} \mid D_i = d^*]$, $\mathbb{E}[|\varepsilon_{i,w_1}|^{2+\zeta} \mid D_i = d^*]$, weighted by $f_{d,+}(d^*)$ (all bounded by M under Assumption 7d).

For the two b_n -localized terms, the same steps apply, but the u -substitution introduces the bandwidth ratio. With $v = \xi u$,

$$\begin{aligned} \frac{1}{h_n} \mathbb{E} \left[K(v)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \left| A_i^{(2)} \right|^{2+\zeta} \right] &= \int_0^\infty K(\xi u)^{2+\zeta} \mathbb{E} \left[\left| A^{(2)} \right|^{2+\zeta} \mid D_i = u h_n + d^* \right] f_d(u h_n + d^*) \, du \\ &= \frac{1}{\xi} \int_0^\infty K(v)^{2+\zeta} \mathbb{E} \left[\left| A^{(2)} \right|^{2+\zeta} \mid D_i = \frac{v}{\xi} h_n + d^* \right] f_d\left(\frac{v}{\xi} h_n + d^*\right) \, dv \\ &\rightarrow q_A^b(\mathbf{t}) < \infty, \end{aligned}$$

and similarly $\frac{1}{h_n} \mathbb{E} \left[K(v)^{2+\zeta} \mathbb{1}(D_i \geq d^*) \left| B_i^{(2)} \right|^{2+\zeta} \right] \rightarrow q_B^b(\mathbf{t}) < \infty$, where q_A^b, q_B^b are finite.

The swap of limit and integral uses the boundedness from Assumptions 7 and 8, and existence of the limits uses continuity in Assumption 7.

Combining, $\frac{1}{h_n} \mathbb{E} \left[\left| \mathbf{t}^\top X_{i,+} \right|^{2+\zeta} \right] \rightarrow 4^{1+\zeta} (q_A^h + q_A^b + q_B^h + q_B^b)(\mathbf{t}) < \infty$. The Lyapunov ratio is then

$$\sum_{i=1}^n \mathbb{E} \left[\left| \frac{1}{\sqrt{n} h_n} \mathbf{t}^\top X_{i,+} \right|^{2+\zeta} \right] = \left(\frac{1}{n h_n} \right)^{\zeta/2} \frac{1}{h_n} \mathbb{E} \left[\left| \mathbf{t}^\top X_{i,+} \right|^{2+\zeta} \right] \rightarrow 0,$$

since $\left(\frac{1}{n h_n} \right)^{\zeta/2} \rightarrow 0$ and $\frac{1}{h_n} \mathbb{E} \left[\left| \mathbf{t}^\top X_{i,+} \right|^{2+\zeta} \right]$ converges to a finite limit. This verifies the Lyapunov condition. Noting $\mathbb{E}[X_{i,+}] = 0$ (which follows from Lemma 3 for the IV blocks and from the definition of $\mu_{+,w_1}(\cdot)$ for the w blocks), we conclude $\mathbf{X}_{n,+} \xrightarrow{d} \mathcal{N}(0, \Sigma_{x,+})$, with $\Sigma_{x,+}$ computed in Lemma F.11.

ii We show $\hat{\mathbf{s}}_+ \xrightarrow{P} \mathbf{s}_+$ termwise.

- $\hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \xrightarrow{P} \mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1}$: standard RDD results (Hahn et al., 2001, Section 4.2) give $\hat{\mathbf{e}}_{+,w} \xrightarrow{P} \mathbf{e}_{+,w}$; Lemma F.1 gives $\hat{\Psi}_{+,1} \xrightarrow{P} \Psi_{+,1}$, and writing $\hat{\Psi}_{+,1}^{-1} - \Psi_{+,1}^{-1} = \hat{\Psi}_{+,1}^{-1}(\Psi_{+,1} - \hat{\Psi}_{+,1})\Psi_{+,1}^{-1}$ gives $\hat{\Psi}_{+,1}^{-1} \xrightarrow{P} \Psi_{+,1}^{-1}$. Slutsky concludes.

- $\xi^3 \mathbf{e}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \xrightarrow{P} \xi^3 \mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1} \Omega_+ \mathbf{e}_2^\top \Psi_{+,2}^{-1}$: $\xi = h_n/b_n$ is constant, $\hat{\Psi}_{+,2} \xrightarrow{P} \Psi_{+,2}$ (Lemma F.8, hence $\hat{\Psi}_{+,2}^{-1} \xrightarrow{P} \Psi_{+,2}^{-1}$), and $\hat{\Omega}_+ \xrightarrow{P} \Omega_+$ (Lemma F.6). Slutsky concludes.
- $(\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \xrightarrow{P} (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1}$ by Lemma F.2 and the inverse expansion above.
- $\xi^3 (\hat{\gamma}_{+,1} - \hat{\gamma}_{-,1}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \xrightarrow{P} \xi^3 (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \Lambda_+ \mathbf{e}_2^\top \Gamma_{+,2}^{-1}$: $\hat{\gamma}_+ \xrightarrow{P} \gamma_+$ (Proposition 2, so $\hat{\gamma}_{+,1} - \hat{\gamma}_{-,1} \xrightarrow{P} \gamma_{+,1} - \gamma_{-,1}$), $\hat{\Lambda}_+ \xrightarrow{P} \Lambda_+$ (Lemma F.5), and $\hat{\Gamma}_{+,2} \xrightarrow{P} \Gamma_{+,2}$ (Lemma F.7). Slutsky concludes.

Hence $\hat{\mathbf{s}}_+ \xrightarrow{P} \mathbf{s}_+$, where

$$\mathbf{s}_+ = \left(\mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1}, -\xi^3 \mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1} \Omega_+ \mathbf{e}_2^\top \Psi_{+,2}^{-1}, (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1}, -\xi^3 (\gamma_{+,1} - \gamma_{-,1}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \Lambda_+ \mathbf{e}_2^\top \Gamma_{+,2}^{-1} \right)^\top.$$

By Slutsky, $\hat{\mathbf{s}}_+^\top \mathbf{X}_{n,+} \xrightarrow{d} \mathcal{N}(0, \mathbf{s}_+^\top \Sigma_{x,+} \mathbf{s}_+)$.

- iii Repeating steps (i)–(ii) below the cutoff gives $\hat{\mathbf{s}}_- \xrightarrow{P} \mathbf{s}_-$ and $\mathbf{X}_{n,-} \xrightarrow{d} \mathcal{N}(0, \Sigma_{x,-})$, with $\mathbf{s}_- = (\mathbf{e}_{+,w}^\top \Psi_{-,1}^{-1}, \xi^3 \mathbf{e}_{+,w}^\top \Psi_{-,1}^{-1} \Omega_- \mathbf{e}_2^\top \Psi_{-,2}^{-1})^\top$, hence $\hat{\mathbf{s}}_-^\top \mathbf{X}_{n,-} \xrightarrow{d} \mathcal{N}(0, \mathbf{s}_-^\top \Sigma_{x,-} \mathbf{s}_-)$. By applying Cramer-Wold to the concatenated plus/minus triangular array with block-diagonal covariance, since the $+/-$ terms are uncorrelated⁸ we can conclude that,

$$\sqrt{nh_n} \left(\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau \right) = \hat{\mathbf{s}}_+^\top \mathbf{X}_{n,+} + \hat{\mathbf{s}}_-^\top \mathbf{X}_{n,-} \xrightarrow{d} \mathcal{N}(0, \mathbf{s}_+^\top \Sigma_{x,+} \mathbf{s}_+ + \mathbf{s}_-^\top \Sigma_{x,-} \mathbf{s}_-).$$

We set $V = \mathbf{s}_+^\top \Sigma_{x,+} \mathbf{s}_+ + \mathbf{s}_-^\top \Sigma_{x,-} \mathbf{s}_-$. We estimate V with,

$$\hat{V} = \hat{\mathbf{s}}_+^\top \left(\frac{1}{nh_n} \sum_i \tilde{X}_{i,+} \tilde{X}_{i,+}^\top \right) \hat{\mathbf{s}}_+ + \hat{\mathbf{s}}_-^\top \left(\frac{1}{nh_n} \sum_i \tilde{X}_{i,-} \tilde{X}_{i,-}^\top \right) \hat{\mathbf{s}}_-,$$

where $\tilde{X}_i$ replaces $\varepsilon_{i,y} = Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+$ with the estimate $\hat{\varepsilon}_{i,y} = Y_i - R_{i,2}^\top \hat{\pi}_+ - W_i^\top \hat{\gamma}_+$, and $\varepsilon_{i,w_1} = W_{i,1} - \mu_{+,w_1}(D_i - d^*)$ with the estimate $\hat{\varepsilon}_{i,w_1} = W_{i,1} - R_{i,2}^\top \hat{\kappa}_+$.

□

⁸Recall that we always have either $\omega_{+,i} = 0$ or $\omega_{-,i} = 0$.

Lemma F.11 (PDD variance). *Under the conditions of Theorem 2, $\frac{1}{h_n} \mathbb{E} [X_{i,+} X_{i,+}^\top] \rightarrow \Sigma_{x,+}$ and $\frac{1}{h_n} \mathbb{E} [X_{i,-} X_{i,-}^\top] \rightarrow \Sigma_{x,-}$, where*

$$\Sigma_{x,+} = \begin{pmatrix} \Sigma_{IV,11}^+ & \Sigma_{IV,12}^+ & \mathbf{P}_{IV,w_1,11}^+ & \mathbf{P}_{IV,w_1,12}^+ \\ (\Sigma_{IV,12}^+)^{\top} & \Sigma_{IV,22}^+ & \mathbf{P}_{IV,w_1,21}^+ & \mathbf{P}_{IV,w_1,22}^+ \\ (\mathbf{P}_{IV,w_1,11}^+)^{\top} & (\mathbf{P}_{IV,w_1,21}^+)^{\top} & \Sigma_{w_1,11}^+ & \Sigma_{w_1,12}^+ \\ (\mathbf{P}_{IV,w_1,12}^+)^{\top} & (\mathbf{P}_{IV,w_1,22}^+)^{\top} & (\Sigma_{w_1,12}^+)^{\top} & \Sigma_{w_1,22}^+ \end{pmatrix} \quad \Sigma_{x,-} = \begin{pmatrix} \Sigma_{IV,11}^- & \Sigma_{IV,12}^- \\ (\Sigma_{IV,12}^-)^{\top} & \Sigma_{IV,22}^- \end{pmatrix}.$$

Writing $\bar{\gamma}_{k,2} = \bar{\gamma}_{k,2}(\xi)$, the diagonal blocks are

$$\begin{aligned} \Sigma_{IV,11}^+ &= f_{d,+}(d^*) \begin{pmatrix} \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \\ \gamma_{1,2} & \gamma_{2,2} \end{bmatrix} \sigma_{+,y}^2(0) & \begin{bmatrix} \gamma_{0,2} \\ \gamma_{1,2} \end{bmatrix} \rho_{+,y^2z}(0)^{\top} \\ \rho_{+,y^2z}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \end{bmatrix} & \gamma_{0,2} \rho_{+,y^2zz^{\top}}(0) \end{pmatrix} \\ \Sigma_{IV,22}^+ &= \frac{1}{\xi} f_{d,+}(d^*) \begin{pmatrix} \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \\ \gamma_{1,2} & \gamma_{2,2} & \gamma_{3,2} \\ \gamma_{2,2} & \gamma_{3,2} & \gamma_{4,2} \end{bmatrix} \sigma_{+,y}^2(0) & \begin{bmatrix} \gamma_{0,2} \\ \gamma_{1,2} \\ \gamma_{2,2} \end{bmatrix} \rho_{+,y^2z}(0)^{\top} \\ \rho_{+,y^2z}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \end{bmatrix} & \gamma_{0,2} \rho_{+,y^2zz^{\top}}(0) \end{pmatrix} \\ \Sigma_{w_1,11}^+ &= f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \\ \gamma_{1,2} & \gamma_{2,2} \end{bmatrix}, \quad \Sigma_{w_1,22}^+ = \frac{1}{\xi} f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \\ \gamma_{1,2} & \gamma_{2,2} & \gamma_{3,2} \\ \gamma_{2,2} & \gamma_{3,2} & \gamma_{4,2} \end{bmatrix}, \end{aligned}$$

and the cross-bandwidth blocks are

$$\begin{aligned} \Sigma_{IV,12}^+ &= f_{d,+}(d^*) \begin{pmatrix} \begin{bmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \\ \bar{\gamma}_{1,2} & \xi \bar{\gamma}_{2,2} & \xi^2 \bar{\gamma}_{3,2} \end{bmatrix} \sigma_{+,y}^2(0) & \begin{bmatrix} \bar{\gamma}_{0,2} \\ \bar{\gamma}_{1,2} \end{bmatrix} \rho_{+,y^2z}(0)^{\top} \\ \rho_{+,y^2z}(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \end{bmatrix} & \bar{\gamma}_{0,2} \rho_{+,y^2zz^{\top}}(0) \end{pmatrix} \\ \Sigma_{w_1,12}^+ &= f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \\ \bar{\gamma}_{1,2} & \xi \bar{\gamma}_{2,2} & \xi^2 \bar{\gamma}_{3,2} \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
\mathbf{P}_{\text{IV},w_1,11}^+ &= f_{d,+}(d^*) \begin{pmatrix} \rho_{+,yw_1}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \\ \gamma_{1,2} & \gamma_{2,2} \end{bmatrix} \\ \rho_{+,yw_1z}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \end{bmatrix} \end{pmatrix}, & \mathbf{P}_{\text{IV},w_1,12}^+ &= f_{d,+}(d^*) \begin{pmatrix} \rho_{+,yw_1}(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \\ \bar{\gamma}_{1,2} & \xi \bar{\gamma}_{2,2} & \xi^2 \bar{\gamma}_{3,2} \end{bmatrix} \\ \rho_{+,yw_1z}(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \end{bmatrix} \end{pmatrix} \\
\mathbf{P}_{\text{IV},w_1,21}^+ &= f_{d,+}(d^*) \begin{pmatrix} \rho_{+,yw_1}(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \bar{\gamma}_{1,2} \\ \xi \bar{\gamma}_{1,2} & \xi \bar{\gamma}_{2,2} \\ \xi^2 \bar{\gamma}_{2,2} & \xi^2 \bar{\gamma}_{3,2} \end{bmatrix} \\ \rho_{+,yw_1z}(0) \begin{bmatrix} \bar{\gamma}_{0,2} & \bar{\gamma}_{1,2} \end{bmatrix} \end{pmatrix}, & \mathbf{P}_{\text{IV},w_1,22}^+ &= \frac{1}{\xi} f_{d,+}(d^*) \begin{pmatrix} \rho_{+,yw_1}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \\ \gamma_{1,2} & \gamma_{2,2} & \gamma_{3,2} \\ \gamma_{2,2} & \gamma_{3,2} & \gamma_{4,2} \end{bmatrix} \\ \rho_{+,yw_1z}(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \end{bmatrix} \end{pmatrix}.
\end{aligned}$$

The blocks of $\Sigma_{x,-}$ are $\Sigma_{\text{IV},11}^-$, $\Sigma_{\text{IV},22}^-$, $\Sigma_{\text{IV},12}^-$, defined as their “+” analogues with all “+” subscripts replaced by “-”.

◀

F.5 Proofs of technical lemmas

Proof of Lemma F.1. Plugging in for $\omega_{+,i}$, $\hat{\Psi}_{+,1} = \frac{1}{nh_n} \sum_i \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{pmatrix} R_{i,1} R_{i,1}^\top & R_{i,1} W_i^\top \\ Z_i R_{i,1}^\top & Z_i W_i^\top \end{pmatrix}$.

We treat a representative entry; the rest are analogous.

- i The $R_{i,1} R_{i,1}^\top$ block converges to $f_{d,+}(d^*) \Gamma_1$ by [Hahn et al. \(1999, Lemma 1\)](#).
- ii For the $R_{i,1} W_{i,j}$ block, set $X_i = \frac{1}{h_n} \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) R_{i,1} W_{i,j} \in \mathbb{R}^2$. For each $\ell \in \{1, 2\}$, we show $\frac{1}{n} \sum_i X_{i,\ell} = \mathbb{E}[X_{i,\ell}] + o_p(1)$ by bounding $\text{Var}(\frac{1}{n} \sum_i X_{i,\ell}) = \frac{1}{n} \text{Var}(X_{1,\ell}) \leq \frac{1}{n} \|X_{1,\ell}\|_{L_2}^2$ and applying Chebyshev’s inequality. Computing the L_2 bound,

$$\begin{aligned}
h_n \mathbb{E}[X_i^2] &= \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right)^2 R_{i,1}^2 W_{i,j}^2 \right] \\
&= \int_0^\infty K(u)^2 \begin{bmatrix} 1 \\ u^2 \end{bmatrix} \mathbb{E} \left[W_{i,j}^2 \mid D_i = u h_n + d^* \right] f_d(u h_n + d^*) \, du \\
&\rightarrow \left(\sigma_{+,w_j}^2(0) + \mu_{+,w_j}(0)^2 \right) f_{d,+}(d^*) \begin{bmatrix} \gamma_{0,2} \\ \gamma_{2,2} \end{bmatrix} < \infty,
\end{aligned}$$

where the squaring is element-wise and the limit/integral swap uses Assumptions 7 and 8. This shows that $\|X_{1,\ell}\|_{L_2}^2 = O(1/h_n)$ and thus $\frac{1}{n} \|X_{1,\ell}\|_{L_2}^2 = O\left(\frac{1}{nh_n}\right) \rightarrow 0$. Then

from Chebyshev's inequality,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n (X_{i,\ell} - \mathbb{E}[X_{i,\ell}])\right| > \epsilon\right) \leq \frac{1}{\epsilon^2} \frac{1}{n} \|X_{1,\ell}\|_{L_2}^2 \rightarrow 0,$$

and the conclusion follows.

iii By Step (ii) it suffices to compute the mean. Using the law of iterated expectations and u -substitution,

$$\begin{aligned} \frac{1}{h_n} \mathbb{E}\left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) R_{i,1} W_{i,j}\right] &= \int_0^\infty K(u) \begin{bmatrix} 1 \\ u \end{bmatrix} \mathbb{E}[W_{i,j} \mid D_i = u h_n + d^*] f_d(u h_n + d^*) \, du \\ &\rightarrow \mu_{+,w_j}(0) f_{d,+}(d^*) \begin{bmatrix} \gamma_{0,1} \\ \gamma_{1,1} \end{bmatrix}. \end{aligned}$$

iv Repeating (ii)–(iii) on the remaining entries gives $\frac{1}{n h_n} \sum_i \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_{i,j} R_{i,1}^\top \xrightarrow{p} \mu_{+,z_j}(0) f_{d,+}(d^*) [\gamma_{0,1}, \gamma_{1,1}]$ and $\frac{1}{n h_n} \sum_i \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) Z_{i,j} W_{i,j} \xrightarrow{p} \mu_{+,z_j w_j}(0) f_{d,+}(d^*) \gamma_{0,1}$.

Combining yields $\hat{\Psi}_{+,1} \xrightarrow{p} \Psi_{+,1}$.

□

Proof of Lemma F.2. Under our conditions this is identical to [Hahn et al. \(1999, Lemma 1\)](#). □

Proof of Lemma F.3. From Equation (5), the first term matches the lemma statement. For the second, since $\hat{\Psi}_{+,1}^{-1} \xrightarrow{p} \Psi_{+,1}^{-1}$ (Lemma F.1), it suffices to show

$$\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \left(g_+(D_i - d^*) - R_{i,1}^\top \alpha_+ \right) \xrightarrow{p} \eta_{+,IV}.$$

Since $\alpha_+ = \mathbf{H}_1(g_+(0), g_+^{(1)}(0))^\top$, we have $R_{i,1}^\top \alpha_+ = g_+(0) + g_+^{(1)}(0)(D_i - d^*)$, the first-order Taylor polynomial of g_+ at 0. By the smoothness of g_+ in Assumption 7, a second-order Taylor expansion with Lagrange remainder gives, for some $\tilde{D}_i - d^* \in [0, D_i - d^*]$,

$$g_+(D_i - d^*) = g_+(0) + g_+^{(1)}(0)(D_i - d^*) + \frac{1}{2} g_+^{(2)}(\tilde{D}_i - d^*)(D_i - d^*)^2,$$

so that $g_+(D_i - d^*) - R_{i,1}^\top \alpha_+ = \frac{1}{2} g_+^{(2)}(\tilde{D}_i - d^*)(D_i - d^*)^2$. Substituting $\omega_{i,+} = \frac{1}{h_n} \mathbb{1}(D_i \geq d^*) K((D_i - d^*)/h_n)$ and writing $(D_i - d^*)^2 = h_n^2 \left(\frac{D_i - d^*}{h_n} \right)^2$, the left-hand side becomes

$$\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \frac{1}{2} g_+^{(2)}(\tilde{D}_i - d^*)(D_i - d^*)^2 = \sqrt{nh_n} h_n^2 \cdot \frac{1}{nh_n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \frac{1}{2} g_+^{(2)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{h_n}\right)^2$$

By a Chebyshev argument identical to Lemma F.1(ii), the average equals its expectation up to $o_p(1)$. By the law of iterated expectations, conditioning on D_i (so that $\mathbb{E}[Z_i | D_i] = \mu_{+,z}(D_i - d^*)$ for $D_i \geq d^*$), and the substitution $u = \frac{D_i - d^*}{h_n}$, with $\tilde{u} = \frac{\tilde{D}_i - d^*}{h_n} \in [0, u]$,

$$\begin{aligned} & \frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \frac{1}{2} g_+^{(2)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{h_n}\right)^2 \right] \\ &= \frac{1}{h_n} \int_{d^*}^{\infty} K\left(\frac{d - d^*}{h_n}\right) \begin{bmatrix} 1 \\ \frac{d - d^*}{h_n} \\ \mu_{+,z}(d - d^*) \end{bmatrix} \frac{1}{2} g_+^{(2)}(\tilde{d} - d^*) \left(\frac{d - d^*}{h_n}\right)^2 f_d(d) \, dd \\ &= \int_0^{\infty} K(u) \begin{bmatrix} 1 \\ u \\ \mu_{+,z}(uh_n) \end{bmatrix} \frac{1}{2} g_+^{(2)}(\tilde{u}h_n) u^2 f_d(uh_n + d^*) \, du \\ &\rightarrow \frac{1}{2} g_+^{(2)}(0) f_{d,+}(d^*) \int_0^{\infty} K(u) \begin{bmatrix} 1 \\ u \\ \mu_{+,z}(0) \end{bmatrix} u^2 \, du \\ &= \frac{1}{2} g_+^{(2)}(0) f_{d,+}(d^*) \begin{bmatrix} \gamma_{2,1} \\ \gamma_{3,1} \\ \gamma_{2,1} \mu_{+,z}(0) \end{bmatrix} = \frac{1}{2} g_+^{(2)}(0) \Omega_+, \end{aligned}$$

where the limit uses $\tilde{u}h_n \rightarrow 0$, so that $g_+^{(2)}(\tilde{u}h_n) \rightarrow g_+^{(2)}(0)$, together with $\mu_{+,z}(uh_n) \rightarrow \mu_{+,z}(0)$ and $f_d(uh_n + d^*) \rightarrow f_{d,+}(d^*)$, and the limit/integral swap is justified by the boundedness in Assumptions 7 and 8. Hence the average converges in probability to $\frac{1}{2} g_+^{(2)}(0) \Omega_+$,

and multiplying by $\sqrt{nh_n}h_n^2 \rightarrow C = c_h^{5/2}$ and applying Slutsky's theorem gives the second term $\xrightarrow{p} \frac{C}{2}g_+^{(2)}(0)\Omega_+ = \eta_{+,IV}$. $\square$

Proof of Lemma F.4. The argument follows Lemma F.3, replacing the IV score by the local-linear w score. A second-order Taylor expansion of μ_{+,w_j} gives the remainder $\frac{1}{2}\mu_{+,w_j}^{(2)}(\tilde{D}_i - d^*)(D_i - d^*)^2$, and the same Chebyshev and u -substitution steps give

$$\frac{1}{h_n}\mathbb{E}\left[\mathbb{1}(D_i \geq d^*)K\left(\frac{D_i - d^*}{h_n}\right)R_{i,1}\frac{1}{2}\mu_{+,w_j}^{(2)}(\tilde{D}_i - d^*)\left(\frac{D_i - d^*}{h_n}\right)^2\right] \rightarrow \frac{1}{2}\mu_{+,w_j}^{(2)}(0)\Lambda_+.$$

Multiplying by $\sqrt{nh_n}h_n^2 \rightarrow C$ and applying Slutsky's theorem gives that $\left(\sqrt{\frac{h_n}{n}}\sum_{i=1}^n\omega_{+,i}R_{i,1}\left\{\mu_{+,w}(D_i - d^*)^\top - \eta_{+,w_j}\right\}\right) \rightarrow 0$. $\square$

Proof of Lemma F.5. We proceed in steps. Some steps of this proof are identical to Lemma F.2 and F.3, so for brevity we only summarize those steps.

i Expanding the sum, note that we can write,

$$\hat{\Lambda}_+ = \frac{1}{nh_n}\sum_{i=1}^n\mathbb{1}(D_i \geq d^*)K\left(\frac{D_i - d^*}{h_n}\right)R_{i,1}\left(\frac{D_i - d^*}{h_n}\right)^2.$$

ii Following the same argument as in Lemma F.2(ii) and applying Chebyshev's inequality, we can show that,

$$\begin{aligned} & \frac{1}{nh_n}\sum_{i=1}^n\mathbb{1}(D_i \geq d^*)K\left(\frac{D_i - d^*}{h_n}\right)R_{i,1}\left(\frac{D_i - d^*}{h_n}\right)^2 \\ &= \frac{1}{h_n}\mathbb{E}\left[\mathbb{1}(D_i \geq d^*)K\left(\frac{D_i - d^*}{h_n}\right)R_{i,1}\left(\frac{D_i - d^*}{h_n}\right)^2\right] + o_p(1). \end{aligned}$$

iii Evaluating the integral in the step above, we find that,

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{h_n}\mathbb{E}\left[\mathbb{1}(D_i \geq d^*)K\left(\frac{D_i - d^*}{h_n}\right)R_{i,1}\left(\frac{D_i - d^*}{h_n}\right)^2\right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{h_n} \int_0^\infty K\left(\frac{d - d^*}{h_n}\right)\left(\frac{1}{\left(\frac{d - d^*}{h_n}\right)}\right)\left(\frac{d - d^*}{h_n}\right)^2 f_d(d) \, dd \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \int_0^\infty K(u) \begin{pmatrix} 1 \\ u \end{pmatrix} u^2 f_d(uh_n + d^*) \, du \quad (\text{I}) \\
&= f_{d,+}(d^*)(\gamma_{2,1}, \gamma_{3,1})^\top,
\end{aligned}$$

where (I) uses u -substitution. The swapping of the limits uses the boundedness from Assumptions 7 and 8, and the existence of the limits uses the continuity conditions in Assumption 7.

iv Combining steps (ii) and (iii), it follows that $\hat{\Lambda}_+ \xrightarrow{P} \Lambda_+$.

□

Proof of Lemma F.6. The argument follows Lemma F.5, with the extra coordinate Z_i . Expanding, $\hat{\Omega}_+ =$

$$\begin{aligned}
&\frac{1}{nh_n} \sum_i \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{pmatrix} R_{i,1} \\ Z_i \end{pmatrix} \left(\frac{D_i - d^*}{h_n}\right)^2. \text{ By the Chebyshev and } u\text{-substitution argument,} \\
&\frac{1}{h_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{pmatrix} R_{i,1} \\ Z_i \end{pmatrix} \left(\frac{D_i - d^*}{h_n}\right)^2 \right] \rightarrow \int_0^\infty K(u) \begin{pmatrix} 1 \\ u \\ \mu_{+,z}(uh_n) \end{pmatrix} u^2 f_d(uh_n + d^*) \, du = \Omega_+,
\end{aligned}$$

so $\hat{\Omega}_+ \xrightarrow{P} \Omega_+$.

□

Proof of Lemma F.7. Plugging in $\delta_{+,i}$, $\hat{\Gamma}_{+,2} = \frac{1}{nb_n} \sum_i \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} R_{i,2}^\top$. By the same Chebyshev and u -substitution argument, now substituting $v = \frac{D_i - d^*}{b_n}$ at the bandwidth b_n ,

$$\frac{1}{b_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} R_{i,2}^\top \right] \rightarrow f_{d,+}(d^*) \int_0^\infty K(v) \begin{pmatrix} 1 & v & v^2 \\ v & v^2 & v^3 \\ v^2 & v^3 & v^4 \end{pmatrix} dv = \Gamma_{+,2},$$

so $\hat{\Gamma}_{+,2} \xrightarrow{P} \Gamma_{+,2}$.

□

Proof of Lemma F.8. The argument follows Lemma F.1, now localized at b_n via $\delta_{+,i}$ and with the additional quadratic coordinate of $R_{i,2}$. Each entry converges by the Chebyshev and u -substitution argument with $v = \frac{D_i - d^*}{b_n}$, giving $\hat{\Psi}_{+,2} \xrightarrow{p} \Psi_{+,2}$. $\square$

Proof of Lemma F.9. We proceed in steps. Recall that we defined,

$$\sqrt{nh_n} \hat{\eta}_{+,IV} = \sqrt{nh_n} h_n^2 \hat{\Omega}_+ \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\pi}_+.$$

i First, we rewrite the estimator $\frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\pi}_+$. We have that,

$$\begin{aligned} & \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\pi}_+ \\ &= \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} Y_i \right) \\ &= \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \\ &\quad + \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \left(g_+(D_i - d^*) - g_+(0) - g_+^{(1)}(0)(D_i - d^*) - \frac{1}{2} g_+^{(2)}(0)(D_i - d^*)^2 \right) \right) \\ &\quad + \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \left(g_+(0) + g_+^{(1)}(0)(D_i - d^*) + \frac{1}{2} g_+^{(2)}(0)(D_i - d^*)^2 + W_i^\top \gamma_+ \right) \right). \end{aligned}$$

The first term matches the term in the lemma statement. We treat the remaining two terms individually.

ii We show that the second term converges to zero in probability. Using the continuous differentiability of $g_+(\cdot)$, for some $\tilde{D}_i - d^* \in [0, D_i - d^*]$, a Taylor expansion gives,

$$g_+(D_i - d^*) = g_+(0) + g_+^{(1)}(0)(D_i - d^*) + \frac{1}{2} g_+^{(2)}(0)(D_i - d^*)^2 + \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*)(D_i - d^*)^3.$$

Substituting into the second term,

$$\mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \left(g_+(D_i - d^*) - g_+(0) - g_+^{(1)}(0)(D_i - d^*) - \frac{1}{2} g_+^{(2)}(0)(D_i - d^*)^2 \right) \right)$$

$$= \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \right).$$

By Lemma F.8, $\hat{\Psi}_{+,2} \xrightarrow{p} \Psi_{+,2}$, and writing $\hat{\Psi}_{+,2}^{-1} - \Psi_{+,2}^{-1} = \hat{\Psi}_{+,2}^{-1}(\Psi_{+,2} - \hat{\Psi}_{+,2})\Psi_{+,2}^{-1}$ gives $\hat{\Psi}_{+,2}^{-1} \xrightarrow{p} \Psi_{+,2}^{-1}$. Hence it suffices to show that

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \xrightarrow{p} 0.$$

Using a Chebyshev argument as in Lemma F.3(ii), the average over the bandwidth b_n equals its expectation up to $o_p(1)$,

$$\begin{aligned} & \frac{1}{nb_n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \\ &= \frac{1}{b_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \right] + o_p(1). \end{aligned}$$

Computing the expectation, for some $\tilde{u} \in [0, u]$ and substituting $u = \frac{d-d^*}{b_n}$,

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{b_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{6} \int_0^\infty K(u) \begin{bmatrix} 1 \\ u \\ u^2 \\ \mu_{+,z}(ub_n) \end{bmatrix} g_+^{(3)}(\tilde{u}b_n) u^3 f_d(ub_n + d^*) du \\ &= f_{d,+}(d^*) \frac{g_+^{(3)}(0)}{6} (\gamma_{3,1}, \gamma_{4,1}, \gamma_{5,1}, \mu_{+,z}(0)^\top \gamma_{3,1})^\top < \infty, \end{aligned}$$

where the limit/integral swap uses Assumptions 7 and 8. Hence the average converges

in probability to a finite constant $\tilde{C}$, and since $b_n \rightarrow 0$, Slutsky's theorem gives

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \frac{1}{6} g_+^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 = b_n \cdot \tilde{C} + o_p(1) \xrightarrow{p} 0.$$

iii We show the third term equals $\frac{1}{2} g_+^{(2)}(0)$. Collecting the polynomial into $R_{i,2}$ and W_i

and using $\mathbf{B}_2 = \text{Diag}(1, b_n, b_n^2)$,

$$\begin{aligned} & \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \left(g_+(0) + g_+^{(1)}(0)(D_i - d^*) + \frac{1}{2} g_+^{(2)}(0)(D_i - d^*)^2 + W_i^\top \gamma_+ \right) \right) \\ &= \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \begin{bmatrix} R_{i,2}^\top & W_i^\top \end{bmatrix} \begin{pmatrix} g_+(0) \\ b_n g_+^{(1)}(0) \\ b_n^2 \frac{1}{2} g_+^{(2)}(0) \\ \gamma_+ \end{pmatrix} \right) \\ &= \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \hat{\Psi}_{+,2} \begin{pmatrix} g_+(0) \\ b_n g_+^{(1)}(0) \\ b_n^2 \frac{1}{2} g_+^{(2)}(0) \\ \gamma_+ \end{pmatrix} = \frac{1}{b_n^2} \mathbf{e}_2^\top \begin{pmatrix} g_+(0) \\ b_n g_+^{(1)}(0) \\ b_n^2 \frac{1}{2} g_+^{(2)}(0) \\ \gamma_+ \end{pmatrix} = \frac{1}{2} g_+^{(2)}(0). \end{aligned}$$

iv Combining steps (i)–(iii),

$$\begin{aligned} \sqrt{nh_n} \hat{\eta}_{+,IV} &= \sqrt{nh_n} h_n^2 \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \\ &\quad + \sqrt{nh_n} h_n^2 \hat{\Omega}_+ \cdot O(b_n) + \sqrt{nh_n} h_n^2 \hat{\Omega}_+ \frac{1}{2} g_+^{(2)}(0). \end{aligned}$$

By Lemma F.6, $\hat{\Omega}_+ \xrightarrow{p} \Omega_+$, and since $h_n \asymp n^{-1/5}$, $\sqrt{nh_n} h_n^2 \rightarrow C = c_h^{5/2}$. Hence by Slutsky's theorem the middle term is $o_p(1)$ and the last term converges to $\frac{C}{2} \Omega_+ g_+^{(2)}(0) = \eta_{+,IV}$. Pre-multiplying by $\hat{\Psi}_{+,1}^{-1} \xrightarrow{p} \Psi_{+,1}^{-1}$ gives the statement.

□

Proof of Lemma F.10. We proceed in steps, following the structure of Lemma F.9. Recall

that we defined,

$$\sqrt{nh_n}\hat{\eta}_{+,w_j} = \sqrt{nh_n}h_n^2\hat{\Lambda}_+ \frac{1}{b_n^2}\mathbf{e}_2^\top \hat{\kappa}_+^{w_j}.$$

i Rewriting the estimator $\frac{1}{b_n^2}\mathbf{e}_2^\top \hat{\kappa}_+^{w_j}$,

$$\begin{aligned} & \frac{1}{b_n^2}\mathbf{e}_2^\top \hat{\kappa}_+^{w_j} \\ &= \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} W_{i,j} \right) \\ &= \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(W_{i,j} - \mu_{+,w_j}(D_i - d^*) \right) \right) \\ & \quad + \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(\mu_{+,w_j}(D_i - d^*) - \mu_{+,w_j}(0) - \mu_{+,w_j}^{(1)}(0)(D_i - d^*) - \frac{1}{2}\mu_{+,w_j}^{(2)}(0)(D_i - d^*)^2 \right) \right) \\ & \quad + \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(\mu_{+,w_j}(0) + \mu_{+,w_j}^{(1)}(0)(D_i - d^*) + \frac{1}{2}\mu_{+,w_j}^{(2)}(0)(D_i - d^*)^2 \right) \right). \end{aligned}$$

The first term matches the term in the lemma statement. We treat the remaining two terms individually.

ii We show the second term converges to zero in probability. Using the continuous differentiability of $\mu_{+,w_j}(\cdot)$, for some $\tilde{D}_i - d^* \in [0, D_i - d^*]$,

$$\mu_{+,w_j}(D_i - d^*) = \mu_{+,w_j}(0) + \mu_{+,w_j}^{(1)}(0)(D_i - d^*) + \frac{1}{2}\mu_{+,w_j}^{(2)}(0)(D_i - d^*)^2 + \frac{1}{6}\mu_{+,w_j}^{(3)}(\tilde{D}_i - d^*)(D_i - d^*)^3.$$

Substituting, the second term equals

$$\mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \frac{1}{6}\mu_{+,w_j}^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \right).$$

By Lemma F.7, $\hat{\Gamma}_{+,2}^{-1} \xrightarrow{P} \Gamma_{+,2}^{-1}$, so it suffices to show the average converges to zero. By a Chebyshev argument as in Lemma F.3(ii) and u -substitution at the bandwidth b_n , for some $\tilde{u} \in [0, u]$,

$$\begin{aligned} & \frac{1}{b_n} \mathbb{E} \left[\mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \frac{1}{6}\mu_{+,w_j}^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 \right] \\ & \rightarrow f_{d,+}(d^*) \frac{\mu_{+,w_j}^{(3)}(0)}{6} (\gamma_{3,1}, \gamma_{4,1}, \gamma_{5,1})^\top < \infty, \end{aligned}$$

so the average converges to a finite constant $\tilde{C}$, and since $b_n \rightarrow 0$, Slutsky's theorem gives

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \frac{1}{6} \mu_{+,w_j}^{(3)}(\tilde{D}_i - d^*) \left(\frac{D_i - d^*}{b_n}\right)^3 = b_n \cdot \tilde{C} + o_p(1) \xrightarrow{p} 0.$$

iii We show the third term equals $\frac{1}{2} \mu_{+,w_j}^{(2)}(0)$. Collecting the polynomial into $R_{i,2}$,

$$\begin{aligned} & \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} \left(\mu_{+,w_j}(0) + \mu_{+,w_j}^{(1)}(0)(D_i - d^*) + \frac{1}{2} \mu_{+,w_j}^{(2)}(0)(D_i - d^*)^2 \right) \right) \\ &= \frac{1}{b_n^2} \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \delta_{+,i} R_{i,2} R_{i,2}^\top \right) \begin{pmatrix} \mu_{+,w_j}(0) \\ b_n \mu_{+,w_j}^{(1)}(0) \\ b_n^2 \frac{1}{2} \mu_{+,w_j}^{(2)}(0) \end{pmatrix} = \frac{1}{b_n^2} \mathbf{e}_2^\top \begin{pmatrix} \mu_{+,w_j}(0) \\ b_n \mu_{+,w_j}^{(1)}(0) \\ b_n^2 \frac{1}{2} \mu_{+,w_j}^{(2)}(0) \end{pmatrix} = \frac{1}{2} \mu_{+,w_j}^{(2)}(0). \end{aligned}$$

iv Combining steps (i)–(iii),

$$\begin{aligned} \sqrt{nh_n} \hat{\eta}_{+,w_j} &= \sqrt{nh_n} h_n^2 \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} (W_{i,j} - \mu_{+,w_j}(D_i - d^*)) \right) \\ &\quad + \sqrt{nh_n} h_n^2 \hat{\Lambda}_+ \cdot O(b_n) + \sqrt{nh_n} h_n^2 \hat{\Lambda}_+ \frac{1}{2} \mu_{+,w_j}^{(2)}(0). \end{aligned}$$

By Lemma F.5, $\hat{\Lambda}_+ \xrightarrow{p} \Lambda_+$, and $\sqrt{nh_n} h_n^2 \rightarrow C$. Hence by Slutsky's theorem the middle term is $o_p(1)$ and the last term converges to $\frac{C}{2} \Lambda_+ \mu_{+,w_j}^{(2)}(0) = \eta_{+,w_j}$. Pre-multiplying by $\hat{\Gamma}_{+,1}^{-1} \xrightarrow{p} \Gamma_{+,1}^{-1}$ gives the statement.

□

Proof of Corollary 2. Starting from the definition of $\hat{\tau}_{\text{pdd}}^{\text{rbc}}$ in Equation (6) and the bias decomposition of $\hat{\tau}_{\text{pdd}} - \tau_{\text{pdd}}$,

$$\begin{aligned} \sqrt{nh_n} (\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau) &= \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_+ - \nu_+\} - \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_- - \nu_-\} + \sum_{j=1}^q (\gamma_{+,j} - \gamma_{-,j}) \sqrt{nh_n} (\hat{\beta}_{+,0}^{w_j} - \beta_{+,0}^{w_j}) \\ &\quad - \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,IV} + \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{-,1}^{-1} \sqrt{nh_n} \hat{\eta}_{-,IV} - \sum_{j=1}^q (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,w_j}. \end{aligned}$$

We substitute the decompositions from Lemmas F.3, F.4, F.9, and F.10. For the “+” IV

terms, Lemmas F.3 and F.9 give

$$\begin{aligned}
& \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\gamma}_+ - \gamma_+\} - \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,IV} \\
&= \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \boldsymbol{\varepsilon}_{i,y}^+ \right) + \hat{\mathbf{e}}_{+,w}^\top \Psi_{+,1}^{-1} \eta_{+,IV} \\
&\quad - \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} h_n^2 \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \boldsymbol{\varepsilon}_{i,y}^+ \right) - \hat{\mathbf{e}}_{+,w}^\top \Psi_{+,1}^{-1} \eta_{+,IV} + o_p(1),
\end{aligned}$$

where $\boldsymbol{\varepsilon}_{i,y}^+ = Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+$. The two $\hat{\mathbf{e}}_{+,w}^\top \Psi_{+,1}^{-1} \eta_{+,IV}$ terms cancel. The “-” IV terms are handled identically, and the bias terms there cancel in the same way.

For the w terms, Lemmas F.4 and F.10 give, for each j ,

$$\begin{aligned}
& (\gamma_{+,j} - \gamma_{-,j}) \sqrt{nh_n} (\hat{\beta}_{+,0}^{w_j} - \beta_{+,0}^{w_j}) - (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \sqrt{nh_n} \hat{\eta}_{+,w_j} \\
&= (\gamma_{+,j} - \gamma_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \boldsymbol{\varepsilon}_{i,w_j}^+ \right) + (\gamma_{+,j} - \gamma_{-,j}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j} \\
&\quad - (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \sqrt{nh_n} h_n^2 \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \left(\frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} R_{i,2} \boldsymbol{\varepsilon}_{i,w_j}^+ \right) - (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j} + o_p(1),
\end{aligned}$$

where $\boldsymbol{\varepsilon}_{i,w_j}^+ = W_{i,j} - \mu_{+,w_j}(D_i - d^*)$. Since $\hat{\gamma}_\pm \xrightarrow{P} \gamma_\pm$ by Proposition 2 and $\mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j}$ is a constant, the difference of the two bias terms is $[(\gamma_{+,j} - \gamma_{-,j}) - (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j})] \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j} = o_p(1)$.

What remains are the main scores at h_n and the bias-correction scores at b_n . For each bias-correction score we apply the identity

$$\begin{aligned}
& \sqrt{nh_n} h_n^2 \cdot \frac{1}{nb_n^2} \sum_{i=1}^n \delta_{+,i} [\dots] \\
&= \frac{h_n^3}{b_n^3} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) [\dots] = \xi^3 \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) [\dots],
\end{aligned}$$

which holds because $\delta_{+,i} = \frac{1}{b_n} \mathbb{1}(D_i \geq d^*) K((D_i - d^*)/b_n)$ and $\sqrt{nh_n} h_n^2 \cdot \frac{\sqrt{nh_n}}{nb_n^3} = h_n^3/b_n^3 = \xi^3$.

Collecting the surviving terms,

$$\begin{aligned}
\sqrt{nh_n}(\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau) &= \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \varepsilon_{i,y}^+ \\
&\quad - \xi^3 \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{+,1}^{-1} \hat{\Omega}_+ \mathbf{e}_2^\top \hat{\Psi}_{+,2}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \varepsilon_{i,y}^+ \\
&\quad + \sum_{j=1}^q (\gamma_{+,j} - \gamma_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{h_n}\right) R_{i,1} \varepsilon_{i,w_j}^+ \\
&\quad - \sum_{j=1}^q \xi^3 (\hat{\gamma}_{+,j} - \hat{\gamma}_{-,j}) \mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \hat{\Lambda}_+ \mathbf{e}_2^\top \hat{\Gamma}_{+,2}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i \geq d^*) K\left(\frac{D_i - d^*}{b_n}\right) R_{i,2} \varepsilon_{i,w_j}^+ \\
&\quad - \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{-,1}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{h_n}\right) \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \varepsilon_{i,y}^- \\
&\quad + \xi^3 \hat{\mathbf{e}}_{+,w}^\top \hat{\Psi}_{-,1}^{-1} \hat{\Omega}_- \mathbf{e}_2^\top \hat{\Psi}_{-,2}^{-1} \cdot \frac{1}{\sqrt{nh_n}} \sum_{i=1}^n \mathbb{1}(D_i < d^*) K\left(\frac{D_i - d^*}{b_n}\right) \begin{bmatrix} R_{i,2} \\ Z_i \end{bmatrix} \varepsilon_{i,y}^- + o_p(1).
\end{aligned}$$

Each coefficient is the corresponding block of $\hat{\mathbf{s}}$ and each average the corresponding block of $\mathbf{X}_n$, so $\sqrt{nh_n}(\hat{\tau}_{\text{pdd}}^{\text{rbc}} - \tau) = \hat{\mathbf{s}}^\top \mathbf{X}_n + o_p(1)$. $\square$

Proof of proposition 3. Using the decomposition in Equation (5), we can express,

$$\sqrt{nh_n}(\hat{\tau}_{\text{pdd}} - \tau_{\text{pdd}}) = \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_+ - \nu_+\} - \hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_- - \nu_-\} + (\gamma_+ - \gamma_-)^\top \sqrt{nh_n} (\hat{\beta}_{+,0}^w - \beta_{+,0}^w),$$

where $\mathbf{e}_{+,w} = (1, 0, \lim_{\epsilon \rightarrow 0^+} \mathbb{E}[W_i^\top \mid D = d^* + \epsilon])^\top \in \mathbb{R}^{2+q}$ and $\hat{\mathbf{e}}_{+,w} = \left(1, 0, \left(\hat{\beta}_{+,0}^w\right)^\top\right)^\top \in \mathbb{R}^{2+q}$.

We focus on each of these three terms individually.

i First, we show that,

$$\mathbb{E} \left[\hat{\Psi}_{+,1}^{-1} \left(\frac{1}{n} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \right] \rightarrow 0.$$

Using Lemma 3 (factuals), we have by the law of iterated expectations,

$$\mathbb{E} \left[\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \right]$$

$$\begin{aligned}
&= \mathbb{E} \left[\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} \mathbb{E} [Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+ \mid D, Z] \right) \right] \\
&= 0.
\end{aligned}$$

Therefore, we can conclude that,

$$\begin{aligned}
&\mathbb{E} \left[\hat{\Psi}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \right] \\
&= \mathbb{E} \left[\Psi_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) \right] + o_p(1) \\
&= o_p(1),
\end{aligned}$$

where formally, the second line uses that $\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{i,+} \begin{bmatrix} R_{i,1} \\ Z_i \end{bmatrix} (Y_i - g_+(D_i - d^*) - W_i^\top \gamma_+) \right) = O(1)$, which is shown in the proof of Theorem 2, and that $\hat{\Psi}_{+,1}^{-1} \xrightarrow{p} \Psi_{+,1}^{-1}$, as argued in the proof of Lemma F.3. Therefore, this term is asymptotically unbiased.

ii Next, we demonstrate that,

$$\mathbb{E} \left[\mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_{i,j} - \mu_{+,w_j}(D_i - d^*)^\top\} \right) \right] \rightarrow 0.$$

Using the law of iterated expectations, it follows that,

$$\begin{aligned}
&\mathbb{E} \left[\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_{i,j} - \mu_{+,w_j}(D_i - d^*)^\top\} \right) \right] \\
&= \mathbb{E} \left[\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \mathbb{E} [W_{i,j} - \mu_{+,w_j}(D_i - d^*)^\top \mid D_i = d] \right) \right] \\
&= 0,
\end{aligned}$$

Therefore, we can conclude that,

$$\mathbb{E} \left[\mathbf{e}_0^\top \hat{\Gamma}_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_i^\top - \mu_{+,w}(D_i - d^*)^\top\} \right) \right]$$

$$\begin{aligned}
&= \mathbb{E} \left[\mathbf{e}_0^\top \Gamma_{+,1}^{-1} \left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_i^\top - \mu_{+,w}(D_i - d^*)^\top\} \right) \right] + o_p(1) \\
&= o_p(1),
\end{aligned}$$

where formally, the second line uses that $\left(\sqrt{\frac{h_n}{n}} \sum_{i=1}^n \omega_{+,i} R_{i,1} \{W_i^\top - \mu_{+,w}(D_i - d^*)^\top\} \right) = O(1)$, which is shown in the proof of Theorem 2, and that $\hat{\Psi}_{+,1}^{-1} \xrightarrow{p} \Psi_{+,1}^{-1}$, as argued in the proof of Lemma F.4. Therefore, this term is asymptotically unbiased.

iii Here, we argue that $\mathbb{E}[\hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_+ - \nu_+\}] \xrightarrow{p} \mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1} \eta_{+,IV}$. Standard results from Hahn et al. (2001, Section 4.1) imply that $\hat{\mathbf{e}}_{+,w} \xrightarrow{p} \mathbf{e}_{+,w}$. Therefore, we can write that,

$$\begin{aligned}
\mathbb{E}[\hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_+ - \nu_+\}] &= \mathbf{e}_{+,w}^\top \mathbb{E}[\sqrt{nh_n} \{\hat{\nu}_+ - \nu_+\}] + o_p(1) \\
&= \mathbf{e}_{+,w}^\top \Psi_{+,1}^{-1} \eta_{+,IV},
\end{aligned}$$

where the last line uses Lemma F.3 and our result from step (i).

iv By symmetry, following the argument from the step above we can conclude that

$$\mathbb{E}[\hat{\mathbf{e}}_{+,w}^\top \sqrt{nh_n} \{\hat{\nu}_- - \nu_-\}] = \mathbf{e}_w^\top \Psi_{-,1}^{-1} \eta_{-,IV}.$$

v Finally, we argue that $(\gamma_{+,j} - \gamma_{-,j}) \mathbb{E}[\sqrt{nh_n} (\hat{\beta}_{+,0}^{w_j} - \beta_{+,0}^{w_j})] \xrightarrow{p} (\gamma_{+,j} - \gamma_{-,j}) \mathbf{e}_0^\top \Gamma_{+,1}^{-1} \eta_{+,w_j}$.

This follows directly from Lemma F.4 and our result from step (ii).

vi Writing,

$$\sqrt{nh_n} (\hat{\beta}_{+,0}^w - \beta_{+,0}^w) = \begin{pmatrix} \sqrt{nh_n} (\hat{\beta}_{+,0}^{w_1} - \beta_{+,0}^{w_1}) \\ \vdots \\ \sqrt{nh_n} (\hat{\beta}_{+,0}^{w_q} - \beta_{+,0}^{w_q}) \end{pmatrix},$$

the result follows. □

Proof of Lemma F.11. We prove $\frac{1}{h_n} \mathbb{E}[X_{i,+} X_{i,+}^\top] \rightarrow \Sigma_{x,+}$; the result for $\Sigma_{x,-}$ follows by symmetry. Each entry is the limit of $\frac{1}{h_n} \mathbb{E}[K(a)K(b) \mathbb{1}(D_i \geq d^*) (\text{poly}_a)(\text{poly}_b)(\text{resid}_a)(\text{resid}_b)]$,

where the kernels are localized at h_n or b_n according to the block. We compute three representative blocks; the remaining blocks follow by the same law-of-iterated-expectations and u -substitution argument, with the residual moments $\sigma_{+,y}^2(\cdot)$, $\rho_{+,y^2z}(\cdot)$, $\rho_{+,y^2zz^\top}(\cdot)$, $\sigma_{+,w_1}^2(\cdot)$, $\rho_{+,yw_1}(\cdot)$, $\rho_{+,yw_1z}(\cdot)$ in place of the generic residual term.

For $\Sigma_{w_1,11}^+$, both factors are localized at h_n . With $u = \frac{D_i - d^*}{h_n}$,

$$\begin{aligned} & \frac{1}{h_n} \mathbb{E} \left[K \left(\frac{D_i - d^*}{h_n} \right)^2 \mathbb{1}(D_i \geq d^*) R_{i,1} R_{i,1}^\top \varepsilon_{i,w_1}^2 \right] \\ &= \int_0^\infty K(u)^2 \begin{bmatrix} 1 & u \\ u & u^2 \end{bmatrix} \sigma_{+,w_1}^2(uh_n) f_d(uh_n + d^*) du \\ &\rightarrow f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} \\ \gamma_{1,2} & \gamma_{2,2} \end{bmatrix}. \end{aligned}$$

For $\Sigma_{w_1,22}^+$, both factors are localized at b_n . Substituting $u = \frac{D_i - d^*}{h_n}$ so that $\frac{D_i - d^*}{b_n} = \xi u$, and then $v = \xi u$,

$$\begin{aligned} \frac{1}{h_n} \mathbb{E} \left[K \left(\frac{D_i - d^*}{b_n} \right)^2 \mathbb{1}(D_i \geq d^*) R_{i,2} R_{i,2}^\top \varepsilon_{i,w_1}^2 \right] &= \int_0^\infty K(\xi u)^2 (1, \xi u, (\xi u)^2)^{\otimes 2} \sigma_{+,w_1}^2(uh_n) f_d(uh_n + d^*) du \\ &= \frac{1}{\xi} \int_0^\infty K(v)^2 (1, v, v^2)^{\otimes 2} \sigma_{+,w_1}^2\left(\frac{v}{\xi} h_n\right) f_d\left(\frac{v}{\xi} h_n + d^*\right) dv \\ &\rightarrow \frac{1}{\xi} f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{bmatrix} \gamma_{0,2} & \gamma_{1,2} & \gamma_{2,2} \\ \gamma_{1,2} & \gamma_{2,2} & \gamma_{3,2} \\ \gamma_{2,2} & \gamma_{3,2} & \gamma_{4,2} \end{bmatrix}. \end{aligned}$$

For $\Sigma_{w_1,12}^+$, one factor is localized at h_n and the other at b_n . With $u = \frac{D_i - d^*}{h_n}$ and $\frac{D_i - d^*}{b_n} = \xi u$, the (a, b) entry, with $a \in \{0, 1\}$ from $R_{i,1}$ and $b \in \{0, 1, 2\}$ from $R_{i,2}$, is

$$\begin{aligned} \frac{1}{h_n} \mathbb{E} \left[K \left(\frac{D_i - d^*}{h_n} \right) K \left(\frac{D_i - d^*}{b_n} \right) \mathbb{1}(D_i \geq d^*) \left(\frac{D_i - d^*}{h_n} \right)^a \left(\frac{D_i - d^*}{b_n} \right)^b \varepsilon_{i,w_1}^2 \right] &= \int_0^\infty K(u) K(\xi u) u^a (\xi u)^b \sigma_{+,w_1}^2(uh_n) f_d(uh_n + d^*) du \\ &\rightarrow \xi^b \bar{\gamma}_{a+b,2}(\xi) \sigma_{+,w_1}^2(0) f_{d,+}(d^*), \end{aligned}$$

which assembles into $\Sigma_{w_1,12}^+ = f_{d,+}(d^*) \sigma_{+,w_1}^2(0) \begin{pmatrix} \bar{\gamma}_{0,2} & \xi \bar{\gamma}_{1,2} & \xi^2 \bar{\gamma}_{2,2} \\ \bar{\gamma}_{1,2} & \xi \bar{\gamma}_{2,2} & \xi^2 \bar{\gamma}_{3,2} \end{pmatrix}$. In every case the limit/integral swap uses the boundedness in Assumptions 7 and 8, and the existence of the limits uses

the continuity in Assumption 7. The IV and IV- w blocks follow identically with the corresponding residual moments, giving the matrix in the lemma statement. $\square$

G Details on partial linearity assumption

G.1 Partial linearity

In order for $\hat{\tau}_{\text{pdd}}^{\text{rbc}}$ to be consistent for the RDD treatment effect τ_0 , the confounding bridges $h_+(\cdot, \cdot)$ and $h_-(\cdot, \cdot)$ must be partially linear functions of the placebo outcome W . In other words, for some continuous, nonlinear functions $g_+(\cdot)$ and $g_-(\cdot)$, the confounding bridges must satisfy

$$h_+(d - d^*, w) = g_+(d - d^*) + w^\top \gamma_+, \quad h_-(d - d^*, w) = g_-(d - d^*) + w^\top \gamma_-.$$

We give sufficient conditions under which this restriction is satisfied.

For simplicity, suppose W is a scalar; the result easily extends for multidimensional W .

Lemma G.1 (Sufficient conditions). *Suppose that for some bounded, nonlinear function $f(\cdot)$ that is continuous in some neighborhood around d^* (excluding d^*), the outcome satisfies*

$$Y = f(D) + \beta_u U + \eta_y, \quad \mathbb{E}[\eta_y \mid D, Z, U] = 0.$$

Furthermore, suppose that the placebo outcome satisfies

$$W = \alpha_0 + \alpha_u U + \eta_w, \quad \mathbb{E}[\eta_w \mid D, Z, U] = 0, \quad \alpha_u \neq 0.$$

Then, the confounding bridges $h_+(\cdot, \cdot)$ and $h_-(\cdot, \cdot)$ satisfy equation (1).

Proof of Lemma G.1.

By taking a conditional expectation and rearranging the equation for W ,

$$U = \frac{1}{\alpha_u} (\mathbb{E}[W \mid D, U] - \alpha_0).$$

Then, taking a conditional expectation in the equation for Y ,

$$\begin{aligned}
\mathbb{E}[Y \mid D, U] &= f(D) + \beta_u U \\
&= f(D) + \frac{\beta_u}{\alpha_u} (\mathbb{E}[W \mid D, U] - \alpha_0) \\
&= f(D) - \frac{\beta_u}{\alpha_u} \alpha_0 + \frac{\beta_u}{\alpha_u} \mathbb{E}[W \mid D, U].
\end{aligned}$$

Consider the confounding bridge

$$h_0(D - d^*, W) = g_0(D - d^*) + W\gamma_0, \quad g_0(D - d^*) = f(D) - \frac{\beta_u}{\alpha_u} \alpha_0, \quad \gamma_0 = \frac{\beta_u}{\alpha_u}.$$

It is immediate that $\mathbb{E}[Y \mid D, U] = \mathbb{E}[h_0(D - d^*, W) \mid D, U]$, satisfying Assumption 5. $\square$